Yet Another First Course in Linear Algebra

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Contents

0.1 Preface for Instructors

0.1.1 Features

There are many free elementary linear algebra textbooks to choose from. Here are some features of this one:

- **Lots of exercises.** Most sections have 20 to 50, including some true/false, basic calculations, easy proofs, and a few hard proofs. There's an emphasis on making connections between a graphical view of linear combinations, systems of linear equations, linear transformations, and matrix-vector multiplication.
- **Open source.** You are free to modify your own copy of the LaTeX code as much or as little as you like, distribute your version freely, as long as you use the CC BY-SA 4.0 license and give me credit (eg, "by Ben Smith and Your Name").
- **Lecture notes/textbook combo.** The file `yafcla_notes.pdf` includes big blank spaces to work out examples and some proofs in class. The file `yafcla_book.pdf` includes the same content but with example solutions and all proofs included. In class, the instructor can either annotate the PDF with tablet/projector, or write on the printed notes with a document camera/projector, or record videos for a flipped classroom. Students can write on the printed notes or annotate their copy of the PDF notes. The landscape orientation was chosen because it's the natural orientation of most screens. Instructors probably want to modify the lecture notes to fit their class, by expanding/contracting blank space, or by adding/removing examples or space for proofs.
- **Flexible rigor.** In my experience, expecting linear algebra students to write some simple proofs in HW and tests shakes them out of their recipe-following habits from earlier math classes, and results in much better understanding of the concepts. The subject matter is ripe for lots of simple proof exercises. It's also a good place to learn *about* mathematical rigor, and one principle of mathematical rigor is that every theorem should have a proof, based on previous definitions and theorem; so, in the textbook, every theorem (with one exception in chapter 3) is proven. That said, depending on your students, it's probably not a good idea to spend class time proving everything. Some of the proofs are much harder than anything I'd expect a student to do, and some are tedious and not instructive. For the proofs I teach in my class, there's a blank space in the lecture notes under the theorem, and other proofs are relegated to the textbook. In your class, you may prefer to cover more or fewer proofs. You can even skip assigning proofs for HW altogether.
- **Motivation first.** Where possible, I have tried to include concrete examples to illustrate definitions and theorems *before* the definition/theorem. Sometimes these give an argument why a theorem should true in general, which I've found is very helpful for students at this level, but shouldn't be confused with a "proof" by example. The exposition in chapters 3 and 4 are unconventional, in a motivation-first kind of way. Chapter 3 introduces determinants as *signed volume* of parallelograms/parallelopipeds, and shows how this leads to the traditional cofactor expansion, an approach inspired by Sergei Treil's excellent *Linear Algebra Done Wrong*. Chapter 4 covers linear difference equations and diagonalizability before eigenstuff, which arises naturally as a way to describe linear difference equations.
- **Brisk pace.** At my school, linear algebra is sadly only a 3-unit class which meets twice a week. This makes it hard to finish the material with many textbooks, which are often divided into too many sections, or repeat too much material. This book covers a standard outline of topics, plus an optional proof-writing section, in 25 sections (or, it *will* have 25 sections, after I write chapter 6). The longer sections are divided into subsections, in case your school has shorter class periods.
- **No calculator needed.** Every exercise and example in the book can be done with a reasonable amount of computation by hand. Fractions and big numbers are avoided. Instead of asking questions like "*diagonalize the following* 3×3 *matrix*" which takes many steps and may require a computer, we'll have questions like "*write the characteristic equation of the following matrix*," or "*given that the eigenvalues of the following matrix are 8, 9, and 10, diagonalize it.*" I have a few reasons for doing this. One is that I can give exams with no calculators allowed, which eliminates the advantage that students students with a pricey graphing calculators have. Another reason is that I believe that doing mental math gives students a better *feel* for things like linear combinations and matrix multiplication.

0.1.2 Possible downsides

Here are some things that you may find are downsides of the book:

- **No technology instruction.** If you want to teach your students to use sympy, sage, or MATLAB, this book won't help you do it. And all of the exercises are doable by hand, so students won't feel motivated to use technology. If you have time, and want to make technology part of your class, I recommend David Austin's excellent free online book *Understanding Linear Algebra*, which teaches sage with fancy embedded code widgets throughout the book. It also has some excellent application sections which you could use to supplement this book.

- **Not a ton of applications.** This book does have some application exercises, and I do try hard to give a sense of how linear algebra is useful for anyone studying or working in STEM. But it doesn't include the wide range of application problems that some books have. Again, you may want to supplement with Austin. And if you have problems you think I should include, feel free to email them to me at smithbt@laccd.edu or with a github pull request.
- **It is a work in progress.** Here is a list of things, from highest to lowest priority, that I still need to do, as of January 8, 2026:
  - A mini-section about the Graham-Schmidt process.
  - Chapter 6 (Symmetric matrices and quadratic forms).
  - More synthesis/review type questions in the last few chapters.
  - Solutions to exercises.
  - More application exercises throughout.

0.1.3 How to compile your own version

There are many good reasons to edit the LaTeX code and compile your own version. Maybe you want your own version of the lecture notes with different examples, or more/less blank space. Maybe you want to add your own exercises. Maybe you want the book to be in portrait orientation. Maybe you want little pictures of cute cats on every page. Go for it! Here's what you need to do:

1. Install a distribution of LaTeX on your computer. Windows users usually use MiKTeX, mac users usually use MacTex, and linux users usually use TeXLive.
2. Make your changes to the yafcla.tex file. If you want to make a version of the lecture notes, make sure the second of the following two lines is commented out like so:

   ```
   \lecturenotestrue  % comment out with % for textbook
   %\lecturenotesfalse % comment out with % for lecture notes
   ```

   This will make it so anything in a `book` environment will be ignored when you compile. If you want to make a version of the textbook, move the `%` to the other line and anything inside a `notes` environment will be ignored.
3. Compile with the `lualatex` command. In a terminal (aka console or command prompt), use `cd` to navigate to the directory where you have your `yafcla.tex` file, and run the command:
   ```
   lualatex --file-line-error --synctex=1
   -interaction=nonstopmode yafcla.tex
   ```
   This may prompt you to install a bunch of latex packages, and may take some time, but it should eventually create `yafcla.pdf`. Then, run the above command again because the first time, labels and references won't be linked up. Then you should rename `yafcla.pdf` to something else, like `yafcla_book.pdf` or `yafcla_notes.pdf`, so that doesn't get overwritten when you compile the other version.

0.2 Preface for Students

What is linear algebra about?

One of the first kinds of equations you learned about in a basic algebra class was linear equations—$y = mx + b$. And in science classes, many fundamental laws and models can be written as linear equations, specifically with $b = 0$:

$$\begin{aligned}\text{Force} &= (\text{Mass})(\text{Acceleration}),\\ \text{Voltage} &= (\text{Current})(\text{Resistance}),\\ \text{Force of spring} &= (\text{Spring constant})(\text{Displacement}),\\ \text{Part} &= (\text{Proportion})(\text{Whole}),\end{aligned}$$

and so on. But the world is often more complicated, with many y variables that depend on many x variables. And some (but not all) of the above equations can have multiple variables, like x_1, x_2, x_3 and y_1, y_2, y_3 instead of just x and y. So instead of just $y = mx$ you might have

$$\begin{aligned}y_1 &= m_{11}x_1 + m_{12}x_2 + m_{13}x_3\\ y_2 &= m_{21}x_1 + m_{22}x_2 + m_{23}x_3\\ y_3 &= m_{31}x_1 + m_{32}x_2 + m_{33}x_3\end{aligned} \tag{1}$$

Linear algebra is fundamentally about *systems of linear equations* like this, which show up in every STEM discipline when you get beyond 100 and 200-level courses. We're going to start by learning how to solve them, which means finding the x's that make all the equations work, given values of the y's and m's, and determine when the system has no solution, one solution, or an infinite number of solutions.

Then, we'll see how we can write the system as a *vector equation*, like

$$\begin{bmatrix} y_1\\ y_2\\ y_3\end{bmatrix} = x_1\begin{bmatrix} m_{11}\\ m_{21}\\ m_{31}\end{bmatrix} + x_2\begin{bmatrix} m_{12}\\ m_{22}\\ m_{32}\end{bmatrix} + x_3\begin{bmatrix} m_{13}\\ m_{23}\\ m_{33}\end{bmatrix}$$

or as a *matrix equation*, like

$$\begin{bmatrix} y_1\\ y_2\\ y_3\end{bmatrix} = M\begin{bmatrix} x_1\\ x_2\\ x_3\end{bmatrix} \text{ where } M = \begin{bmatrix} m_{11} & m_{12} & m_{13}\\ m_{21} & m_{22} & m_{23}\\ m_{31} & m_{32} & m_{33}\end{bmatrix}$$

In Chapter 2 we'll see how matrices themselves are really interesting mathematical objects, which can be combined with addition and multiplication just like numbers. And just like $F = ma$ can be combined with $a = \frac{\Delta v}{t}$ to make $F = m\frac{\Delta v}{t}$, we can combine matrix equations $\vec{y} = M\vec{x}$ and $\vec{x} = R\vec{z}$ to make $\vec{y} = (MR)\vec{z}$. We'll see how the properties of matrices M and R are related to properties of MR and what that all says about existence and uniqueness of solutions to the related systems of equations like equation 1.

In Chapter 3 we'll look at *determinants* which sort of describe how "big" a matrix is.

Chapter 4 is about eigenvectors and eigenvalues, which are important for describing what happens when matrices are raised to powers: $\vec{y} = M^t\vec{x}$. This is kind of like a multivariable version of exponential functions like $y = ae^{kt}$.

A lot of the time, a solution to a system like equation (1) doesn't exist, but you can still find a *least-squares solution*, which basically means getting the right sides of the equations as close as possible to the left sides. In Chapter 5 we'll learn how to find least-squares solutions using dot products and related theory.

In Chapter 6 we'll learn about symmetric matrices and principal component analysis. Results from this chapter are widely used in many kinds of data analysis, from signal processing to computer vision to machine learning to genetics.

Finally, Chapter 7 is about how most everything we've learned about vectors actually applies to many kinds of mathematical objects such as functions, polynomials, matrices, and solutions to differential equations.

Chapter 1

Existence and Uniqueness of Solutions

1.1 Systems of Linear Equations

In this class, we're going to study:

- two-dimensional space, aka the xy-plane, aka $\mathbb{R}^2$,
- three-dimensional space, aka xyz-space, aka $\mathbb{R}^3$,
- and in general, n-dimensional space, aka $\mathbb{R}^n$, where we write points in the form $(x_1, x_2, \dots, x_n)$.

Definition. A **system of linear equations** can be written in the form:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1c}x_c &= y_1 \\ &\vdots \\ a_{r1}x_1 + a_{r2}x_2 + \cdots + a_{rc}x_c &= y_r \end{aligned}$$

where all of the **coefficients** $a_{11}, \dots, a_{rc}$ and **constant coefficients** $y_1, \dots, y_r$ are real numbers. A **solution** of the system is a list of numbers $(x_1, \dots, x_c)$ that makes all of the equations true when plugged in. The set of all solutions of the system is called the **solution set** and two systems are **equivalent** if they have the same solution set.

Example 1. Solve the system $\begin{aligned} x_1 + x_2 &= 3 \\ -x_1 + 2x_2 &= 0 \end{aligned}$ by graphing both equations, and also by the Gaussian elimination method that you may remember from previous classes.

Even though you've probably had experience with the elimination method in previous classes, you might not have thought very hard about *why* it works. The proof of the next theorem goes into this in (maybe too much) depth. *Fun Fact:* The first written record of Gaussian elimination comes from China, circa 150 BC. It was independently invented in several cultures. In Europe, it was first widely published by Isaac Newton and well-known before Gauss was born.

Theorem 1.1: Gaussian elimination doesn't change solutions.
If one system of linear equations is transformed into another system using one of the operations below, then the two systems are equivalent.

- **Swapping:** Switch the positions of two equations. (aka *interchange*.)
- **Scaling:** Multiply both sides of an equation by a non-zero real number.
- **Multaddiplacement:** Re*place* each side of one equation by the *sum* of itself and a *mult*iple of another equation's sides. (aka simply *replacement* in most textbooks.)

Example 2. Write and sketch a system of two equations in two variables with:

(a) No solution

(b) An infinite number of sol'ns

(c) Exactly two sol'ns

The above example shows why any system of two linear equations with two variables has either zero solutions, one solution, or an infinite number of solutions. It turns out this is also true for any number of equations with any number of variables, which is hard to visualize, but we'll see why it's true by analyzing the systems themselves.

Definition. A system is called **consistent** if it has at least one solution; if it has exactly one solution, the solution is **unique**. A system with no solutions is called **inconsistent**.

Example 3. Is each system consistent? If so, is the solution unique?

(a)
$$\begin{aligned} x_1 &= 5 \\ x_2 + x_3 &= 13 \\ -2x_2 - x_3 &= -19 \end{aligned}$$

(b)
$$\begin{aligned} x_1 + 2x_3 &= 3 \\ x_2 &= 5 \\ -5x_1 - 10x_3 &= 22 \end{aligned}$$

(c)
$$\begin{aligned} x_1 + 2x_3 &= 3 \\ x_2 &= 5 \\ -5x_1 - 10x_3 &= -15 \end{aligned}$$

Here's a rough strategy for Gaussian elimination that we'll be using throughout the class:

1. If the top equation doesn't have x_1, use swaps to get it there first. Scale the top equation so the coefficient of x_1 is 1.
2. Use multaddiplacements to clear all the x_1 terms from the other equations.
3. If the second equation doesn't have x_2, use swaps to get it there. Scale the second equation so the coefficient of x_2 is 1. (If x_2 isn't in any of the equations below equation 1, skip this and the next step.)
4. Use multaddiplacements to clear all the x_2 terms from the other equations.
5. Repeat with the rest of the variables until each variable shows up in only one equation, if possible.

Note that in Example 3 (b) and (c), it's not possible to get x_3 in an equation by itself because x_3 got canceled out along with x_2. And then it's not possible to clear out the x_3 from the first equation. In general, when there's an infinite number of solutions or no solution, you can't get each equation to have a single variable, but the strategy above will get you as close as possible.

Example 4. Solve the system:
$$\begin{aligned} x_1 - 2x_2 &= -7 \\ -2x_1 + 4x_2 + x_3 &= 18 \\ x_1 - 2x_2 + x_3 &= -3 \end{aligned}$$

1.1 Exercises

For exercises 1 through 6, use Gaussian elimination to determine if the system is consistent or not, and if so, if the solution is unique.

1. $$\begin{aligned} 2x_1 + 6x_2 &= 18 \\ -2x_1 - 5x_2 &= -14 \end{aligned}$$

2. $$\begin{aligned} x_1 + x_2 &= 4 \\ -2x_1 - 2x_2 &= 0 \end{aligned}$$

3. $$\begin{aligned} x_1 + 2x_2 &= 0 \\ 3x_1 + 4x_2 &= 0 \end{aligned}$$

4. $$\begin{aligned} x_1 + 2x_2 &= 0 \\ 3x_1 + 6x_2 &= 0 \end{aligned}$$

5. $$\begin{aligned} -2x_1 + 0x_2 - 7x_3 &= 1 \\ -2x_1 + 0x_2 - 6x_3 &= -2 \end{aligned}$$

6. $$\begin{aligned} 2x_1 + x_2 &= -3 \\ 2x_1 - x_2 &= 11 \\ x_1 - x_2 &= 9 \end{aligned}$$

For exercises 7 through 12, Is the statement true or false? Justify your answer. If true, either reference the definition/theorem/paragraph where it's stated in the text, or explain why it's true. If false, maybe the statement is an incorrect version of something in the text which you can reference; or you can give a counterexample that shows it's not always true. (These instructions apply to T/F problems throughout the book.)

7. (T/F?) Suppose you use Gaussian elimination to reduce a system of two equations and get $\begin{aligned} x_1 &= 1 \\ x_2 &= 2 \end{aligned}$. This equation has two solutions.

8. (T/F?) A *solution* of a system of equations is a list of numbers $(x_1, \dots, x_n)$ that makes at least one of the equations true when plugged into the variables.

9. (T/F?) Each of the elimination operations (swapping, scaling, multaddiplacement), can be reversed using an elimination operation.

10. (T/F?) If two systems of linear equations have the same solution sets, then they're consistent with each other.

11. (T/F?) If two linear systems have the same solution set, then they're equivalent to each other.

12. (T/F?) If you scale an equation in a system by multiplying both sides of the equation by zero, the resulting system is equivalent to the first one.

For exercises 13 through 16, use Gaussian elimination to determine if the system is consistent or not, and if so, determine if the solution is unique. If the system has a unique solution, find it.

13. $$\begin{aligned} x_1 - 3x_2 + 5x_3 &= 23 \\ -3x_2 + 6x_3 &= 21 \\ x_3 &= 8 \end{aligned}$$

14. $$\begin{aligned} 4x_1 - x_2 + 3x_3 &= 4\pi \\ -3x_1 + x_2 - 3x_3 &= -3\pi \\ -4x_1 + 2x_2 - 6x_3 &= 2 - 4\pi \end{aligned}$$

15. $$\begin{aligned} x_1 + 2x_3 &= 3 \\ x_2 - 4x_4 &= 0 \\ x_3 + 2x_4 &= -5 \\ 2x_1 - 3x_2 + 5x_3 + 14x_4 &= 1 \end{aligned}$$

16. $$\begin{aligned} x_1 + 2x_2 &= 3 \\ 4x_1 + 8x_2 &= 12 \\ -3x_1 - 6x_2 &= -9 \end{aligned}$$

17. (T/F?) A system of three linear equations with two variables is always inconsistent. *(Remember to always justify T/F answers.)*

18. (T/F?) A system of two linear equations with three variables is always consistent with an infinite number of solutions.

19. (T/F?) If a system of two linear equations with three variables is consistent, then it has an infinite number of solutions.

20. (T/F?) If a system of linear equations has all of the constant coefficients (on the right sides of the equal sign) equal to zero, then the system is consistent.

21. Find the value(s) of p that make the following system consistent:
$$\begin{aligned} x_1 + x_2 + 5x_3 &= -4 \\ -3x_1 - 2x_2 - 13x_3 &= 8 \\ -7x_1 - 5x_2 - 31x_3 &= p \end{aligned}$$

22. What can you conclude about c and d if the following system has an infinite number of solutions: $$\begin{aligned} x_1 + 2x_2 &= 3 \\ cx_1 + dx_2 &= 6 \end{aligned}$$

23. What can you conclude about c and d if the following system has a unique solution?

$$\begin{aligned} x_1 + 2x_2 &= 3 \\ cx_1 + dx_2 &= 6 \end{aligned}$$

24. What can you conclude about c and d if the following system is inconsistent?

$$\begin{aligned} x_1 + 2x_2 &= 3 \\ cx_1 + dx_2 &= 6 \end{aligned}$$

25. What can you conclude about a, b, c and d if the following system is inconsistent?

$$\begin{aligned} ax_1 + bx_2 &= 0 \\ cx_1 + dx_2 &= 5 \end{aligned}$$

1.2 Row Echelon Form and Pivots

Exercise 1. (*Warm-up*) Use Gaussian elimination to determine if the system is consistent or not, and if so, determine if the solution is unique. If the system has a unique solution, find it.

(a)
$$\begin{aligned} 2x_1 + 4x_2 + 6x_3 &= 8 \\ x_1 + 2x_2 + 3x_3 &= -5 \end{aligned}$$

(b)
$$\begin{aligned} 2x_1 + 4x_2 + 6x_3 &= 8 \\ x_1 + 2x_2 + 3x_3 &= 4 \end{aligned}$$

(c)
$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ x_2 - x_3 &= -1 \\ x_3 &= 6 \end{aligned}$$

Definition. The **coefficient matrix** of a system of r linear equations with c variables is an array with r rows and c columns, where each entry in the matrix corresponds with the coefficient of a variable in the equation. Each row of the matrix represents a row of the equation and each column of the matrix represents a variable.
So a system of linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1c}x_c &= y_1 \\ &\vdots \\ a_{r1}x_1 + a_{r2}x_2 + \cdots + a_{rc}x_c &= y_r \end{aligned}$$

has coefficient matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1c} \\ \vdots & \vdots & \vdots & \vdots \\ a_{r1} & a_{r2} & \cdots & a_{rc} \end{bmatrix}$$

The **augmented matrix** of a system has an extra column on the right with the constant coefficients, like so:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1c} & y_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{r1} & a_{r2} & \cdots & a_{rc} & y_r \end{array}\right]$$

If a matrix has r rows and c columns, we say its **size** is $r \times c$. So the size of this augmented matrix is $r \times (c+1)$.

Example 1. For the system in exercise 1(c), write the augmented matrix of the original system, and the augmented matrix of the reduced system that tells you the solution. What is the size of the augmented matrix?

Definition. A **leading entry** of a row in a matrix is the leftmost non-zero entry in the row.
We say a matrix is in **row echelon form** (or simply **echelon form**) if it satisfies:

(i) Any rows with all zeros are below rows with non-zero entries.

(ii) Every leading entry is to the right of the leading entry in row above. (This doesn't apply to the top row.)

A matrix is in **reduced row echelon form** if it's in row echelon form and also satisfies:

(iii) Every leading entry is the only non-zero entry in its column.

(iv) Every leading entry is 1.

The Gaussian elimination operations from theorem 1.1 also have matching **row operations**:

- **Swapping:** Switch the positions of two rows. (aka *interchange*.)
- **Scaling:** Multiply all entries in a row by a non-zero real number.
- **Multaddiplacement:** Re*place* a row by the *sum* of itself and a *mult*iple of another row. (aka simply *replacement* in most textbooks.)

We say that two matrices are **row-equivalent** if one can be transformed to the other by a series of these row operations.

By theorem 1.1, if the augmented matrices of two systems are row-equivalent, then the two systems are equivalent (have the same solution sets).

Example 2. Circle which of the following matrices are in row echelon form, and double-circle the ones that are in reduced row echelon form.

$$\begin{bmatrix} -5 & 7 \\ 0 & -3 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -7 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 7 & 12 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 4 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Definition. If there is a reduced row echelon matrix equivalent to A with a leading entry in row i, column j, we say i, j is a **pivot position** of A, and column j is a **pivot column**.

Note: As we'll see in theorem 1.7, every matrix is row-equivalent to a *unique* reduced row echelon matrix, which means we don't have to worry about situations where, say $2, 2$ is a leading entry in one reduced row echelon form of A, but not a leading entry in another reduced row echelon form of A.

Example 3. Suppose each of the following are augmented matrices for a system. Is each consistent, and if so, how many solutions does it have?

(a) $\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -7 \end{array}\right]$ (b) $\left[\begin{array}{c|c} -5 & 7 \\ 0 & -3 \\ 0 & 0 \end{array}\right]$ (c) $\left[\begin{array}{ccc|c} 1 & 4 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{array}\right]$

Definition. Variables which correspond to pivot columns are called **basic variables**. Any non-basic variables are called **free variables**. When you solve for the basic variables in terms of the free variables, you have a *parametric* description of the solution set. Any choice of the free variables determines a different solution.

Example 4. Reduce the following augmented matrix for a system to reduced row echelon form, and describe the solution set.

$$\left[\begin{array}{ccc|c} -3 & 12 & 1 & -5 \\ -4 & 16 & 1 & -5 \\ -1 & 4 & 1 & -5 \end{array}\right]$$

Theorem 1.2: Existence and uniqueness basics.

(i) A system is consistent if and only if the augmented matrix does *not* have a pivot in the rightmost column.

(ii) If the system is consistent and there is at least one non-pivot column in the coefficient matrix, then there are an infinite number of solutions.

(iii) If the system is consistent and there are no non-pivot columns in the coefficient matrix, then the solution is unique.

1.2 Exercises

For exercises 2 through 7, is the matrix in row echelon form? What about reduced row echelon form? If it's not in reduced row echelon form, use Gaussian elimination to make it so.

2. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 8 & \pi \\ 0 & 0 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 6 \\ 0 & 1 & 0 & 7 \end{bmatrix}$

5. $\begin{bmatrix} 2 & 3 \\ 6 & 9 \\ -2 & -3 \end{bmatrix}$

6. $\begin{bmatrix} 0 & 1 & 2 & 4 \\ -2 & 3 & 0 & 26 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 2 & 0 & 5 \\ 1 & 3 & 0 & 9 \\ -2 & -4 & 0 & -7 \end{bmatrix}$

For exercises 8 through 13, an augmented matrix of a system is given. Solve the system (either write *no solution*; or if has a unique solution, write it; or if it has an infinite number of solutions, write the parametric form).

8. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 6 \\ 0 & 1 & 0 & 7 \end{array}\right]$

9. $\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right]$

10. $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \end{array}\right]$

11. $\left[\begin{array}{ccc|c} -2 & 4 & -4 & -6 \\ 2 & -4 & 5 & 11 \\ -3 & 6 & -6 & -11 \end{array}\right]$

12. $\left[\begin{array}{ccc|c} -2 & 4 & -4 & -6 \\ 2 & -4 & 5 & 11 \\ -3 & 6 & -6 & -9 \end{array}\right]$

13. $\left[\begin{array}{cccc|c} 1 & 1 & 5 & 1 & 0 \\ -2 & -1 & -7 & -2 & 0 \\ -1 & -1 & -5 & 2 & 0 \end{array}\right]$

Remember to give some justification for all T/F answers.

14. (T/F?) A system of 2 equations with 4 variables has a 4×2 coefficient matrix.

15. (T/F?) A system of 2 equations with 4 variables has a 2×5 augmented matrix.

16. (T/F?) Every matrix in reduced row echelon form is also in row echelon form.

17. (T/F?) One example of a valid row operation is replacing row 2 with row 2 minus row 2, resulting in a row of all zeros.

18. (T/F?) If there's a leading entry in column 3 of an echelon form coefficient matrix, then x_3 is a basic variable.

19. (T/F?) If an augmented matrix has a row equal to $\begin{bmatrix} 0 & 0 & 0 & -2 \end{bmatrix}$, then the system is inconsistent.

20. (T/F?) If a coefficient matrix has non-pivot columns, then the corresponding system has an infinite number of solutions.

21. (T/F?) If a coefficient matrix has non-pivot columns, and the rightmost column of the corresponding augmented matrix is not a pivot column, then the system has an infinite number of solutions.

22. Find all the values of y_1 and y_2 so that the system with this augmented matrix is inconsistent, if possible: $\left[\begin{array}{cc|c} -1 & 2 & y_1 \\ 3 & -6 & y_2 \end{array}\right]$

23. Find all the values of y_1, y_2, and y_3 so that the system with this augmented matrix is inconsistent, if possible: $\left[\begin{array}{ccc|c} -1 & -3 & -4 & y_1 \\ 2 & 5 & 7 & y_2 \\ 0 & -1 & -2 & y_3 \end{array}\right]$

24. Find all the values of y_1, y_2, and y_3 so that the system with this augmented matrix is inconsistent, if possible: $\left[\begin{array}{ccc|c} -1 & -3 & -4 & y_1 \\ 2 & 5 & 7 & y_2 \\ 0 & -1 & -1 & y_3 \end{array}\right]$

25. Show that scaling (multiplying) a row by zero can change the solution set using an example.

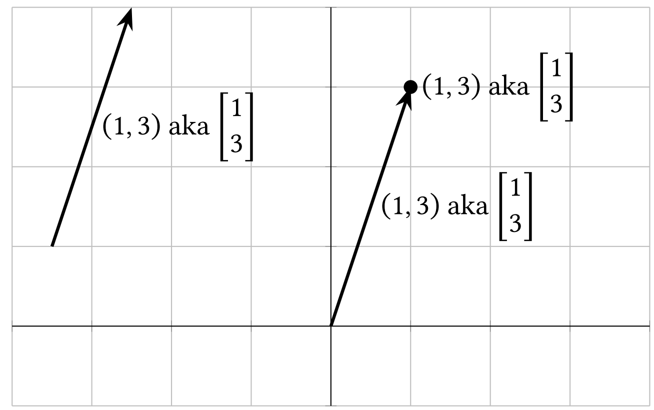
26. Give a 3×3 coefficient matrix A, not in echelon form, so that the system with augmented matrix $\left[\begin{array}{c|c} & y_1 \\ A & y_2 \\ & y_3 \end{array}\right]$ is consistent for any values of y_1, y_2, and y_3.

27. Give a 3×3 coefficent matrix A, not in echelon form, so that the system with augmented matrix $\left[\begin{array}{c|c} & y_1 \\ A & y_2 \\ & y_3 \end{array}\right]$ is consistent for some values of y_1, y_2, and y_3, and inconsistent for others. Give an example of values (y_1, y_2, y_3) that makes it consistent and also an example (y_1, y_2, y_3) that makes it inconsistent.

28. Find values of a, b, and c so that $f(x) = c + bx + ax^2$ has $f(0) = -3$, $f(1) = -2$, and $f(2) = -3$.

1.3 Vectors, Linear Combinations, and Span

Suppose $(x_1, \ldots, x_c)$ is any point in $\mathbb{R}^c$. We will often represent such a point as a $c \times 1$ matrix $\begin{bmatrix} x_1 \\ \vdots \\ x_c \end{bmatrix}$, call it a **vector**, and write it as $\vec{x}$. In fact, we'll use the terms *point* and *vector* interchangeably, and the notations $(x_1, \ldots, x_c)$ and $\begin{bmatrix} x_1 \\ \vdots \\ x_c \end{bmatrix}$ interchangeably. Sometimes it will be helpful to think of a vector as any arrow with the same magnitude and direction as the arrow from $(0, \ldots, 0)$ to the point $(x_1, \ldots, x_c)$.



But, don't go overboard: we will never write a vector/point as a single-row matrix like $\begin{bmatrix} x_1 & \ldots & x_c \end{bmatrix}$.

Vectors can be scaled (multiplied) by any real number, or added together like so:

$$k \begin{bmatrix} u_1 \\ \vdots \\ u_c \end{bmatrix} = \begin{bmatrix} ku_1 \\ \vdots \\ ku_c \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} u_1 \\ \vdots \\ u_c \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_c \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_c + v_c \end{bmatrix}$$

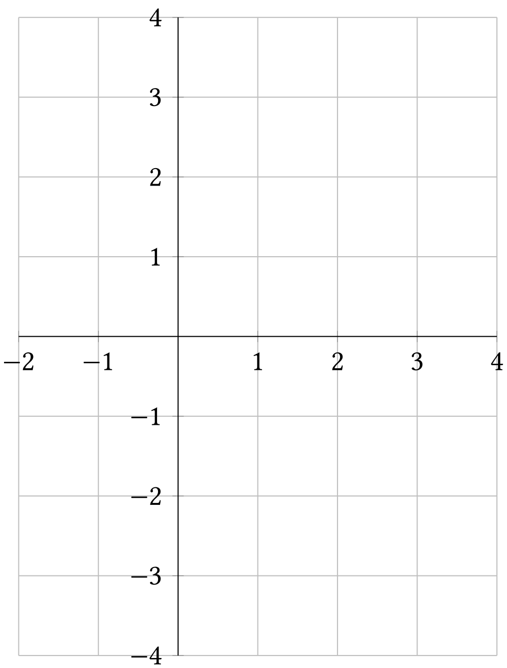
It's important to understand what these operations look like with points/arrows:

Example 1. Suppose $\vec{w} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\vec{z} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Find and graph:

(a) $\vec{w}$ and $\vec{z}$

(b) $\vec{w} + \vec{z}$

(c) $k\vec{w}$ for $k = -2, -1, 0, 0.5, 2$



Theorem 1.3: Properties of vector operations. For all vectors $\vec{x}, \vec{y}$, and $\vec{z}$ in $\mathbb{R}^c$ and all scalars r and t:

(i) $\vec{x} + \vec{y} = \vec{y} + \vec{x}$ (vector addition is **commutative**)

(ii) $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$ (vector addition is **associative**)

(iii) $\vec{x} + \vec{0} = \vec{0} + \vec{x} = \vec{x}$

(iv) $\vec{x} + (-\vec{x}) = \vec{0}$

(v) $(r + t)\vec{x} = r\vec{x} + t\vec{x}$

(vi) $r(\vec{x} + \vec{y}) = r\vec{x} + r\vec{y}$

(vii) $r(t\vec{x}) = (rt)\vec{x}$

Each of these is true because of a corresponding property of real numbers which can be applied to each entry of the vectors. Note that we will study mathematical objects in this class where some of these rules are not true.

Let's prove (i) and leave some others for exercises.

Definition. A **list** of vectors is a finite set of vectors, that all have the same number of entries, written in some order. Sometimes we'll write a list with parentheses, sometimes without. So $\vec{a}_1, \vec{a}_2, \vec{a}_3$ means the same thing as $(\vec{a}_1, \vec{a}_2, \vec{a}_c)$ but not the same thing as $\vec{a}_3, \vec{a}_2, \vec{a}_1$.
If $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_c$ is a list of vectors in $\mathbb{R}^n$ and $x_1, x_2, \dots, x_c$ are in $\mathbb{R}$, then

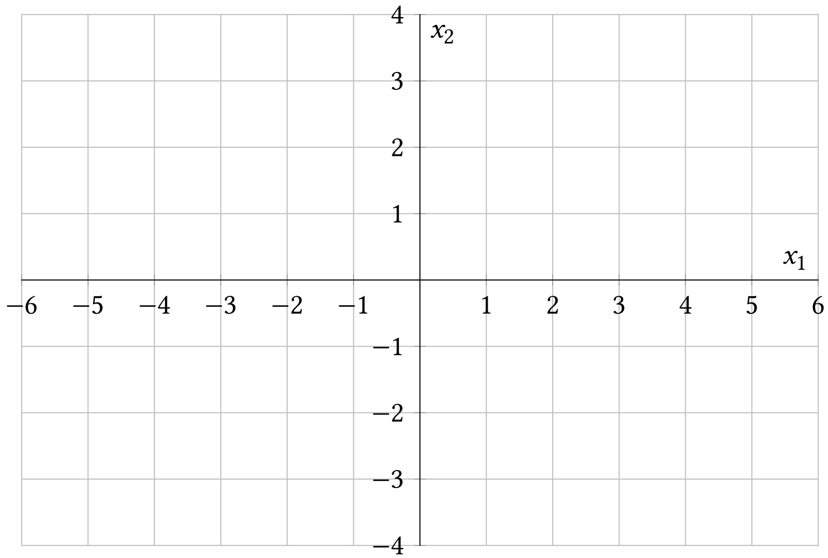
$$x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_c\vec{a}_c$$

is called a **linear combination** of $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_c$ with coefficients (aka weights) $x_1, x_2, \dots, x_c$.

Example 2. Given $\vec{u} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, graph some vectors in the set:

$$\{t\vec{u} + r\vec{v} \,:\, t, r \text{ are integers}\}$$

(Read this as "the set of all vectors of the form $t\vec{u}+r\vec{v}$ such that t, r are any integers.")



Example 3. Find coefficients that make $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ a linear combination of $\vec{u} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. First use the graph to make an estimate, then set up and solve a system of equations.

> **Definition.** The **span** of $\vec{a}_1, \ldots, \vec{a}_c$, written $\mathrm{span}(\vec{a}_1, \ldots, \vec{a}_c)$, is the set of *all* linear combinations of $\vec{a}_1, \ldots, \vec{a}_c$.
> The span of the empty list () is defined to be $\{\vec{0}\}$.

Example 4. Let $\vec{u} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} -6 \\ -2 \end{bmatrix}$.

(a) Graph span$(\vec{u}, \vec{v})$.

(b) How many vectors are in the list $(\vec{u}, \vec{v})$?

(c) How many vectors are in span$(\vec{u}, \vec{v})$?

(d) Is span$(\vec{u}, \vec{v}) = \mathbb{R}^2$?

(e) For what value(s) of k is $\begin{bmatrix} k \\ 5 \end{bmatrix}$ in span$(\vec{u}, \vec{v})$?

(f) How many solutions does $x_1\vec{u} + x_2\vec{v} = \vec{0}$ have?

Example 5. Let $\vec{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$.

(a) Show that span$(\vec{u}, \vec{v}) = \mathbb{R}^2$. (In this case we say the list $(\vec{u}, \vec{v})$ *spans* $\mathbb{R}^2$.)

(b) How many solutions does $x_1\vec{u} + x_2\vec{v} = \vec{0}$ have?

Example 6. Let $\vec{u} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$.

(a) Graph span$(\vec{u}, \vec{v})$.

(b) For what value(s) of k is $\vec{y} = \begin{bmatrix} h \\ 2 \\ 4 \end{bmatrix}$ in span$(\vec{u}, \vec{v})$?

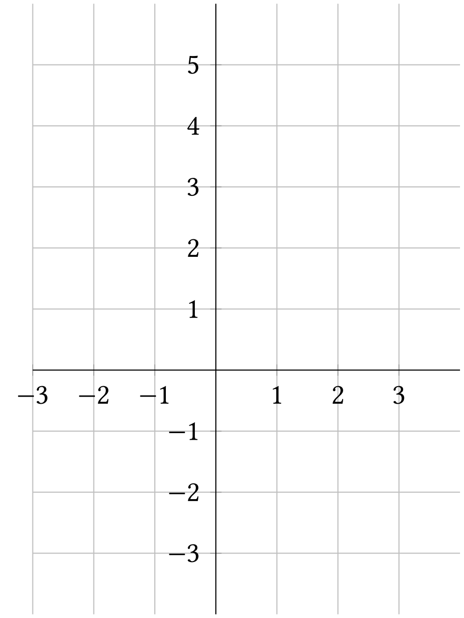
(c) How many solutions does $x_1\vec{u} + x_2\vec{v} = \vec{0}$ have?

1.3 Exercises

1. Simplify $-3\begin{bmatrix} 0 \\ -7 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 3 \end{bmatrix}$.

2. Simplify $3\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix} - \begin{bmatrix} 9 \\ 16 \\ -1 \end{bmatrix}$.

3. Let $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$. Find and graph $\vec{u}, \vec{v}, \vec{u} + \vec{v}, \vec{u} - \vec{v}, \vec{v} - \vec{u}$, and $k\vec{u}$ for $k = -2, -1, 0, 1, 2$.



4. Using $\vec{u}$ and $\vec{v}$ from exercise 3, use the graph to estimate solutions (x_1, x_2) to each vector equation: (a) $x_1\vec{u} + x_2\vec{v} = \begin{bmatrix} 2.5 \\ 0 \end{bmatrix}$

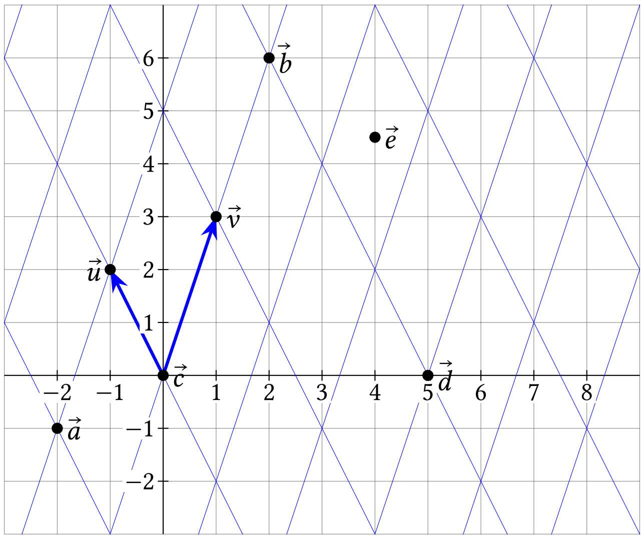
(b) $x_1\vec{u} + x_2\vec{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

5. Using $\vec{u}$ and $\vec{v}$ from exercise 3, translate each vector equation into a system of equations and solve:

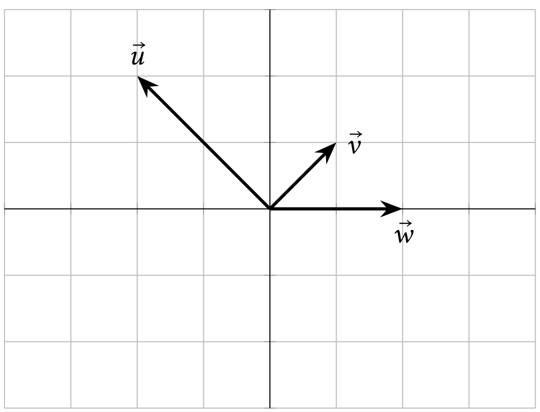
(a) $x_1\vec{u} + x_2\vec{v} = \begin{bmatrix} 2.5 \\ 0 \end{bmatrix}$ (b) $x_1\vec{u} + x_2\vec{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

6. The graph below shows $\vec{u}, \vec{v}$, and other points. Use it to solve each vector equation for (x_1, x_2):

(a) $x_1\vec{u} + x_2\vec{v} = \vec{a}$ (b) $x_1\vec{u} + x_2\vec{v} = \vec{b}$
(c) $x_1\vec{u} + x_2\vec{v} = \vec{c}$ (d) $x_1\vec{u} + x_2\vec{v} = \vec{d}$
(e) $x_1\vec{u} + x_2\vec{v} = \vec{e}$



7. Use the graph below (and no calculation) to estimate three different solutions of $x_1\vec{u} + x_2\vec{v} + x_3\vec{w} = \vec{0}$.



For exercises 8 to 16, convert the vector equation to a system of equations, write its augmented matrix, and solve it (either write *no solution*; or if has a unique solution, write it; or if it has an infinite number of solutions, write the parametric form).

8. $x_1\begin{bmatrix}5\\-3\end{bmatrix} + x_2\begin{bmatrix}-11\\7\end{bmatrix} = \begin{bmatrix}-4\\4\end{bmatrix}$

9. $x_1\begin{bmatrix}-2\\4\end{bmatrix} + x_2\begin{bmatrix}-6\\13\end{bmatrix} + x_3\begin{bmatrix}2\\0\end{bmatrix} = \begin{bmatrix}0\\7\end{bmatrix}$

10. $x_1\begin{bmatrix}1\\2\end{bmatrix} + x_2\begin{bmatrix}4\\8\end{bmatrix} = \begin{bmatrix}1\\7\end{bmatrix}$

11. $x_1\begin{bmatrix}1\\2\end{bmatrix} + x_2\begin{bmatrix}4\\8\end{bmatrix} = \begin{bmatrix}1\\2\end{bmatrix}$

12. $x_1\begin{bmatrix}-2\\2\end{bmatrix} + x_2\begin{bmatrix}1\\1\end{bmatrix} + x_3\begin{bmatrix}2\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$ (Note: these are the vectors from exercise 7. Does your answer agree with that answer?)

13. $x_1\begin{bmatrix}1\\2\\-2\end{bmatrix} + x_2\begin{bmatrix}0.5\\1\\-1\end{bmatrix} + x_3\begin{bmatrix}0\\1\\1\end{bmatrix} = \begin{bmatrix}-2\\1\\9\end{bmatrix}$

14. $x_1\begin{bmatrix}1\\2\\-2\end{bmatrix} + x_2\begin{bmatrix}0.5\\1\\-1\end{bmatrix} + x_3\begin{bmatrix}0\\1\\1\end{bmatrix} = \begin{bmatrix}-2\\1\\12\end{bmatrix}$

15. $x_1\begin{bmatrix}-2\\5\\-1\end{bmatrix} + x_2\begin{bmatrix}6\\-14\\3\end{bmatrix} + x_3\begin{bmatrix}-3\\11\\-2\end{bmatrix} = \begin{bmatrix}8\\-25\\4\end{bmatrix}$

16. $x_1\begin{bmatrix}-2\\5\\-1\end{bmatrix} + x_2\begin{bmatrix}6\\-14\\3\end{bmatrix} + x_3\begin{bmatrix}-3\\11\\-2\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$

17. Write four vectors that are in span($\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}0\\10\\0\end{bmatrix}$).

18. Let M be the list $\begin{bmatrix}5\\-3\end{bmatrix}, \begin{bmatrix}-11\\7\end{bmatrix}$. (Hint: see your answer to exercise 8.)

(a) Is $\vec{y} = \begin{bmatrix}-4\\4\end{bmatrix}$ in M? How many vectors are in M?

(b) Is $\vec{y} = \begin{bmatrix}-4\\4\end{bmatrix}$ in span(M)? How many vectors are in span(M)?

19. Let W be the list $\left(\begin{bmatrix}1\\2\\-2\end{bmatrix}, \begin{bmatrix}0.5\\1\\-1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}\right)$.

(a) Does W span $\mathbb{R}^3$? (ie, is span(W) $= \mathbb{R}^3$?) Justify your answer. (Hint: see exercise 14.)

(b) For what values of k is $\begin{bmatrix}k\\3\\3\end{bmatrix}$ in span(W)?

Remember to justify any true/false answers, maybe by citing something from the section, or writing your own reasoning, or giving a counterexample.

20. (T/F?) If $\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ then $2\vec{a} + \vec{b} = \begin{bmatrix} 2 & 3 \\ 4 & 4 \end{bmatrix}$

21. (T/F?) $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ is another way to write the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

22. (T/F?) If $\vec{u} \neq \vec{0}$, then scalar multiples of $\vec{u}$ are always points on the line through the origin and $\vec{u}$.

23. (T/F?) The system of linear equations with augmented matrix $\left[\vec{a}_1 \quad \vec{a}_2 \mid \vec{y}\right]$ is consistent if and only if $\vec{y}$ is in $\operatorname{span}(\vec{a}_1, \vec{a}_2)$.

24. (T/F?) If $\vec{u} \neq \vec{0}$, then $\operatorname{span}(\vec{u})$ contains an infinite number of vectors.

25. (T/F?) If $\vec{u} = \vec{0}$, then $\operatorname{span}(\vec{u})$ has an infinite number of vectors.

26. (T/F?) If $\vec{u}, \vec{v}$ are any two vectors in $\mathbb{R}^2$, then the list $(\vec{u}, \vec{v})$ spans $\mathbb{R}^2$.

27. (T/F?) For any vectors $\vec{a}_1, \dots, \vec{a}_c$, the origin is always contained in $\operatorname{span}(\vec{a}_1, \dots, \vec{a}_c)$.

28. Prove theorem 1.3 (ii).

29. Prove theorem 1.3 (v).

30. Prove theorem 1.3 (vii).

31. Give an example of two vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^3$, where $\operatorname{span}(\vec{u}, \vec{v})$ is *not* a plane.

32. Suppose $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ is in $\operatorname{span}\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 3 \end{bmatrix}\right)$. What can you conclude about y_1 and y_2?

33. Suppose $\operatorname{span}\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ k \end{bmatrix}\right) = \mathbb{R}^2$. What can you conclude about k?

34. Suppose $\vec{u}_1, \vec{u}_2, \vec{u}_3$ spans $\mathbb{R}^3$. Prove that the list $\vec{u}_1 - \vec{u}_2, \vec{u}_2 - \vec{u}_3, \vec{u}_3$ also spans $\mathbb{R}^3$.

1.4 Homogeneous Systems and Linear Independence

Exercise 1. (*Warm-up*) Write a vector equation that is equivalent to the system. (Don't solve, just change the notation.)

$$\begin{aligned} 4x_1 - x_2 + 3x_3 &= 4 \\ -3x_1 + x_2 - 3x_3 &= -3 \\ -4x_1 + 2x_2 - 6x_3 &= 2 \end{aligned}$$

Exercise 2. (*Warm-up*) Choose the correct geometric description of $\operatorname{span}(\vec{v}_1, \vec{v}_2)$, where $\vec{v}_1 = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$.

(A) The two points $(3, 0, 2)$ and $(-2, 0, 3)$.

(B) The line through the two points $(3, 0, 2)$ and $(-2, 0, 3)$.

(C) The plane through the three points $(0, 0, 0)$, $(3, 0, 2)$, and $(-2, 0, 3)$.

(D) All of $\mathbb{R}^3$.

Exercise 3. (*Warm-up*) Choose the correct geometric description of $\operatorname{span}(\vec{v}_1)$, where $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

(A) The point $(1, 2)$.

(B) The line through $(0, 0)$ and $(1, 2)$.

(C) All of $\mathbb{R}^2$.

Exercise 4. (*Warm-up*) Choose the correct geometric description of $\operatorname{span}(\vec{v}_1, \vec{v}_2)$, where $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$.

(A) The points $(1, 2)$ and $(3, 6)$.

(B) The line through $(0, 0)$ and $(1, 2)$.

(C) All of $\mathbb{R}^2$.

Exercise 5. (*Warm-up*) Suppose that $x_1\vec{a}_1+x_2\vec{a}_2+x_3\vec{a}_3=\vec{0}$ and that $x_3\neq 0$. Write $\vec{a}_3$ as a linear combination of $\vec{a}_1$ and $\vec{a}_2$.

Exercise 6. (*Warm-up*) Suppose that $y_1\vec{a}_1+y_2\vec{a}_2=\vec{a}_3$. Write $\vec{0}$ as a linear combination of $\vec{a}_1, \vec{a}_2$, and $\vec{a}_3$.

The last two exercises show that for any vectors $\vec{a}_1, \vec{a}_2$, and $\vec{a}_3$, the following two statements are equivalent:

- It is possible to write $\vec{a}_3$ as a linear combination of the other two vectors.
- There exist numbers x_1, x_2, x_3, with $x_3\neq 0$, so that $x_1\vec{a}_1+x_2\vec{a}_2+x_3\vec{a}_3=\vec{0}$.

1.4.1 Linear Independence: Main Ideas

Definition. A linear system of equations that can be written in the form

$$x_1\vec{a}_1+\cdots+x_c\vec{a}_c=\vec{0}$$

is called **homogeneous**. Homogeneous systems are always consistent since $\vec{x}=(x_1,\ldots,x_c)=\vec{0}$ is always a solution, called the **trivial solution.** A **non-trivial** solution is any solution $\vec{x}\neq\vec{0}$.
When a non-trivial solution exists, we say the list $\vec{a}_1,\ldots,\vec{a}_c$ is **linearly dependent**. When the only solution is the trivial solution, we say the list is **linearly independent.** The empty list () is defined to be linearly independent as well.
If $A=\begin{bmatrix}\vec{a}_1 & \cdots & \vec{a}_c\end{bmatrix}$, the solution set of $x_1\vec{a}_1+\cdots+x_c\vec{a}_c=\vec{0}$ is called the **null space** of A and is written null A.

Example 1. Let $A=\begin{bmatrix}\vec{a}_1 & \vec{a}_2\end{bmatrix}=\begin{bmatrix}3 & 4\\ 6 & 8\end{bmatrix}$.

(a) Is the list of vectors $\vec{a}_1, \vec{a}_2$ linearly dependent or independent?

(b) Can you write null A as the span of some vector(s)?

(c) Can you write column 2 as a scalar multiple of column 1?

(d) Is $\operatorname{span}(\vec{a}_1, \vec{a}_2)=\mathbb{R}^2$?

Example 2. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 \end{bmatrix} = \begin{bmatrix} -1 & -3 & 1 \\ 2 & 5 & 2 \\ -1 & -2 & -1 \end{bmatrix}$.

(a) Is the list of vectors $(\vec{a}_1, \vec{a}_2, \vec{a}_3)$ linearly dependent or independent?

(b) Can you write null A as the span of some vector(s)?

(c) Can you write one of the columns as a linear combination of the preceding columns?

(d) Is $\operatorname{span}(\vec{a}_1, \vec{a}_2, \vec{a}_3) = \mathbb{R}^3$?

Example 3. Let $A = \begin{bmatrix} \vec{a}_1 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$.

(a) Is the list of one vector $(\vec{a}_1)$ linearly dependent or independent?

(b) Can you write null A as the span of some vector(s)?

(c) Does $(\vec{a}_1)$ span $\mathbb{R}^2$?

Example 4. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 0 & 2 \end{bmatrix}$.

(a) Is the list of vectors $\vec{a}_1, \vec{a}_2$ linearly dependent or independent?

(b) Can you write null A as the span of some vector(s)?

(c) Can you write column 2 as a scalar multiple of column 1?

(d) Does $(\vec{a}_1, \vec{a}_2)$ span $\mathbb{R}^2$?

Theorem 1.4: If a list contains only a single vector and it's not $\vec{0}$, then the list is linearly independent.

Theorem 1.5: If a list of vectors contains $\vec{0}$, then it's linearly dependent.

Examples 3 and 4 should help you understand why these theorems are true, but they aren't rigorous proof. The proofs are left as exercises.

Logic Fun Corner: A statement of the form "If P, then Q" is false if there exists a *counterexample*, which makes P true and Q false. The *converse* of "If P, then Q" is "If Q, then P". If both are true then we write "P if and only if Q" or abbreviate it "P iff Q."

For example, the statement "If $x^2 = 9$, then $x = 3$" is false because there's a counterexample: __________.

Example 5. Is the converse of theorem 1.5 true?

Strategy to write nul A as a span: This is the strategy we used in examples 1 through 4. After reducing $\left[A \mid \vec{0}\right]$ to reduced echelon form, and solving for *all* the (basic *and* free) variables in terms of the free variables, we can write

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_c \end{bmatrix} = \begin{bmatrix} 0 + \text{maybe some combination of free variable terms} \\ 0 + \text{maybe some combination of free variable terms} \\ \vdots \\ 0 + \text{maybe some combination of free variable terms} \end{bmatrix}$$

without constants on the right sides. This means that any solution of a homogeneous system can be written as a linear combination of some vectors, with the free variables as weights. So the solution set is the span of these vectors. You can also write the solution set in **parametric vector form**, eg $\left\{x_3\begin{bmatrix}5\\6\end{bmatrix} : x_3 \text{ is in } \mathbb{R}\right\}$ instead of $\operatorname{span}\left(\begin{bmatrix}5\\6\end{bmatrix}\right)$.

Theorem 1.6: Characterization of linear dependence.
A list of vectors $W = \vec{a}_1, \ldots, \vec{a}_c$ is linearly dependent iff one of the vectors can be written as a linear combination of the others.
Specifically, if W is linearly dependent, and $\vec{a}_1 \neq \vec{0}$, then some $\vec{a}_i$ with $i > 1$ is a linear combination of the preceding vectors:

$$x_1\vec{a}_1 + \cdots + x_{i-1}\vec{a}_{i-1} = \vec{a}_i$$

Warm-up exercises 5 and 6 as well as examples 1 through 4 give some justification. We'll skip proving this in class, but there's a proof in the textbook.

1.4.2 Theorems About Pivot Positions

Maybe we've been a little sloppy talking about "the" basic variables and free variables, and "the" pivot positions of a matrix. But is it possible you could row-reduce a matrix using two different sequences of row operations, resulting in different reduced echelon matrices, with different pivot columns? Could a single column be both a pivot column and a non-pivot column, depending on the row operations used? Fortunately, no:

Theorem 1.7: Uniqueness of reduced row echelon form.
Every matrix is row-equivalent to a unique reduced row echelon matrix.

Lemma. Suppose A and B are augmented matrices for two systems of homogeneous equations, with each matrix the same size, and each in reduced row echelon form, and $A \neq B$. Then the two systems don't have the same solution sets.

The lemma might seem intuitive but the proof is pretty technical and has been relegated to the textbook. Let's use it to prove the theorem.

Example 6. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \vec{a}_4 \end{bmatrix} = \begin{bmatrix} 1 & -5 & 4 & 2 \\ 2 & -9 & 11 & 4 \\ -1 & 5 & -3 & 2 \end{bmatrix}$.

(a) Is the list of vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ linearly dependent or independent?

(b) Can you write null A as the span of some vector(s)?

(c) Can you write one of the columns as a linear combo of the preceding columns?

(d) Is $\text{span}(\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4) = \mathbb{R}^3$?

Theorem 1.8: The existence theorem. Suppose $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ is an $r \times c$ matrix. The following are equivalent:

(i) A has a pivot position in every row.

(ii) Every system of equations that has A as its coefficient matrix is consistent.

(iii) $\operatorname{span}(\vec{a}_1, \dots, \vec{a}_c) = \mathbb{R}^r$.

Theorem 1.9: The uniqueness theorem. Suppose $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ is an $r \times c$ matrix. The following are equivalent:

(i) A has a pivot position in every column.

(ii) $\operatorname{null} A = \{\vec{0}\}$.

(iii) The columns $\vec{a}_1, \dots, \vec{a}_c$ are a linearly independent list of vectors.

Theorem 1.10: If a list of vectors has more vectors than there are entries in the vectors, then it's linearly dependent.

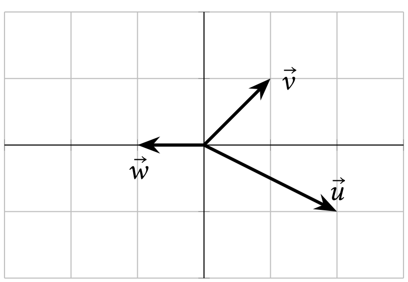
1.4 Exercises

1.4.1 Linear Independence: Main Ideas

7. Use the graph of $\mathbb{R}^2$ below (and no calculation) to answer:

(a) Give one or two non-trivial solutions of $x_1\vec{u} + x_2\vec{v} + x_3\vec{w} = \vec{0}$ if possible.

(b) Write $\vec{w}$ as a linear combination of $\vec{u}, \vec{v}$ if possible.



8. Using the same $\vec{u}, \vec{v}$ in the graph of $\mathbb{R}^2$ in the previous exercise:

(a) Give one or two non-trivial solutions of $x_1\vec{u} + x_2\vec{v} = \vec{0}$ if possible.

(b) Write $\vec{v}$ as a linear combination of $\vec{u}$ if possible.

9. Each of the following are augmented matrices for linear systems. Which system(s) are consistent?

(a) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & 10 \end{array}\right]$ (b) $\left[\begin{array}{ccc|c} 1 & 0 & 3 & -20 \\ 0 & 1 & 17 & 0 \end{array}\right]$

(c) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 10 \end{array}\right]$ (d) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 0 \end{array}\right]$

10. Each of the following are augmented matrices for linear systems. Out of the system(s) that are consistent, which have an infinite number of solutions?

(a) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & 10 \end{array}\right]$ (b) $\left[\begin{array}{ccc|c} 1 & 0 & 3 & -20 \\ 0 & 1 & 17 & 0 \end{array}\right]$

(c) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 10 \end{array}\right]$ (d) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 0 \end{array}\right]$

11. Each of the following are *coefficient* matrices for linear systems. In other words, each is a matrix A for a matrix equation $A\vec{x} = \vec{b}$. Which system(s) are always consistent, no matter vector $\vec{b}$ is on the other side of the equation?

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 17 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

12. Write your own example of a matrix A that makes $A\vec{x} = \vec{0}$ have an infinite number of solutions. Justify your answer.

13. Which of the following is the solution set of $\begin{aligned} x_1 + 2x_2 &= 0 \\ 3x_1 + 6x_2 &= 0 \end{aligned}$?

(A) $\operatorname{span}\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$ (B) $\operatorname{span}\left(\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right)$ (C) $\mathbb{R}^2$ (D) $\{\vec{0}\}$

14. Describe the solution set of $x_1 + 3x_2 = 0$ as the span of some vectors. H

Remember to give some justification for all True/False answers:

15. (T/F?) The system of equations $\begin{aligned} x_1 + x_2 &= 3 \\ -x_1 + 2x_2 &= 0 \end{aligned}$ is homogeneous because of the number 0 on the right side.

H14. Hint: $\left[\begin{array}{cc|c} 1 & 3 & 0 \end{array}\right]$ is already in reduced row echelon form.

16. (T/F?) Every homogeneous linear system of equations is consistent.

17. (T/F?) The solution set of every homogeneous system can be written as the span of some vectors.

18. (T/F?) If $\vec{a}_1, \vec{a}_2, \vec{a}_3$ are in $\mathbb{R}^3$, and $\vec{a}_1, \vec{a}_2$ is a linearly dependent list, then the list $\vec{a}_1, \vec{a}_2, \vec{a}_3$ is linearly dependent.

19. (T/F?) If $\vec{a}_1, \vec{a}_2, \vec{a}_3$ are in $\mathbb{R}^3$, and $\vec{a}_1, \vec{a}_2$ is a linearly *independent* list, then the list $\vec{a}_1, \vec{a}_2, \vec{a}_3$ is linearly *independent*.

20. (T/F?) The list of vectors $\vec{u}, \vec{v}, 3\vec{u} - 2\vec{v}$ is linearly dependent.

21. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 4 \\ 1 & 0 \end{bmatrix}$.

(a) Is the list of vectors $\vec{a}_1, \vec{a}_2$ linearly dependent or independent?
(b) Write null A as the span of some vector(s).
(c) Can you write one of the columns as a linear combination of the preceding columns?
(d) Does the list $(\vec{a}_1, \vec{a}_2)$ span $\mathbb{R}^3$?

22. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 \end{bmatrix} = \begin{bmatrix} 3 & 1 & -5 \\ 2 & 4 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

(a) Is the list of vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3$ linearly dependent or independent?
(b) Write null A as the span of some vector(s).
(c) Can you write one of the columns as a linear combination of the preceding columns?
(d) Does the list $(\vec{a}_1, \vec{a}_2, \vec{a}_3)$ span $\mathbb{R}^3$?

23. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

(a) Is the list of vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3$ linearly dependent or independent?
(b) Write null A as the span of some vector(s).
(c) Can you write one of the columns as a linear combination of the preceding columns?
(d) Is $\operatorname{span}(\vec{a}_1, \vec{a}_2, \vec{a}_3) = \mathbb{R}^2$?

24. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \vec{a}_4 \end{bmatrix} = \begin{bmatrix} 4 & 8 & -18 & 6 \\ -1 & -2 & 5 & -3 \\ -1 & -2 & 4 & 0 \end{bmatrix}$.

(a) Is the list of vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ linearly dependent or independent?

(b) Write null A as the span of some vector(s).

(c) Can you write one of the columns as a linear combination of the preceding columns?

(d) Is $\operatorname{span}(\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4) = \mathbb{R}^3$?

25. Let $A = \begin{bmatrix} 10 & 20 & 30 \\ -9 & -18 & -27 \\ 8 & 16 & 24 \end{bmatrix}$

(a) Row reducing A and using theorem 1.8, is the span of the columns of A equal to $\mathbb{R}^3$?

(b) Note that the second and third columns are scalar multiples of the first. Is the span of the columns a single point, a line, a plane, or all of $\mathbb{R}^3$?

(c) Note that the third column is the sum of the first two columns. Use this fact to write a solution of $x_1\vec{a}_1 + x_2\vec{a}_2 + x_3\vec{a}_3 = \vec{0}$.

26. Find a counterexample that proves the converse of theorem 1.4 is false.

27. Prove or give a counterexample: If $\vec{a}_1, \dots, \vec{a}_c$ is a linearly independent list of at least two vectors, and you remove one vector, then the remaining list is still independent.

28. Prove or give a counterexample: If $\vec{a}_1, \dots, \vec{a}_c$ is a linearly dependent list of at least two vectors, and you remove one vector, then the remaining list is still dependent.

29. Prove or give a counterexample: If $\vec{a}_1, \dots, \vec{a}_c$ is a linearly independent list of vectors in $\mathbb{R}^n$ and if k is a real number with $k \neq 0$, then the list $k\vec{a}_1, \dots, k\vec{a}_c$ is also linearly independent.

30. Prove or give a counterexample: If $\vec{a}_1, \dots, \vec{a}_c$ and $\vec{b}_1, \dots, \vec{b}_c$ are each linearly independent lists in $\mathbb{R}^n$, then the list $\vec{a}_1 + \vec{b}_1, \dots, \vec{a}_c + \vec{b}_c$ is also linearly independent.

31. Suppose $\vec{u}_1, \vec{u}_2, \vec{u}_3$ is linearly independent in $\mathbb{R}^c$. Prove that the list $\vec{u}_1 - \vec{u}_2, \vec{u}_2 - \vec{u}_3, \vec{u}_3$ is also linearly independent.

32. (*Challenge*) Suppose $\vec{u}_1, \dots, \vec{u}_c$ is a linearly independent list of vectors in $\mathbb{R}^n$, and $\vec{w}$ is in $\mathbb{R}^n$. Prove that if $\vec{u}_1 + \vec{w}, \dots, \vec{u}_c + \vec{w}$ is linearly dependent, then $\vec{w}$ is in $\operatorname{span}(\vec{u}_1, \dots, \vec{u}_c)$.

H39. Hint: Be careful.

1.4.2 Theorems About Pivot Positions

33. (T/F?) If there is a pivot position in every column of an $r \times c$ matrix A, then the columns of A span $\mathbb{R}^r$.

34. Let $\vec{v}_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ -1 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

(a) Before doing any calculations, what is the maximum possible number of pivots that $\begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 \end{bmatrix}$ could have? What does theorem 1.8 say about whether $\operatorname{span}(\vec{v}_1, \vec{v}_2, \vec{v}_3) = \mathbb{R}^4$ or not?

(b) What must be true about $\vec{y} = (y_1, y_2, y_3, y_4)$ in order for $\vec{y}$ to be in $\operatorname{span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$? (Row-reduce $\left[\begin{array}{ccc|c} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 & \vec{y} \end{array}\right]$.)

35. Is the list $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$ linearly dependent or independent?

36. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 \end{bmatrix} = \begin{bmatrix} 2 & 4 & -6 \\ 1 & 4 & -1 \\ 1 & 3 & -2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Does $x_1\vec{a}_1 + x_2\vec{a}_2 + x_3\vec{a}_3 = \vec{b}$ have a solution for all possible $\vec{b}$? If not, describe the set of all $\vec{b}$ for which it does have a solution.

37. Let $\begin{bmatrix} 2 & 4 & -5 \\ 1 & 4 & -1 \\ 1 & 3 & -2 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Does $x_1\vec{a}_1 + x_2\vec{a}_2 + x_3\vec{a}_3 = \vec{b}$ have a solution for all possible $\vec{b}$? If not, describe the set of all $\vec{b}$ for which it does have a solution.

38. (T/F?) A system of equations with more equations than variables cannot have an infinite number of solutions.

39. (T/F?) If $\vec{u}$ and $\vec{v}$ are in $\mathbb{R}^c$, and $\vec{v}$ can't be written as a multiple of $\vec{u}$, then the list $\vec{u}, \vec{v}$ is linearly independent. H

40. (T/F?) If a 10×12 matrix has 10 pivot columns, then the span of the columns equals $\mathbb{R}^{10}$.

41. Write a list of two $\mathbb{R}^3$ vectors $\vec{a}_1, \vec{a}_2$ so that $x_1\vec{a}_1 + x_2\vec{a}_2 = \vec{0}$ has a unique solution. Then write a list of two $\mathbb{R}^3$ vectors $\vec{b}_1, \vec{b}_2$ so that $x_1\vec{b}_1 + x_2\vec{b}_2 = \vec{0}$ has an infinite number of solutions.

42. Prove theorem 1.4. Remember that your proof should cover *any* list containing a single non-zero vector, not just one example. So start with something like "Suppose $W = (\vec{a})$ is a list with just one vector, and $\vec{a} \neq \vec{0}$."

43. Prove theorem 1.5.

44. Explain why a system of linear equations with more variables than equations cannot have a unique solution.

45. Explain, using theorem 1.8, why a set of fewer than r vectors cannot span $\mathbb{R}^r$.

46. Suppose A is an $r \times c$ matrix, and for all $\vec{y}$ in $\mathbb{R}^r$, the equation $A\vec{x} = \vec{y}$ has at most one solution. Show that the columns of A are linearly independent, using the definition.

47. Suppose A is an $r \times c$ matrix with c pivot columns. Show that the columns of A are linearly independent.

48. Find a number k so that $\left\{ \begin{bmatrix} 2 \\ -4 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ -6 \end{bmatrix}, \begin{bmatrix} -3 \\ 9 \\ k \end{bmatrix} \right\}$ is linearly independent.

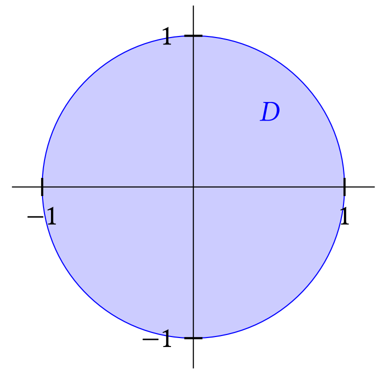
1.5 Subspaces of $\mathbb{R}^c$

Exercise 1. (*Warm-up*) The graph shows the unit disk $D = \{(x_1, x_2) : x_1^2 + x_2^2 \leq 1\}$.

(a) (T/F?) $\vec{0}$ is in D.

(b) (T/F?) For any two points $\vec{x}$ and $\vec{y}$ in D, the vector $\vec{x}+\vec{y}$ is in D. (If you're stuck, a good approach to this kind of problem is to try to find a counterexample: $\vec{x}, \vec{y}$ so that $\vec{x} + \vec{y}$ is *not* in D.)

(c) (T/F?) For any point $\vec{x}$ in D and any scalar k, $k\vec{x}$ is in D.

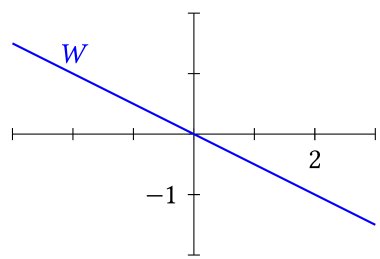


Exercise 2. (*Warm-up*) The graph below shows $W = \text{span}((2, -1))$

(a) (T/F?) $\vec{0}$ is in W.

(b) (T/F?) For any two points $\vec{x}$ and $\vec{y}$ in W, the vector $\vec{x} + \vec{y}$ is in W.

(c) (T/F?) For any point $\vec{x}$ in W and any scalar k, $k\vec{x}$ is in W.



Definition. A **subspace** U of $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ for which:

(i) The zero vector of $\mathbb{R}^n$ is in U.

(ii) If $\vec{p}$ and $\vec{q}$ are in U, then $\vec{p}+\vec{q}$ is in U. (U is **closed under vector addition**.)

(iii) If $\vec{p}$ is in U and c is in $\mathbb{R}$, then $c\vec{p}$ is in U. (U is **closed under scalar multiplication**.)

Example 1. Show that the set $U = \left\{\vec{0}\right\}$ is a subspace of $\mathbb{R}^n$. This is called the *trivial subspace.*

Note: We say set V is a *subset* of W if everything in V is also in W. This means that W is a subset of itself. We say V is a *proper subset* of W if it's a subset of W and there's something in W but not V. So W is not a proper subset of itself.

Example 2. Show that $U = \mathbb{R}^n$ is a subspace of $\mathbb{R}^n$.

Example 3. Is $M = \{(a,b,c) : abc = 0\}$ a subspace of $\mathbb{R}^3$?

Example 4. Is $P = \{(a,b,c) : a+b+c = 0\}$ a subspace of $\mathbb{R}^3$?

Theorem 1.11: Null spaces are subspaces.
For any $r \times c$ vector $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$, the null space of A is a subspace of $\mathbb{R}^c$.

Theorem 1.12: Spans are subspaces.
If $\vec{a}_1, \ldots, \vec{a}_c$ is a list of vectors in $\mathbb{R}^n$, then $\operatorname{span}(\vec{a}_1, \ldots, \vec{a}_c)$ is a subspace of $\mathbb{R}^n$.

Definition. The **column space** of a matrix A, written $\operatorname{col} A$, is the span of the columns of A.
The **row space** of A, written $\operatorname{row} A$, is the span of the rows of A, written as vectors.

Example 5. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. Find a few vectors in each of the following:

(a) $\operatorname{col} A$

(b) $\operatorname{row} A$

(c) $\operatorname{null} A$

Example 6. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$.

(a) Is $\begin{bmatrix} 12 \\ 21 \end{bmatrix}$ in col A?

(b) Is $\begin{bmatrix} 5 \\ 7 \\ 10 \end{bmatrix}$ in row A?

(c) Is $\begin{bmatrix} -\sqrt{7} \\ 2\sqrt{7} \\ -\sqrt{7} \end{bmatrix}$ in null A?

Theorem 1.13: Subspaces are spans.
If W is a subspace of $\mathbb{R}^n$, then $W = \text{span}(\vec{v}_1, \dots, \vec{v}_c)$ for some linearly independent list of vectors $\vec{v}_1, \dots, \vec{v}_c$ in $\mathbb{R}^n$, with $c \leq n$.

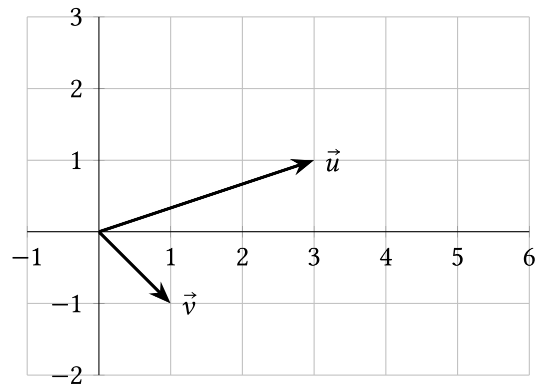
Combining theorems 1.12 and 1.13 we get:

Corollary. A subset W of $\mathbb{R}^n$ is a subspace if and only if it can be written as the span of a list of vectors.

Example 7. Let $\vec{u} = (3, 1)$ and $\vec{v} = (1, -1)$.

(a) Draw each of the following points:

$\vec{u} + (0)\vec{v}$
$\vec{u} + \vec{v}$
$\vec{u} + 2\vec{v}$
$\vec{u} + 2.5\vec{v}$
$\vec{u} + (-1)\vec{v}$
$\vec{u} + (-2)\vec{v}$



(b) Is the set $W = \{\vec{u} + t\vec{v} \,:\, t \text{ is in } \mathbb{R}\}$ a subspace of $\mathbb{R}^2$?

Note: In general, the set of points $\{\vec{u} + t\vec{v} \,:\, t$ is any real number$\}$ describes a line including the point $\vec{u}$, parallel to the vector $\vec{v}$, as an arrow.

1.5 Exercises

3. Suppose A is an $r \times c$ matrix, $\vec{u}$ is in null A, and $\vec{v}$ is in col A. Fill in each blank with $\vec{u}$ or $\vec{c}$.

(a) The entries of _____ are weights of a linear combination of columns of A.
(b) _____ is a linear combination of columns of A.
(c) _____ has r entries.
(d) _____ has c entries.
(e) If _____ is not $\vec{0}$, it means the columns of A are linearly dependent vectors.

4. Is the set $W = \{(x_1, x_2, x_3) \,:\, 7x_1 - 3x_2 + \pi x_3 = 0\}$ a subspace of $\mathbb{R}^3$?

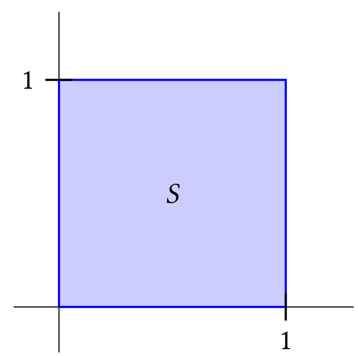
5. Is the set $Z = \{(x_1, x_2, x_3) \,:\, 7x_1 - 3x_2 + \pi x_3 = 8\}$ a subspace of $\mathbb{R}^3$?

6. Is the set $U = \{(x_1, x_2, x_3) \,:\, x_1 = 3x_3\}$ a subspace of $\mathbb{R}^3$?

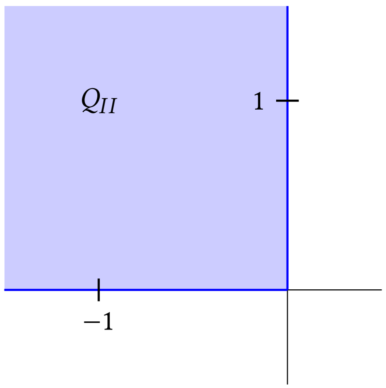
7. Is the set $R = \{(x + 3y, z + x, y + 2z, \pi x) \,:\, x, y,$ and z are in $\mathbb{R}\}$ a subspace of $\mathbb{R}^4$?

8. Is the set $X = \{(x_1, x_2, x_3) \,:\, x_1 = x_3^2\}$ a subspace of $\mathbb{R}^3$?

9. Is the unit square, $S = \{(x_1, x_2) \,:\, 0 \le x_1 \le 1, 0 \le x_2 \le 1\}$, a subspace of $\mathbb{R}^2$?



10. Is Quadrant II, $Q_{II} = \{(x_1, x_2) \,:\, x_1 \le 0, x_2 \ge 0\}$, a subspace of $\mathbb{R}^2$?



11. Let $A = \begin{bmatrix} 1 & 3 & -7 \\ -3 & -9 & 23 \end{bmatrix}$. Find a few vectors in each of these subspaces:

(a) col A

(b) row A

(c) null A

12. Let $A = \begin{bmatrix} 1 & 3 & -7 \\ -3 & -9 & 23 \end{bmatrix}$.

(a) Is $(-6, 20)$ in col A?

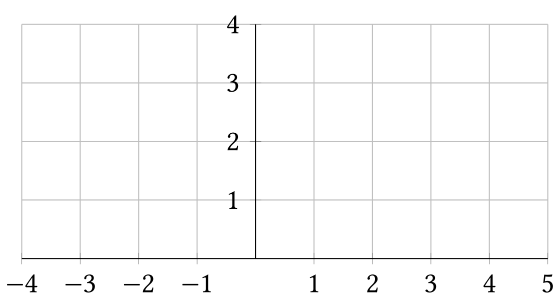
(b) Is $(0, 5, 0)$ in row A?

(c) Is $(6, -2, 0)$ in null A?

13. Let $\vec{a} = (1, 2)$ and $\vec{b} = (-2, 1)$.

(a) Draw the following points.

$\vec{a} + 0\vec{b}$
$\vec{a} + 1\vec{b}$
$\vec{a} + 2\vec{b}$
$\vec{a} - \vec{b}$
$\vec{a} - 1.8\vec{b}$



(b) Is the set of points $\{\vec{a} + t\vec{b} \,:\, t \text{ is in } \mathbb{R}\}$ a subspace of $\mathbb{R}^2$?

14. (T/F?) The set $\{\vec{a} + t\vec{b} \,:\, t \text{ is in } \mathbb{R}\}$ describes a line including the point $\vec{b}$ parallel to the vector $\vec{a}$ (as an arrow).

15. (T/F?) The null space of $[\vec{a}_1 \ \cdots \ \vec{a}_c]$ is the set of solutions $(x_1, \dots, x_c)$ of the equation $x_1\vec{a}_1 + \cdots + x_c\vec{a}_c = \vec{0}$.

16. (T/F?) The column space of $[\vec{a}_1 \ \cdots \ \vec{a}_c]$ is the set of vectors $\vec{y}$ for which the equation $x_1\vec{a}_1 + \cdots + x_c\vec{a}_c = \vec{y}$ is consistent.

17. (T/F?) If A is an $m \times n$ matrix, then the row space of A is a subspace of $\mathbb{R}^m$

18. (T/F?) If A is a matrix, $\vec{u}$ is in the null space of A and $\vec{x}$ is in the row space of A, then $\vec{u}$ and $\vec{x}$ have the same number of entries.

19. Let k be a real number. Prove that $U = \{(x_1, x_2) \,:\, x_1 = 3x_2 + k\}$ is a subspace of $\mathbb{R}^2$ if and only if $k = 0$.

20. Prove that if the set $W = \{\vec{u} + t\vec{v} \,:\, t \text{ is in } \mathbb{R}\}$ is a subspace then the list of vectors $\vec{u}, \vec{v}$ is linearly dependent.

21. Prove that if the list of vectors $\vec{u}, \vec{v}$ is linearly dependent, then the set $W = \{\vec{u} + t\vec{v} \,:\, t \text{ is in } \mathbb{R}\}$ is a subspace. [H]

22. Give an example of a non-empty subset of $\mathbb{R}^2$ that is closed under scalar multiplication, but is not a subspace of $\mathbb{R}^2$.

23. Give an example of a non-empty subset of $\mathbb{R}^2$ that is closed under vector addition but is not a subspace of $\mathbb{R}^2$.

24. Suppose A is an $r \times c$ matrix, $\vec{u} = (u_1, u_2, \dots, u_c)$ is in the null space of A and $(b_1, b_2, \dots, b_c)$ is one of the rows of A.

(a) Explain why $u_1b_1 + u_2b_2 + \cdots + u_cb_c = 0$.

(b) Suppose $\vec{x} = (x_1, x_2, \dots, x_c)$ is in the row space of A. Explain why $u_1x_1 + u_2x_2 + \cdots + u_cx_c = 0$.

25. The **intersection** of two sets P and Q is defined as $P \cap Q = \{x \,:\, x \text{ is in } P \text{ and } x \text{ is in } Q\}$. For example, $\text{span}(1,2) \cap \text{span}(3,0) = \{\vec{0}\}$. And $\{(x_1, x_2) \,:\, x_1 > 0\} \cap \{(x_1, x_2) \,:\, x_2 > 0\}$ is quadrant I. Prove that if P and Q are subspaces of $\mathbb{R}^c$, then $P \cap Q$ is a subspace too.

26. *(Challenge)* Give an example of a 4×4 matrix A so that col A = null A.

[H]21. Hint: Consider the two separate cases: $\vec{u} = \vec{0}$ or $\vec{u} \neq \vec{0}$.

1.6 Bases and Coordinates

Exercise 1. (*Warm-up*) Let $\vec{a}_1, \vec{a}_2$ be the columns of $A = \begin{bmatrix} 1 & 3 \\ 0 & 4 \\ 0 & 2 \end{bmatrix}$.

(a) Is the list $\vec{a}_1, \vec{a}_2$ linearly independent? Why or why not?

(b) Is $\operatorname{span}(\vec{a}_1, \vec{a}_2) = \mathbb{R}^3$? In other words, does $(\vec{a}_1, \vec{a}_2)$ span $\mathbb{R}^3$? Why or why not?

Exercise 2. (*Warm-up*) Let $\vec{b}_1, \vec{b}_2, \vec{b}_3$ be the columns of $B = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 2 \end{bmatrix}$.

(a) Is $\vec{b}_1, \vec{b}_2, \vec{b}_3$ linearly independent? Why or why not?

(b) Is $\operatorname{span}(\vec{b}_1, \vec{b}_2, \vec{b}_3) = \mathbb{R}^2$? In other words, does $(\vec{b}_1, \vec{b}_2, \vec{b}_3)$ span $\mathbb{R}^2$? Why or why not?

Exercise 3. (*Warm-up*) Let $\vec{e}_1, \vec{e}_2, \vec{e}_3$ be the columns of $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(a) Is the list $\vec{e}_1, \vec{e}_2, \vec{e}_3$ linearly independent? Why or why not?

(b) Is $\operatorname{span}(\vec{e}_1, \vec{e}_2, \vec{e}_3) = \mathbb{R}^3$? In other words, does $(\vec{e}_1, \vec{e}_2, \vec{e}_3)$ span $\mathbb{R}^3$? Why or why not?

1.6.1 Bases and Coordinates: Main Ideas

Definition. Let U be a subspace of $\mathbb{R}^n$. (Note that since $\mathbb{R}^n$ is a subspace of itself, this definition also applies to $U = \mathbb{R}^n$.) A list of vectors $\mathscr{B} = (\vec{b}_1, \ldots, \vec{b}_c)$ is called a **basis** for U if:

(i) $\mathscr{B}$ is linearly independent.

(ii) $\operatorname{span} \mathscr{B} = \mathbb{R}^n$.

Example 1. Is $\mathscr{B} = \left(\begin{bmatrix} 0 \\ \pi \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} \right)$ a basis of $\mathbb{R}^3$?

Example 2. Is $\mathscr{B} = \left(\begin{bmatrix} 0 \\ \pi \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} \right)$ a basis of $\operatorname{span} \mathscr{B}$?

Example 3. Is $\mathscr{E} = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right)$ a basis of $\mathbb{R}^3$? (This is called the **standard basis** of R^3 and $I_3 = \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix}$ is called the 3×3 **identity matrix**.)

Example 4. Is $\mathscr{B} = \left(\begin{bmatrix}2\\3\end{bmatrix}, \begin{bmatrix}7\\-1\end{bmatrix}, \begin{bmatrix}0\\10\end{bmatrix}\right)$ a basis of $\mathbb{R}^2$?

Example 5. Is $\mathscr{B} = \left(\begin{bmatrix}2\\3\end{bmatrix}, \begin{bmatrix}7\\-1\end{bmatrix}, \begin{bmatrix}0\\10\end{bmatrix}\right)$ a basis of span $\mathscr{B}$?

Definition. Suppose $\mathscr{B} = (\vec{b}_1, \ldots, \vec{b}_c)$ is a basis for a subspace U of $\mathbb{R}^n$, and that $\vec{y}$ is a vector in U. If

$$x_1\vec{b}_1 + \cdots + x_c\vec{b}_c = \vec{y}$$

then the vector $(x_1, \ldots, x_c)$ is called the **$\mathscr{B}$-coordinate vector of $\vec{y}$** and we write $(x_1, \ldots, x_c) = \left[\vec{y}\right]_{\mathscr{B}}$.

Example 6. $\mathscr{B} = \{\vec{b}_1, \vec{b}_2\} = \{(3,1), (-4,2)\}$ is a basis of $\mathbb{R}^2$. Use the graph to:

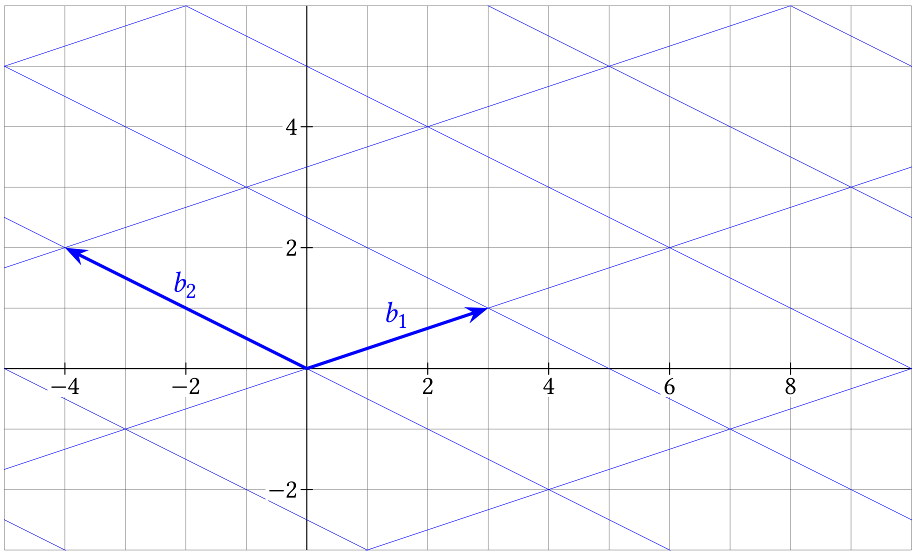
(a) Find $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = (1,2)$

(b) Find $[(-3,-1)]_{\mathscr{B}}$.

(c) Find $[(10,0)]_{\mathscr{B}}$.

(d) Find $[(7,-1)]_{\mathscr{B}}$.

(e) Find $\left[\vec{b}_1\right]_{\mathscr{B}}$



Theorem 1.14: Unique representation theorem.
Let $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_c)$ be a basis for a subspace U of $\mathbb{R}^n$. Then for each $\vec{y}$ in U, there is only one $[\vec{y}]_{\mathscr{B}}$.

Partial proof. Suppose $\vec{x} = (x_1, \dots, x_c) = [\vec{y}]_{\mathscr{B}}$ and $\vec{z} = (z_1, \dots, z_c) = [\vec{y}]_{\mathscr{B}}$. In other words, suppose $x_1\vec{b}_1 + \cdots + x_c\vec{b}_c = \vec{y}$ and $z_1\vec{b}_1 + \cdots + z_c\vec{b}_c = \vec{y}$.

Exercise 28 asks you to fill in the gaps below.

Since $\vec{y} - \vec{y} = \vec{0}$, we can write:

$$\begin{aligned} \text{(missing steps here!)} &= \vec{0} \\ \underline{\hspace{3em}}\,\vec{b}_1 + \cdots + \underline{\hspace{3em}}\,\vec{b}_c &= \vec{0} \end{aligned}$$

Since the list $(\vec{b}_1, \dots, \vec{b}_c)$ is linearly independent, we know all of the coefficients are zero: $\underline{\hspace{3em}} = 0, \dots, \underline{\hspace{3em}} = 0$. Thus $\vec{x} = \vec{z}$ and $[\vec{y}]_{\mathscr{B}}$ is unique. ■

Theorem 1.15: Removing dependent vectors doesn't change span.
Suppose $L = (a_1, \dots, a_n)$ is a list of vectors, and $U = \operatorname{span}(a_1, \dots, a_n)$. If one of the vectors $\vec{a}_j$ is a linear combination of the other vectors, then removing $\vec{a}_j$ from the list doesn't change the span. In other words, $U = \operatorname{span}(\vec{a}_1, \dots, \vec{a}_{j-1}, \vec{a}_{j+1}, \dots, \vec{a}_n)$.

Partial proof. Suppose $\vec{a}_j$ is a linear combination of the other vectors:

$$\vec{a}_j = x_1\vec{a}_1 + \cdots + x_{j-1}\vec{a}_{j-1} + x_{j+1}\vec{a}_{j+1} + \cdots + x_n\vec{a}_n$$

To show that the span doesn't change, suppose that $\vec{u} = z_1\vec{a}_1 + \cdots + z_n\vec{a}_n$ is in U. Exercise 29 is to finish rest of the proof: show that $\vec{u}$ is in $\operatorname{span}(\vec{a}_1, \dots, \vec{a}_{j-1}, \vec{a}_{j+1}, \dots, \vec{a}_n)$.

1.6.2 Bases for Column and Row Spaces

The next two examples will help justify theorem 1.16 below.

Example 7. Let $F = \begin{bmatrix} \vec{f}_1 & \vec{f}_2 & \vec{f}_3 & \vec{f}_4 & \vec{f}_5 \end{bmatrix} = \begin{bmatrix} 1 & -5 & 0 & 0 & 6 \\ 0 & 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) Explain why the list $(\vec{f}_1, \vec{f}_3, \vec{f}_4)$ is linearly independent.

(b) Explain why $\operatorname{col} F = \operatorname{span}(\vec{f}_1, \vec{f}_3, \vec{f}_4)$.

Example 8. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \vec{a}_4 & \vec{a}_5 \end{bmatrix} = \begin{bmatrix} 1 & -5 & 0 & -1 & -2 \\ 1 & -5 & 0 & -1 & -2 \\ -2 & 10 & 0 & 3 & 12 \\ 0 & 0 & 1 & 0 & 7 \end{bmatrix}$.

A is row-equivalent to F from the previous exercise, which you can check if you want.

(a) Explain why the list $(\vec{a}_1, \vec{a}_3, \vec{a}_4)$ is linearly independent.

(b) Explain why $\operatorname{col} A = \operatorname{span}(\vec{a}_1, \vec{a}_3, \vec{a}_4)$.

Theorem 1.16: Basis for column space.
The pivot columns of A are a basis for the column space of A. (*Note:* This doesn't say anything about "pivot columns of an echelon matrix equivalent to A.")

Example 9. Label the rows of F from example 7 as:

$$\vec{\phi}_1, \vec{\phi}_2, \vec{\phi}_3, \vec{\phi}_4 = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 8 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

And label the rows of A from example 8 as:

$$\vec{\alpha}_1, \vec{\alpha}_2, \vec{\alpha}_3, \vec{\alpha}_4 = \begin{bmatrix} 1 \\ -5 \\ 0 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -5 \\ 0 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} -2 \\ 10 \\ 0 \\ 3 \\ 12 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 7 \end{bmatrix}$$

(a) Explain why row F is a subset of row A.

(b) Explain why row A is a subset of row F.

(c) Explain why $\vec{\phi}_1, \vec{\phi}_2, \vec{\phi}_3$ is a basis for row F and thus row A.

Theorem 1.17: Basis for row space.
If a matrix A is row-equivalent to a row echelon matrix F, then row A = row F, and the pivot rows of F make a basis for both row A and row F.

1.6 Exercises

1.6.1 Bases and Coordinates: Main Ideas

4. Is the list of vectors $\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}-2\\-4\end{bmatrix}, \begin{bmatrix}3\\0\end{bmatrix}$ a basis of $\mathbb{R}^2$?

5. Is the list of vectors $\begin{bmatrix}7\\1\end{bmatrix}, \begin{bmatrix}3\\0\end{bmatrix}$ a basis of $\mathbb{R}^2$?

6. Is the list of one vector $\begin{bmatrix}2\\5\end{bmatrix}$ a basis of $\mathbb{R}^2$? Is it a basis of a subspace of $\mathbb{R}^2$?

7. Is the list of vectors $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}0\\-4\\5\end{bmatrix}, \begin{bmatrix}-2\\0\\-1\end{bmatrix}$ a basis of $\mathbb{R}^3$? Is it a basis of a subspace of $\mathbb{R}^3$?

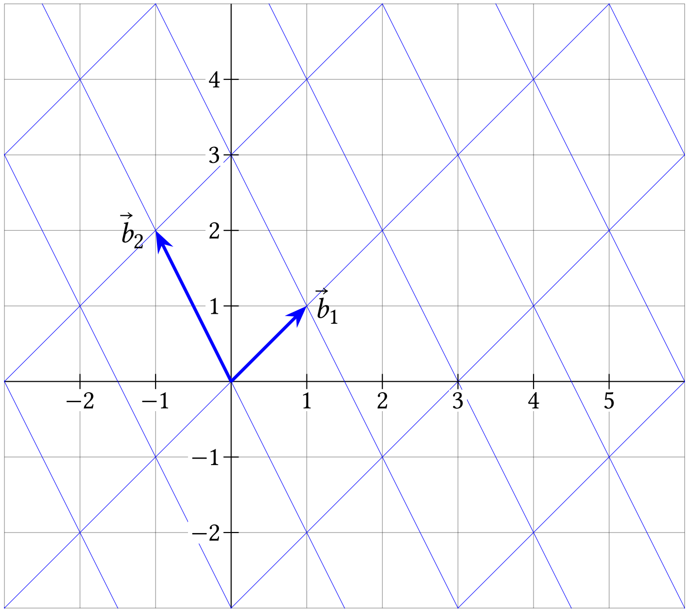
8. Is the list of vectors $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}0\\-4\\5\end{bmatrix}, \begin{bmatrix}-2\\0\\0\end{bmatrix}$ a basis of $\mathbb{R}^3$? Is it a basis of a subspace of $\mathbb{R}^3$?

9. Write your own list of vectors that spans $\mathbb{R}^4$ but isn't a basis of $\mathbb{R}^4$.

10. Write your own list of vectors in $\mathbb{R}^4$ that is linearly independent, but isn't a basis of $\mathbb{R}^4$.

11. The grid below shows $\mathscr{B} = (\vec{b}_1, \vec{b}_2)$. Find:

(a) $[(1,4)]_{\mathscr{B}}$ (b) $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = (1,4)$
(c) $[(5,-1)]_{\mathscr{B}}$ (d) $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = (5,-1)$



12. Recall that $\vec{e}_1$ means $(1,0)$ and $\vec{e}_2$ means $(0,1)$. In the grid above, with $\mathscr{B} = (\vec{b}_1, \vec{b}_2)$:

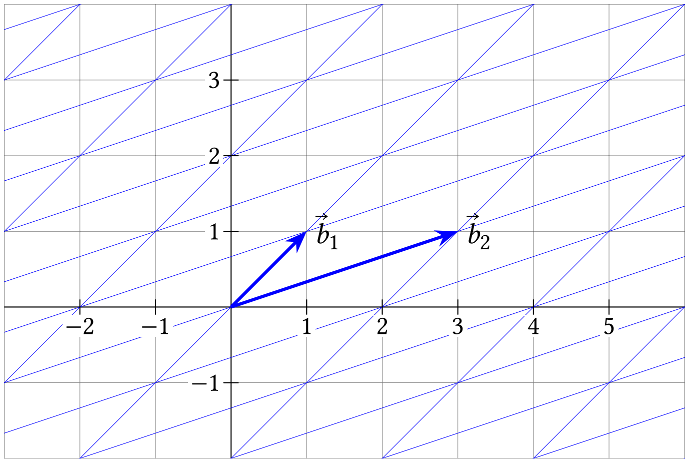
(a) Note that $2\vec{b}_1 - \vec{b}_2 = (3,0)$. Use this to find $[\vec{e}_1]_{\mathscr{B}}$.

(b) Note that $\vec{b}_1 + \vec{b}_2 = (0,3)$. Use this to find $[\vec{e}_2]_{\mathscr{B}}$.

(c) Use your results from (a) and (b) to find $[(4,2)]_{\mathscr{B}}$.

13. The grid below shows $\mathscr{B} = (\vec{b}_1, \vec{b}_2)$. Find:

(a) $[(-1,1)]_{\mathscr{B}}$ (b) $\vec{y}$ if $\left[\vec{y}\right]_{\mathscr{B}} = (-1,1)$
(c) $[(5,3)]_{\mathscr{B}}$ (d) $\vec{y}$ if $\left[\vec{y}\right]_{\mathscr{B}} = (5,3)$



14. Recall that $\vec{e}_1$ means $(1,0)$ and $\vec{e}_2$ means $(0,1)$. In the grid above, with $\mathscr{B} = (\vec{b}_1, \vec{b}_2)$:

(a) Note that $-\vec{b}_1 + \vec{b}_2 = (2,0)$. Use this to find $\left[\vec{e}_1\right]_{\mathscr{B}}$.

(b) Note that $3\vec{b}_1 - \vec{b}_2 = (0,2)$. Use this to find $\left[\vec{e}_2\right]_{\mathscr{B}}$.

(c) Use your results from (a) and (b) to find $\left[(\pi, \sqrt{2})\right]_{\mathscr{B}}$.

15. Let $\mathscr{B} = (\vec{b}_1, \vec{b}_2) = \left(\begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \end{bmatrix}\right)$. Find:

(a) $[(-3,5)]_{\mathscr{B}}$

(b) $\vec{y}$ if $\left[\vec{y}\right]_{\mathscr{B}} = (-3,5)$

(c) $[(6,0)]_{\mathscr{B}}$

(d) $\vec{y}$ if $\left[\vec{y}\right]_{\mathscr{B}} = (6,0)$

16. Let $\mathscr{B} = (\vec{b}_1, \vec{b}_2) = \left(\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 4 \end{bmatrix}\right)$.

(a) Explain why $\mathscr{B}$ is a basis of a subspace of $\mathbb{R}^3$.

(b) Find $[(2,-2,12)]_{\mathscr{B}}$.

(c) Find $\vec{y}$ if $\left[\vec{y}\right]_{\mathscr{B}} = (0,1)$.

(d) Find $[(-3,-5,2)]_{\mathscr{B}}$.

(e) Find $\vec{y}$ if $\left[\vec{y}\right]_{\mathscr{B}} = (\pi, \sqrt{2})$.

17. (T/F?) A basis of U is a list of vectors that spans U and is as small as possible.

18. (T/F?) A basis of U is a linear independent list of vectors in U that is as small as possible.

19. (T/F?) Every basis must contain $\vec{0}$.

20. (T/F?) Every linearly independent list of vectors is the basis of some subspace.

21. (T/F?) If $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_p)$ is a basis of a subspace U of $\mathbb{R}^q$, and $\vec{y}$ is in U, then $\left[\vec{y}\right]_{\mathscr{B}}$ is in $\mathbb{R}^q$.

22. (T/F?) If a list of vectors spans a subspace H, then some subset of the list is a basis of H.

23. (T/F?) If a list of vectors has *more* vectors than entries in each vector, then it cannot be a basis of a subspace of $\mathbb{R}^n$.

24. (T/F?) If a list of vectors has *fewer* vectors than entries in each vector, then it cannot be a basis of a subspace of $\mathbb{R}^n$.

25. Prove that every basis of $\mathbb{R}^n$ must have exactly n vectors.

26. Prove or find a counterexample: Every subspace U of $\mathbb{R}^n$ has a basis.

27. The $n \times n$ **identity matrix** is

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

with 1's in the main diagonal and 0's everywhere else, and it's columns, $\mathscr{E} = (\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n)$ are called the standard basis of $\mathbb{R}^n$. Suppose $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_n)$ is another basis of $\mathbb{R}^n$.

(a) Explain why $\left[\vec{b}_1\right]_{\mathscr{B}} = \vec{e}_1, \dots, \left[\vec{b}_n\right]_{\mathscr{B}} = \vec{e}_n$.

(b) Explain why $\left[\vec{b}_1\right]_{\mathscr{E}} = \vec{b}_1, \dots, \left[\vec{b}_n\right]_{\mathscr{E}} = \vec{b}_n$.

28. Write the proof of theorem 1.14 with the gaps filled in.

29. Write the finished proof of 1.15.

1.6.2 Bases for Column and Row Spaces

For problems 30 through 35, find bases for null A, col A, and row A for the given A.

30. $A = \begin{bmatrix} 3 & 6 & 9 \\ -1 & -2 & -2 \end{bmatrix}$

31. $A = \begin{bmatrix} 1 & 3 & -2 \\ -1 & -2 & 7 \\ -2 & -4 & 14 \end{bmatrix}$

32. $A = \begin{bmatrix} 0 & 0 \\ \pi & -2\pi \\ 3 & -6 \end{bmatrix}$

33. $A = \begin{bmatrix} 1 & 3 & -2 \\ -1 & -2 & 7 \\ -2 & -4 & 12 \end{bmatrix}$

34. $A = \begin{bmatrix} 2 & -6 & 4 \\ 1 & -3 & 2 \\ -1 & 4 & 1 \\ -2 & 7 & -1 \end{bmatrix}$

35. $A = \begin{bmatrix} 2 & 1 & -1 & -2 \\ -6 & -3 & 4 & 7 \\ 4 & 2 & 1 & -1 \end{bmatrix}$

36. Consider the matrices F and A which are row-equivalent in examples 7 and 8. Are the pivot columns of F, $(\vec{f}_1, \vec{f}_3, \vec{f}_4)$, a basis of col A? Explain.

37. Consider the matrices F and A which are row-equivalent in examples 7 and 8. Are the first three rows of A, the pivot rows, a basis of row A? Explain.

38. (T/F?) If a matrix A is row equivalent to a row echelon matrix M, then the pivot columns of M are a basis of col A.

39. (T/F?) If a matrix A is row equivalent to a row echelon matrix M, then the pivot rows of M are a basis of col A.

1.7 Dimension

Exercise 1. (*Warm-up*) Explain why $\mathscr{B} = \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)$ is a not basis of $\mathbb{R}^2$.

Exercise 2. (*Warm-up*) Explain why $\mathscr{B} = \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -7 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \end{bmatrix}\right)$ is a not basis of $\mathbb{R}^2$.

Exercise 3. (*Warm-up*) Let $\mathscr{B} = \left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}\right)$. Note that since $\mathscr{B}$ is linearly independent, $\mathscr{B}$ is a basis of span$(\mathscr{B})$.

(a) Find $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$.

(b) Find $[\vec{y}]_{\mathscr{B}}$ if $\vec{y} = \begin{bmatrix} 1 \\ 6 \\ 7 \end{bmatrix}$.

(c) What do you think: is it possible to have a linearly independent list of 3 vectors in span$(\mathscr{B})$?

Theorem 1.18: Properties of coordinates.
Suppose $\mathscr{B} = (\vec{b}_1, \ldots, \vec{b}_n)$ is a basis of U, a subspace of $\mathbb{R}^c$. Then, for any $\vec{y}, \vec{z}$ in U and any real number k:

(i) $\left[k\vec{y}\right]_{\mathscr{B}} = k\left[\vec{y}\right]_{\mathscr{B}}$

(ii) $\left[\vec{y} + \vec{z}\right]_{\mathscr{B}} = \left[\vec{y}\right]_{\mathscr{B}} + \left[\vec{z}\right]_{\mathscr{B}}$

(iii) If $\left[\vec{y}\right]_{\mathscr{B}} = \left[\vec{z}\right]_{\mathscr{B}}$, then $\vec{y} = \vec{z}$ (We say $f(\vec{y}) = \left[\vec{y}\right]_{\mathscr{B}}$ is an *injective* or *one-to-one* function.)

(iv) If $\vec{x}$ is any vector in $\mathbb{R}^n$, then there is a $\vec{y}$ in U so that $\left[\vec{y}\right]_{\mathscr{B}} = \vec{x}$ (We say $f(\vec{y}) = \left[\vec{y}\right]_{\mathscr{B}}$ is a *surjective* or *onto* function.)

Proofs of (i) and (iv) are exercises, so let's prove (ii) and (iii).

Theorem 1.19: More vectors than a basis are dependent. If a subspace U of $\mathbb{R}^c$ has a basis with n vectors in it, then any list with more than n vectors in U is linearly dependent.

Theorem 1.20: If U is a subspace of $\mathbb{R}^c$, then every basis of U has the same number of vectors.

Definition. The **dimension** of a subspace U of $\mathbb{R}^c$, written $\dim U$, is the number of vectors in a basis of U. If $\dim U = n$, we say U is n-dimensional.

Example 1. Find the dimension of $\mathbb{R}^3$.

Example 2. Find the dimension of the solution set of $x_1 + 8x_2 - x_3 = 0$ in $\mathbb{R}^3$.

Example 3. Find the dimension of $U = \left\{ \begin{bmatrix} 2p+t \\ -7p+4q-2r \\ 6q-3r+5t \\ 8t \end{bmatrix} : p,q,r,t \text{ in } \mathbb{R} \right\}$.

We'll need the following lemma for the "buy two, get one free" theorem below.

Lemma. Suppose U is a subspace of $\mathbb{R}^c$, with $\dim U = n$, and $\mathscr{V} = \vec{v}_1, \ldots, \vec{v}_n$ is a list of n vectors in U. Then $\mathscr{V}$ is linearly dependent if and only if $\mathscr{V}$ doesn't span U.

Theorem 1.21: Buy two, get one free.
Suppose $\vec{v}_1, \ldots, \vec{v}_n$ is a list of n vectors in a subspace U of $\mathbb{R}^c$. If any two of the following are true, then so is the third:

(i) $\dim U = n$.

(ii) $\vec{v}_1, \ldots, \vec{v}_n$ is linearly independent.

(iii) $\vec{v}_1, \ldots, \vec{v}_n$ spans U.

Definition. The dimension of the column space of A is called the **rank** of A.

Recall the pivot columns of A are a basis of col A, and the pivot rows of an echelon matrix equivalent to A are a basis of row A. And the way echelon form works, you can't have more than one pivot in single row or column. Thus the number of pivot rows of A equals the number of pivot columns of A, and $\dim(\text{row } A) = \dim(\text{col } A) = \text{rank } A$.

Example 4. Find rank A and dim(null A) for $A = \begin{bmatrix} 1 & 3 & -5 & 1 & 2 \\ -2 & -6 & 7 & 0 & -10 \end{bmatrix}$.

The next theorem is quite useful. Note that dim(null A) is sometimes called the *nullity* of A.

Theorem 1.22: The rank-nullity theorem.
If A has c columns, then:

$$\text{rank } A + \dim(\text{null } A) = c$$

1.7 Exercises

4. (T/F?) Since all vectors in $\mathbb{R}^4$ have four entries, every subspace of $\mathbb{R}^4$ is 4-dimensional.

5. (T/F?) If A is a 7×5 matrix, then $\dim(\text{null } A) = 5$.

6. (T/F?) For any matrix A, $\dim(\text{col } A) = \dim(\text{row } A)$.

7. (T/F?) For any matrix A, $\dim(\text{null } A) = \dim(\text{row } A)$.

8. (T/F?) Every linearly independent list of 3 vectors in $\mathbb{R}^5$ is the basis of some 3-dimensional subspace of $\mathbb{R}^5$.

For exercises 9 through 15, find $\dim(\text{col } A)$, $\dim(\text{row } A)$, and $\dim(\text{null } A)$.

9. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

10. $A = \begin{bmatrix} 1 & 3 & -2 \\ 4 & 12 & -8 \end{bmatrix}$

11. $A = \begin{bmatrix} 3 & 7 \\ 0 & -8 \\ 3 & -1 \\ 1 & 0 \end{bmatrix}$

12. $A = \begin{bmatrix} -1 & 4 & 8 & 34 \\ -1 & 5 & 11 & 43 \\ 1 & -2 & -2 & -16 \end{bmatrix}$

13. $A = \begin{bmatrix} -1 & -1 & 1 \\ 4 & 5 & -2 \\ 8 & 11 & -2 \\ 34 & 43 & -16 \end{bmatrix}$

14. $A = \begin{bmatrix} 3 & -2 & -4 \\ -1 & 1 & 2 \\ 6 & -2 & -4 \end{bmatrix}$

15. $A = \begin{bmatrix} 3 & -2 & -4 \\ -1 & 1 & 2 \\ 6 & -2 & -5 \end{bmatrix}$

16. If A is 7×5, what is the largest possible $\dim(\text{row } A)$?

17. If A is $7{\times}5$, and $\dim(\text{null } A) = 1$, how many rows of A can be removed without changing the row space?

18. Write a 4×3 matrix A so that $\dim(\text{null } A) = 0$ and rank $A = 3$.

19. Write a 4×3 matrix A so that $\dim(\operatorname{null} A) = 2$ and $\operatorname{rank} A = 1$.

20. Write a 4×3 matrix A so that $\dim(\operatorname{null} A) = 3$ and $\operatorname{rank} A = 0$.

21. (T/F?) If U is a subspace of $\mathbb{R}^c$, and the list $\vec{a}_1, \ldots, \vec{a}_n$ spans U, then any list of $n+1$ vectors in U is linearly dependent.

22. (T/F?) Every 1-dimensional subspace of $\mathbb{R}^2$ has either $(\vec{e}_1)$ or $(\vec{e}_2)$ as its basis.

23. (T/F?) If A is a *square* $n \times n$ matrix, then the columns of A span $\mathbb{R}^n$ if and only if they are linearly independent.

24. (T/F?) If U is a subspace of $\mathbb{R}^n$, and $\vec{u}_1, \ldots, \vec{u}_j$ is a list of j linearly independent vectors in U, then $\dim U \geq j$.

25. (T/F?) If U is a subspace of $\mathbb{R}^n$, and $\vec{u}_1, \ldots, \vec{u}_j$ is a list of j linearly dependent vectors in U, then $\dim U \leq j$.

26. (T/F?) If U is a subspace of $\mathbb{R}^n$, $\vec{u}_1, \ldots, \vec{u}_j$ is a list of j vectors in U, and $\operatorname{span}(\vec{u}_1, \ldots, \vec{u}_j) = U$, then $\dim U \leq j$.

27. (T/F?) If U is a subspace of $\mathbb{R}^n$, $\vec{u}_1, \ldots, \vec{u}_j$ is a list of j vectors in U that *doesn't* span U, then $\dim U \geq j$.

28. Frank solves a homogeneous system of 30 equations with 31 variables, and finds that every solution can be written as a multiple of a single vector. Can Frank be sure that, if he were to replace all the zeros on the right sides with any list of 30 real numbers, that the new system would always have a solution?

29. Recall that $\{(0,0,0)\}$ is a subspace of $\mathbb{R}^3$. What is the dimension of this subspace? (Find a basis for it. You may need to look up the definitions of span and linear independence.)

30. Prove the lemma before the "buy two, get one free" theorem, for the edge case where $n = 0$ and $V = ()$.

31. Prove theorem 1.18 part (i).

32. Prove theorem 1.18 part (iv).

33. Prove that we can add part (iv) onto the existence theorem below. We already proved that (i), (ii), and (iii) are equivalent, so it would be sufficient to show that (i) is equivalent to (iv).

> **Theorem 1.23: The existence theorem v2.**
> Suppose $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ is an $r \times c$ matrix. The following are equivalent:
> (i) A has a pivot position in every row.
> (ii) Every system of equations that has A as its coefficient matrix is consistent.
> (iii) $\operatorname{span}(\vec{a}_1, \ldots, \vec{a}_c) = \mathbb{R}^r$.
> (iv) $\dim(\operatorname{null} A) = c - r$.

34. Explain why we can add part (iv) to the uniqueness theorem below. We already proved that (i), (ii), and (iii) are equivalent, so it would be sufficient to show that (i) is equivalent to (iv).

> **Theorem 1.24: The uniqueness theorem v2.**
> Suppose $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ is an $r \times c$ matrix. The following are equivalent:
> (i) A has a pivot position in every column.
> (ii) $\operatorname{null} A = \{\vec{0}\}$.
> (iii) The columns $\vec{a}_1, \ldots, \vec{a}_c$ are a linearly independent list of vectors.
> (iv) $\dim(\operatorname{col} A) = c$

35. Exercise 26 in 1.5 asked you to find a 4×4 matrix A with $\operatorname{col} A = \operatorname{null} A$. Show that it's impossible to find a 5×5 matrix A with $\operatorname{col} A = \operatorname{null} A$.

36. Explain why there does not exist a 2×4 matrix whose null space is $\{(x_1, x_2, x_3, x_4 \,:\, x_1 = x_3)\}$.

37. (*challenge*) Suppose U and V are each 3-dimensional subspaces of $\mathbb{R}^5$. Show that there is a nonzero vector that is in both U and V.

38. (*challenge*) Prove or give a counterexample: If $\vec{u}_1, \vec{u}_2, \vec{u}_3$ are linearly independent vectors in $\mathbb{R}^3$, V is a subspace of $\mathbb{R}^3$ that does not contain $\vec{u}_1$ or $\vec{u}_2$, then $V = \operatorname{span}(\vec{u}_3)$.

Chapter 2

Matrices and Transformations

Chapter 1 was mostly about one system of equations at a time, or one linear combination at a time. But there are lots of physical phenomena in the world that can be described as a *composition* of linear combinations: take a linear combination of the columns of one matrix B, and then use the entries from the result as weights in a linear combination of another matrix A. Using the notation in chapter 1 to describe this would be a big ugly mess.

We'll see in 2.1 and 2.2 how we can write this composition simply as $A(B\vec{x})$ and then $(AB)\vec{x}$. In 2.2, we'll also look at other important matrix operations like matrix addition, scaling, and *transposes*. Section 2.3 is about the *inverse* of A, which is sort of like "matrix division." These operations form the basics of *matrix algebra*–we can manipulate equations where the objects being added, multiplied, etc, are matrices. These simple-looking equations give us powerful techniques to understand very complicated phenomena across the sciences.

2.1 Matrix-Vector Products and Transformations

Exercise 1. (*Warm-up*) Let $f(x) = 5x + 1$.

(a) Find $f(4 \cdot 2)$ and $4 \cdot f(2)$. Are they equal?

(b) Find $f(3 - 2)$ and $f(3) + f(-2)$. Are they equal?

Exercise 2. (*Warm-up*) Let $g(x) = 5x$.

(a) Find $g(4 \cdot 2)$ and $4 \cdot g(2)$. Are they equal?

(b) Find $g(3 - 2)$ and $g(3) + g(-2)$. Are they equal?

We'll see later that g in exercise 2 is an example of a *linear transformation* but f in exercise 1 is not.

2.1.1 Matrix-Vector Product

Definition. Matrix-vector product.
Suppose A is an $r \times c$ matrix with columns $\vec{a}_1, \ldots, \vec{a}_c$ and $\vec{x}$ is a vector in $\mathbb{R}^c$. Then we define the product $A\vec{x}$ to be the linear combination of the columns of A using the corresponding entries in $\vec{x}$ as coefficients:

$$A\vec{x} = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_c \end{bmatrix} = x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_c\vec{a}_c$$

Example 1. Find each product:

(a) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \\ \frac{1}{3} \end{bmatrix}$ (b) $\begin{bmatrix} \sqrt{e} & 5 \\ \frac{113}{17} & 6 \\ \sin\frac{\pi}{12} & 7 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

Example 2. Write the system as a vector equation and as a matrix equation in the form $A\vec{x} = \vec{y}$:

$$\begin{aligned} \pi x_1 + 2x_2 - x_3 &= 4 \\ -6x_2 + 3x_3 &= 5 \end{aligned}$$

In general, if A is an $r \times c$ matrix with columns $\vec{a}_1, \ldots, \vec{a}_c$, then the following are different ways of saying the same thing:

- The vector equation $x_1\vec{a}_1 + \cdots + x_c\vec{a}_c = \vec{y}$
- The matrix equation $A\vec{x} = \vec{y}$
- The system of equations with augmented matrix $\left[\begin{array}{ccc|c} \vec{a}_1 & \cdots & \vec{a}_c & \vec{y} \end{array}\right]$

Much of what we learned in chapter 1 can be re-stated in terms of $A\vec{x}$. Here are some definitions and theorems that have been amended, with additions in red:

Definition. If $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$, a linear system of equations that can be written in the form

$$x_1\vec{a}_1 + \cdots + x_c\vec{a}_c = \vec{0} \quad \text{or} \quad A\vec{x} = \vec{0} \tag{2.1}$$

is called **homogeneous**. The solution set of equation 2.1 is called the **null space** of A and is written null A:

$$\text{null } A = \left\{(x_1, \ldots, x_c) \;:\; x_1\vec{a}_1 + \cdots + x_c\vec{a}_c = \vec{0}\right\} = \left\{\vec{x} \;:\; A\vec{x} = \vec{0}\right\}$$

The **column space** of$A$, written col A, is the span of the columns of A:

$$\text{col } A = \left\{x_1\vec{a}_1 + \cdots + x_c\vec{a}_c \;:\; x_1, \ldots, x_n \text{ in } \mathbb{R}\right\} = \left\{A\vec{x} \;:\; \vec{x} \text{ in } \mathbb{R}^c\right\}$$

Theorem 2.1: The existence theorem, v3.
Suppose $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ is an $r \times c$ matrix. The following are equivalent:

(i) A has a pivot position in every row.

(ii) Every system of equations that has A as its coefficient matrix is consistent.

(iii) col $A = \mathbb{R}^r$.

(iv) dim(null A) $= c - r$.

(v) For every $\vec{y}$ in $\mathbb{R}^r$, the equation $A\vec{x} = \vec{y}$ has a solution.

Theorem 2.2: The uniqueness theorem, v3.
Suppose $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ is an $r \times c$ matrix. The following are equivalent:

(i) A has a pivot position in every column.

(ii) null $A = \{\vec{0}\}$.

(iii) The columns $\vec{a}_1, \ldots, \vec{a}_c$ are a linearly independent list of vectors.

(iv) dim(col A) $= c$.

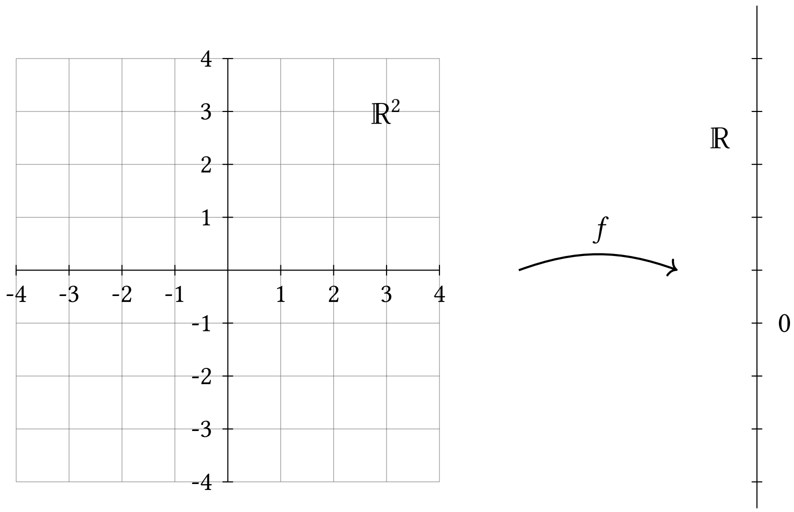
(v) If $A\vec{x} = \vec{0}$, then $\vec{x} = \vec{0}$.

2.1.2 Linear Transformations

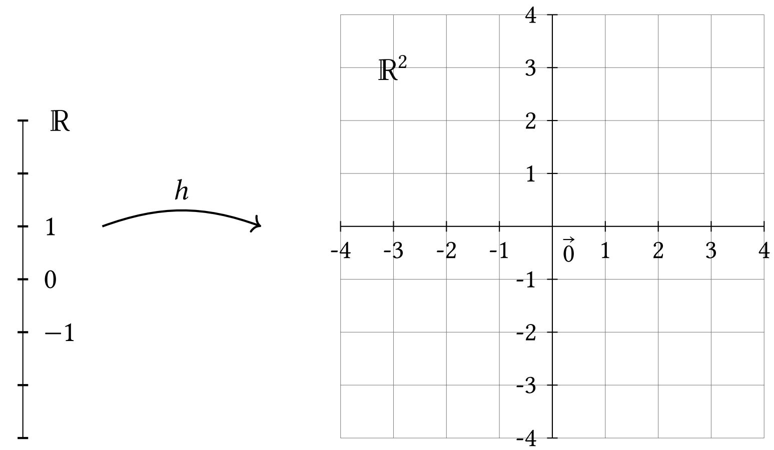
Let's take a short detour to go over some newish terminology.

Definition. A **transformation** (sometimes called a *map* or simply *function*) T from $\mathbb{R}^c$ to $\mathbb{R}^r$ takes any input vector $\vec{x}$ in $\mathbb{R}^c$ and gives a unique output vector $T(\vec{x})$ in $\mathbb{R}^r$.
The set of all inputs, $\mathbb{R}^c$, is called the **domain** of T, and $\mathbb{R}^r$, where the outputs live, is called the **codomain**. The set of all outputs $\{T(\vec{x}) : \vec{x} \text{ is in } \mathbb{R}^r\}$ is called the **range** of T. So, the range is a *subset* of the codomain.
The notation $T : \mathbb{R}^c \to \mathbb{R}^r$ specifies the domain and codomain.

Example 3. If $f : \mathbb{R}^2 \to \mathbb{R}$ has formula $f(x_1, x_2) = x_1^2 + x_2^2$, find the domain, codomain, and range of f.



Example 4. Define $h(x) = \begin{bmatrix} 3x \\ 2x \end{bmatrix}$. Find the domain, codomain, and range of h.



Example 5. Let $A = \begin{bmatrix} 2 & -\pi & 3 & \sqrt{2} \\ 4 & 0 & 1 & -8 \end{bmatrix}$ and $T(\vec{x}) = A\vec{x}$. Find $T(0, 1, -2, 0)$. Describe the domain, codomain, range of T.

Theorem 2.3: Linearity of $A\vec{x}$.
Suppose A is an $r \times c$ matrix, $\vec{x}, \vec{z}$ are in $\mathbb{R}^c$, and k is a scalar. Then:

(i) $A(\vec{x} + \vec{z}) = A\vec{x} + A\vec{z}$

(ii) $A(k\vec{x}) = k(A\vec{x})$

Proof. (i) Exercise 25 asks you to fill in the blanks:

$$\begin{aligned}
A(\vec{x}+\vec{z}) &= A\begin{bmatrix} x_1 + z_1 \\ \vdots \\ x_c + z_c \end{bmatrix} \\
&= \underline{\qquad\quad}\vec{a}_1 + \cdots + \underline{\qquad\quad}\vec{a}_c && \text{(def'n of matrix-vector product)} \\
&= \underline{\qquad} + \underline{\qquad} + \cdots + \underline{\qquad} + \underline{\qquad} && \text{(theorem 1.3 part (v))} \\
&= x_1\vec{a}_1 + \cdots + x_c\vec{a}_c + z_1\vec{a}_1 + \cdots + z_c\vec{a}_c && \text{(theorem } \underline{\qquad\qquad\qquad}\text{)} \\
&= \underline{\qquad\quad} + \underline{\qquad\quad} && \text{(def'n of matrix-vector product)}
\end{aligned}$$

(ii) Exercise 26 asks you to fill in the blanks:

$$\begin{aligned}
A(k\vec{x}) &= A\begin{bmatrix} kx_1 \\ \vdots \\ kx_c \end{bmatrix} \\
&= (\underline{\qquad\quad})\vec{a}_1 + \cdots + (\underline{\qquad\quad})\vec{a}_c && \text{(def'n of matrix-vector product)} \\
&= k(\underline{\qquad\quad}) + \cdots + k(\underline{\qquad\quad}) && \text{(theorem 1.3 part (vii))} \\
&= k(\underline{\qquad\qquad\qquad\qquad\qquad}) && \text{(theorem 1.3 part (vi))} \\
&= \underline{\qquad\qquad} && \text{(def'n of matrix-vector product)}
\end{aligned}$$

■

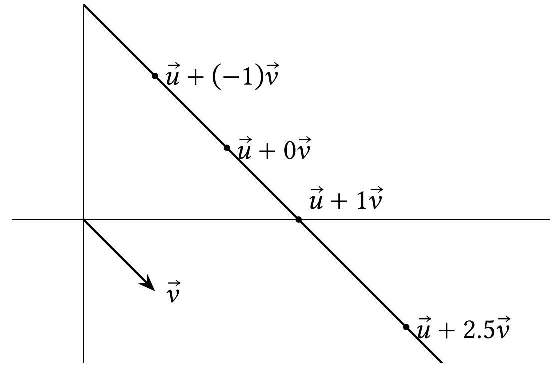
Definition. A transformation T is called **linear** if for all real numbers k, and all vectors $\vec{x}$ and $\vec{z}$ in the domain,

(i) $T(\vec{x} + \vec{z}) = T(\vec{x}) + T(\vec{z})$

(ii) $T(k\vec{x}) = kT(\vec{x})$

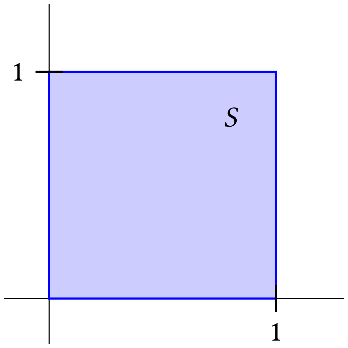
If a transformation is of the form $T(\vec{x}) = A\vec{x}$, theorem 2.3 shows it's linear. We'll soon see that *all* linear transformations $\mathbb{R}^c \to \mathbb{R}^r$ are of this form.

Example 6. Recall from section 1.5 that, given any two vectors $\vec{u}, \vec{v}$ in $\mathbb{R}^c$, the set of points $\{\vec{u} + t\vec{v} \,:\, t \text{ in } \mathbb{R}\}$ is a line through the point $\vec{u}$ parallel to $\vec{v}$ (as an arrow). Explain why linear transformations *map lines onto lines.*



Example 7. Recall $I = \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is called the 3×3 identity matrix. If $T(\vec{x}) = I\vec{x}$, find $T(9, 8, 7)$.

Example 8. The set of points $S = \{(x_1, x_2) : 0 \le x_1 \le 1, 0 \le x_2 \le 1\}$ is called the unit square. Define $T(\vec{x}) = A\vec{x}$ for $A = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$. Describe the set $\{T(\vec{x}) : \vec{x} \text{ in } S\}$. Note: this is called a *shear transformation*.



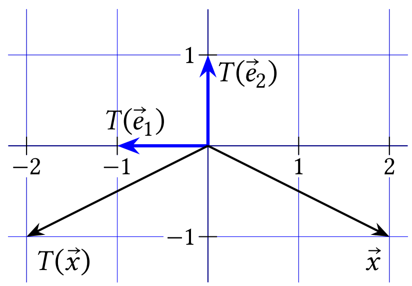
The next theorem generalizes the idea from this example.

Theorem 2.4: Standard matrix of a transformation.
If $T : \mathbb{R}^c \to \mathbb{R}^r$ is a linear transformation, then:

$$T(\vec{x}) = \begin{bmatrix} T(\vec{e}_1) & \cdots & T(\vec{e}_c) \end{bmatrix} \vec{x} = x_1 T(\vec{e}_1) + \cdots + x_c T(\vec{e}_c)$$

The matrix $A = \begin{bmatrix} T(\vec{e}_1) & \cdots & T(\vec{e}_c) \end{bmatrix}$ is called the **standard matrix** of T.

Example 9. $T : \mathbb{R}^2 \to \mathbb{R}^2$ reflects points across the x_2-axis. Find the standard matrix of T and check that it works with a couple points.



Solution. The graph shows T reflects the point $(0, 1)$, aka $\vec{e}_1$, across the x_2-axis to $(-1, 0)$. And T doesn't move $\vec{e}_2$, so $T(\vec{e}_2) = \vec{e}_2$. Thus the standard matrix of T is:

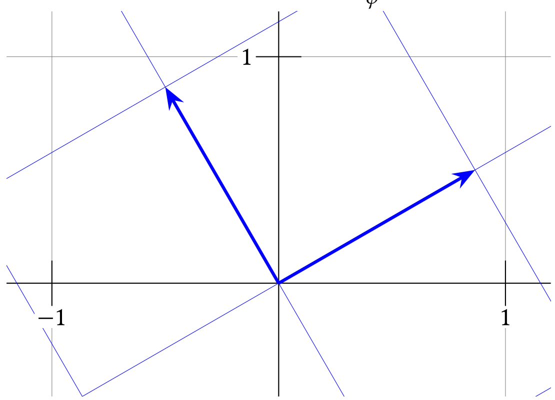
$$\begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Checking that it works with $\vec{x} = (2, -1)$:

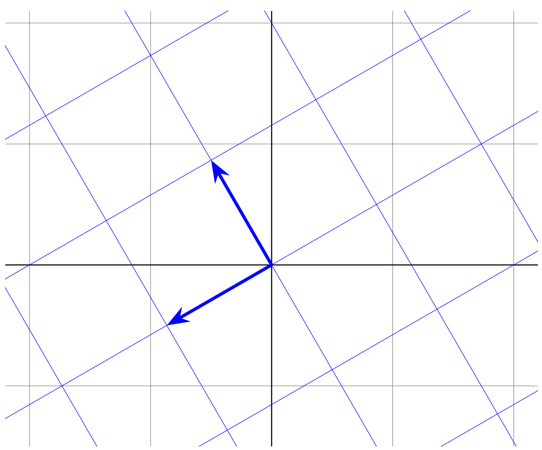
$$T(\vec{x}) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix} \quad \blacksquare$$

Example 10. $R_\phi : \mathbb{R}^2 \to \mathbb{R}^2$ rotates points an angle of ϕ counterclockwise around the origin.

(a) Find the standard matrix for R_ϕ.



(b) Check the matrix for $R_{\frac{2\pi}{3}}$ works for $\vec{x} = (1, \sqrt{3})$.



2.1 Exercises

2.1.1 Matrix-Vector Product

3. Multiply: $\begin{bmatrix} 7 & 3 \\ 0 & 2 \\ -5 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

4. Multiply: $\begin{bmatrix} 8 & 7 & 6 \\ 5 & 4 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$.

5. Multiply: $\begin{bmatrix} 8 & -3 & 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 3 \\ 0.5 \end{bmatrix}$

6. Multiply (write as a single vector): $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

7. Write as a system of linear equations (no need to solve): $\begin{bmatrix} 8 & 2 & 0 \\ 3 & \pi & 7 \\ 0.5 & -1 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ -7 \end{bmatrix}$.

8. Write as a vector equation (in the form $x_1\vec{a}_1 + \cdots + x_c\vec{a}_c = \vec{y}$) and as a matrix equation (in the form $A\vec{x} = \vec{y}$). No need to solve: $\begin{aligned} x_1 + x_2 &= 3 \\ -x_1 + 2x_2 &= 0 \end{aligned}$

9. Suppose $A = \begin{bmatrix} a & b & c \\ x & y & z \end{bmatrix}$, and $A\vec{p}$ exists.

(a) Is $\vec{p}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$?

(b) Is $A\vec{p}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$?

10. Suppose A is a 4×3 matrix. Show that there must be a vector $\vec{y}$ in $\mathbb{R}^4$ that makes $A\vec{x} = \vec{y}$ inconsistent.

11. Construct a 3×3 matrix A, with some non-zero entries, such that $\vec{x} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ is a solution of $A\vec{x} = \vec{0}$.

Remember to justify all True/False answers. Maybe cite something in the text that shows it's true, or if it's false give a counterexample or write a corrected version of it.

12. (T/F?) If $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ and $\vec{v}$ is in $\mathbb{R}^c$, then $A\vec{v}$ is in $\operatorname{span}(\vec{a}_1, \dots, \vec{a}_c)$.

13. (T/F?) If $A\vec{x} = \vec{0}$ and $\vec{x} \neq \vec{0}$ then the columns of A are linearly dependent.

14. (T/F?) If $A\vec{x} = \vec{0}$ and $\vec{x} = \vec{0}$ then the columns of A are linearly independent.

15. (T/F?) The i^{th} entry of $A\vec{x}$ is the dot product of row i of A and $\vec{x}$. Note: the dot product of $(u_1, u_2, \dots, u_c)$ and $(v_1, v_2, \dots, v_c)$ is defined to be $u_1v_1 + u_2v_2 + \cdots + u_cv_c$.

2.1.2 Linear Transformations

16. (T/F?) The words "range" and "codomain" mean exactly the same thing.

17. (T/F?) The words "transformation," "map," and "function" all mean exactly the same thing.

18. (T/F?) Every transformation that can be written $T(\vec{x}) = A\vec{x}$ is a linear transformation.

19. (T/F?) If $T : \mathbb{R}^c \to \mathbb{R}^r$ is a linear transformation, then $T(a\vec{x} + b\vec{y}) = aT(\vec{x}) + bT(\vec{y})$ for all a, b in $\mathbb{R}$ and all $\vec{x}, \vec{y}$ in $\mathbb{R}^c$.

20. Match each geometric description of a transformation with its standard matrix.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$E = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \quad F = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Does nothing: ____

Horizontal stretch: ____

Reflection across the x_1-axis: ____

Reflection across the line $x_1 = x_2$: ____

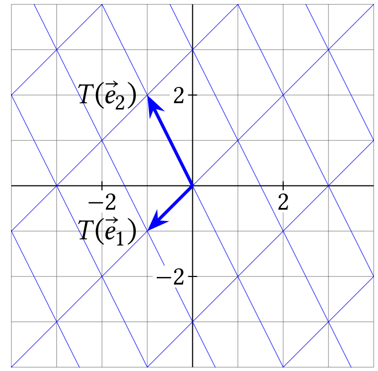
Projection onto the x_2-axis: ____

Horizontal shear (points move horizontally depending on x_2): ____

Vertical shear (points move vertically depending on x_1): ____

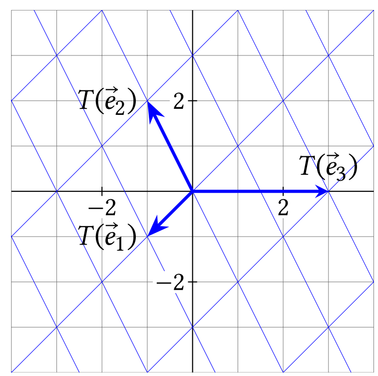
21. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ with $T(\vec{e}_1)$ and $T(\vec{e}_2)$ shown below.

(a) Find $T(-2, -1)$ and $T(0, 2)$.
(b) Find $\vec{x}$ so that $T(\vec{x}) = (3, 0)$.
(c) Find $\vec{x}$ so that $T(\vec{x}) = \vec{0}$. Explain why the solution is unique.
(d) Is the range of T a single point, a line, or all of $\mathbb{R}^2$?



22. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ with $T(\vec{e}_1), T(\vec{e}_2)$ and $T(\vec{e}_3)$ shown below.

(a) Find $T(1, 0, 1)$ and $T(3, 1, 2)$.
(b) Find a non-zero $\vec{x}$ so that $T(\vec{x}) = \vec{0}$.
(c) If $T(\vec{x}) = A\vec{x}$, find a basis for null A.
(d) Is the range of T a single point, a line, or all of $\mathbb{R}^2$?



23. Write your own example of a matrix A that is the standard matrix for $T : \mathbb{R}^2 \to \mathbb{R}^2$, with the property that the range of T is a line in $\mathbb{R}^2$. Is it possible to find a non-zero $\vec{x}$ so that $T(\vec{x}) = \vec{0}$?

24. Write your own example of a matrix A that is the standard matrix for $T : \mathbb{R}^3 \to \mathbb{R}^3$, with the property that the range of T is all of $\mathbb{R}^3$. Is it possible to find a non-zero $\vec{x}$ so that $T(\vec{x}) = \vec{0}$?

25. Write a finished proof of theorem 2.3 part (i) with the blanks filled in.

26. Write a finished proof of theorem 2.3 part (ii) with the blanks filled in.

27. Find the standard matrix for an $\mathbb{R}^2 \to \mathbb{R}^2$ linear transformation that rotates points 90° counterclockwise, and then stretches all points away from the origin by a factor of 2.

28. Find the standard matrix for an $\mathbb{R}^3 \to \mathbb{R}^3$ linear transformation that projects points onto the nearest point on the yz-plane.

29. Find the standard matrix for an $\mathbb{R}^2 \to \mathbb{R}^2$ linear transformation that reflects points across the line $x_1 = x_2$.

30. Find the standard matrix for an $\mathbb{R}^3 \to \mathbb{R}^3$ linear transformation that reflects points across the xy-plane and then rotates them counterclockwise by 45° around the x-axis.

31. Find the standard matrix for $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(x_1, x_2) = (-x_2, 3x_1 + x_2, 8x_1)$.

32. Find the standard matrix for $T : \mathbb{R}^4 \to \mathbb{R}^2$ defined by $T(x_1, x_2, x_3, x_1) = (-2x_2 + \pi x_4, x_1 + x_2 + x_3 + x_4)$.

33. Find the standard matrix for an $\mathbb{R}^2 \to \mathbb{R}^2$ that vertically shears $\vec{e}_1$ to $\vec{e}_1 + 0.5\vec{e}_2$ but leaves the x_2-axis unchanged.

34. (T/F?) The transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(\vec{x}) = \begin{bmatrix} 3 \\ -2 \end{bmatrix} + \vec{x}$ is a linear transformation.

35. (T/F?) The transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x_1, x_2) = \begin{bmatrix} \sin x_1 \\ |x_2| \end{bmatrix}$ is a linear transformation.

36. (T/F?) If A has a pivot position in every column, then the range of $T(\vec{x}) = A\vec{x}$ equals its codomain.

37. (T/F?) If A has a pivot position in every column, and $T(\vec{x}) = A\vec{x}$, then the only solution of $T(\vec{x}) = \vec{0}$ is $\vec{x} = \vec{0}$.

38. (T/F?) If A has a pivot position in every row, then the range of $T(\vec{x}) = A\vec{x}$ equals its codomain.

39. (T/F?) If A has a pivot position in every row, and $T(\vec{x}) = A\vec{x}$, then the only solution of $T(\vec{x}) = \vec{0}$ is $\vec{x} = \vec{0}$.

40. Show that if $T : \mathbb{R}^c \to \mathbb{R}^r$ is a linear transformation, then $T(\vec{0}) = \vec{0}$.

41. Suppose $T : \mathbb{R}^c \to \mathbb{R}^r$ is a linear transformation, and let $\vec{x}, \vec{y}, \vec{z}$ be a linearly dependent list in $\mathbb{R}^c$. Prove that $T(\vec{x}), T(\vec{y}), T(\vec{z})$ is linearly dependent.

42. Find a counterexample to the following false statement: If $T : \mathbb{R}^c \to \mathbb{R}^r$ is a linear transformation, and $\vec{x}, \vec{y}, \vec{z}$ is a linearly independent list in $\mathbb{R}^c$, then $T(\vec{x}), T(\vec{y}), T(\vec{z})$ is linearly independent. [H]

43. Suppose $T : \mathbb{R}^c \to \mathbb{R}^r$ and $S : \mathbb{R}^r \to \mathbb{R}^n$ are linear transformations. Show that $Q : \mathbb{R}^c \to \mathbb{R}^n$ defined by $Q(\vec{x}) = S(T(\vec{x}))$ is linear.

44. Suppose A and B are $r \times c$ matrices, and that $A\vec{x} = B\vec{x}$ for all $\vec{x}$ in $\mathbb{R}^c$. Prove that $A = B$. (Note: you can't just divide by $\vec{x}$. [H]

45. Suppose $\vec{x}$ and $\vec{z}$ are two points in $\mathbb{R}^c$, and $T : \mathbb{R}^c \to \mathbb{R}^r$ is linear.

(a) Let $\vec{w}$ be the point that is 70% of the way from $\vec{x}$ to $\vec{z}$ along a straight line. Write a formula for $\vec{w}$ in terms of $\vec{x}$ and $\vec{z}$. [H]

(b) Show that $T(\vec{w})$ is 70% of the way from $T(\vec{x})$ to $T(\vec{z})$ along a straight line.

46. Show that every linear transformation $T : \mathbb{R} \to \mathbb{R}$ has a formula $T(x) = kx$ for some real number k.

47. *(Challenge)* Find the matrix of the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ that projects points onto the nearest point on the line $x_2 = 2x_1$. [H]

48. *(Challenge)* Find an example of a function $f : \mathbb{R}^2 \to \mathbb{R}$ such that $f(k\vec{v}) = kf(\vec{v})$ for all real numbers k and all $\vec{v}$ in $\mathbb{R}^2$.

[H]42. Hint: You need an independent list $\vec{x}, \vec{y}, \vec{z}$, and a linear T that makes $T(\vec{x}), T(\vec{y}), T(\vec{z})$ dependent.

[H]44. Hint: What happens if $\vec{x} = \vec{e}_1$ or $\vec{e}_2$?

[H]45. Hint: Draw a picture. What does $\vec{z} - \vec{x}$ look like?

[H]47. Hint: Let θ be the angle from the x_1-axis up to the line. Start by rotating points by $-\theta$, which brings the line to the x_1-axis. Then project points onto the x_1-axis, and then rotate them back by θ.

2.2 Matrix Multiplication

Exercise 1. (*Warm-up*) Multiply: (a) $\begin{bmatrix} -2 & 3 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 7 \end{bmatrix}\begin{bmatrix} \pi \\ e \\ 0 \end{bmatrix}$

Exercise 2. (*Warm-up*) Let $A = \begin{bmatrix} 0 & -4 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & -0.5 \end{bmatrix}$. Multiply:

(a) $A\left(B\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right)$

(b) $\begin{bmatrix} -4 & -12 & 2 \\ -2 & -3 & 7 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Note: It turns out that $A(B\vec{x}) = \begin{bmatrix} 0 & 4 & 2 \\ 2 & -10 & -18 \end{bmatrix}\vec{x}$ for all $\vec{x}$ in $\mathbb{R}^3$. Later in this section we'll figure out a formula for how to multiply two matrices that makes $AB = \begin{bmatrix} 0 & 4 & 2 \\ 2 & -10 & -18 \end{bmatrix}$ and in general, $A(B\vec{x}) = (AB)\vec{x}$.

Exercise 3. (*Warm-up*) Recall that the *dot product* of $(u_1, \dots, u_c)$ and $(v_1, \dots, v_c)$ is $u_1v_1 + \cdots + u_cv_c$.

(a) Multiply, and write as a single vector (even though you can't combine like terms):

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

(b) (T/F?) In general, the i^{th} entry in $A\vec{x}$ is the dot product of row i of A and $\vec{x}$.

Matrix scaling and addition work the same as with vectors, which are really just single-column matrices. If the matrices are different sizes, their sum is undefined. Also, we'll recycle the symbol 0 to stand for a matrix where all the entries equal the number 0.

Example 1. Simplify:

(a) $3I_2 + 0$

(b) $\begin{bmatrix} 0 & 4 & 8 \\ 12 & 16 & 20 \end{bmatrix} - 2\begin{bmatrix} \pi & e \\ 7 & 6 \\ 2 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 10 & 11 & 12 \\ 13 & 14 & 15 \end{bmatrix} - \left[3I_2 \mid \vec{0}\right]$

Theorem 2.5: If A, B, and F are equal-sized matrices, and p, q are real numbers, then:

(i) $A + B = B + A$

(ii) $A + (B + F) = (A + B) + F$

(iii) $0 + A = A$

(iv) $(p + q)A = pA + qA$

(v) $p(A + B) = pA + pB$

(vi) $p(qA) = (pq)A$

Each of these are true because of a corresponding property of real number addition/multiplication that applies to each of the entries in the matrices. Proofs are left as exercises.

Matrix multiplication is where it gets complicated. The goal is to define (AB) so that $(AB)\vec{x} = A(B\vec{x})$. Suppose $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_c \end{bmatrix}$ is a $r \times c$ matrix and $B = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \cdots & \vec{b}_d \end{bmatrix}$ is a $c \times d$ matrix. Let's fill in the blanks to check our understanding:

B defines a linear transformation that takes any input vector $\vec{x}$ in $\mathbb{R}^{\_\_}$ and gives an output vector $B\vec{x} =$ ________________ in $\mathbb{R}^{\_\_}$.

Since the number of entries in $B\vec{x}$ equals the number of columns of A, we know

$A(B\vec{x})$ is defined:

$$\begin{aligned} A(B\vec{x}) &= A(\qquad\qquad\qquad\qquad\qquad) && \text{(defn of matrix-vector product)} \\ &= && \text{(linearity of matrix-vector product)} \\ &= && \text{(linearity of matrix-vector product)} \\ &= \begin{bmatrix} \qquad\qquad\qquad\qquad\qquad \end{bmatrix}\vec{x} && \text{(defn of matrix-vector product)} \end{aligned}$$

Thus, the following definition makes it so for any $\vec{x}$ in $\mathbb{R}^d$, we have $(AB)\vec{x} = A(B\vec{x})$.

Definition. If A is an $r \times c$ matrix, and $B = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \cdots & \vec{b}_d \end{bmatrix}$ is a $c \times d$ matrix, then the product is the $r \times d$ matrix:

$$AB = \begin{bmatrix} A\vec{b}_1 & A\vec{b}_2 & \cdots & A\vec{b}_d \end{bmatrix}$$

Example 2. Let $A = \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 4 & -2 \\ 0 & 3 & 1 \end{bmatrix}$. Multiply:

(a) AB

(b) BA

Example 3. Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$. Multiply:

(a) AB

(b) BA

Recall from exercise 3 that the i^{th} entry in $A\vec{x}$ is the dot product of row i of A and $\vec{x}$. If we apply this rule to the i^{th} entry in $A\vec{b}_j$ (the j^{th} column in AB) we get the following rule:

Theorem 2.6: Row-column rule. The entry in row i, column j of AB is the dot product of row i of A and column j of B.
In other words, if we write $(AB)_{ij}$ for the entry in row i, column j of AB, then:

$$(AB)_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j} + \cdots + A_{ic}B_{cj}$$

The original definition of AB is very useful for conceptual understanding, proofs, and computations with GPUs that can do lots of multiplications in parallel. The row-column rule is often more comfortable for hand calculations.

Example 4. Repeat the last example, but use the row-column rule.

Example 5. True/False?

(a) If $MA = MB$, and $M \neq 0$, then $A = B$.

(b) If $AB = 0$ then either $A = 0$ or $B = 0$.

The last example shows you can't assume that properties of real numbers apply to matrices as well. But fortunately, numbers and matrices do have a lot in common.

> **Theorem 2.7: Properties of matrix multiplication.** If A, B, and C have sizes for which the products and sums are defined, then:
>
> (i) $A(BC) = (AB)C$ (matrix multiplication is *associative*)
>
> (ii) If A is $r \times c$, then $I_r A = A I_c = A$
>
> (iii) $A(B + C) = AB + AC$
>
> (iv) $(A + B)C = AC + BC$
>
> (v) If k is a real number, then $k(AB) = (kA)B = A(kB)$

Proofs of (ii) through (v) are left as exercises.

> **Definition.** The **transpose** of an $r \times c$ matrix A, written A^T, is the $c \times r$ matrix with rows and columns swapped. In other words, $(A^T)_{ij} = A_{ji}$.

Example 6. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}^T =$

Example 7. Suppose $A = \begin{bmatrix} a_{11} & \cdots & a_{1c} \\ a_{21} & \cdots & a_{2c} \\ \vdots & \vdots & \vdots \\ a_{r1} & \cdots & a_{rc} \end{bmatrix}$, and we write the rows of A as $\vec{\alpha}_1 = \begin{bmatrix} a_{11} \\ \vdots \\ a_{1c} \end{bmatrix}, \vec{\alpha}_2 = \begin{bmatrix} a_{21} \\ \vdots \\ a_{2c} \end{bmatrix}, \ldots, \vec{\alpha}_r = \begin{bmatrix} a_{r1} \\ \vdots \\ a_{rc} \end{bmatrix}$. Then,

$$\begin{aligned} \vec{\alpha}_1^T &= \\ \vec{\alpha}_2^T &= \\ &\vdots \\ \vec{\alpha}_r^T &= \end{aligned}$$

So we can write A in terms of its rows:

$$A =$$

Theorem 2.8: If A and B are equal-sized matrices, and k is a real number, then:

(i) $(A^T)^T = A$

(ii) $(kA)^T = kA^T$

(iii) $(A+B)^T = A^T + B^T$

The proof is straightforward and left for exercises.

Example 8. Suppose A and B have sizes that make AB defined. Is A^TB^T necessarily defined? Is B^TA^T necessarily defined?

Theorem 2.9: If AB is defined, then $(AB)^T = B^TA^T$.

Example 9. Let $\vec{x} = (1, 2, 3)$. Multiply:

(a) $\vec{x}^T\vec{x}$

(b) $\vec{x}\vec{x}^T$

2.2 Exercises

For exercises 4 through 12, either multiply the matrices, or if it's undefined, say so.

4. $\begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0.5 \\ -1 & 2 & 0 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 0 & 0.5 \\ -1 & 2 & 0 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.25 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

7. $\begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}^T\begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}$

8. $\begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}\begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}^T$

9. $\begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 4 & \pi \end{bmatrix}$

10. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 4 & \pi \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

11. $\begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$

12. $\begin{bmatrix} 7 \\ -2 \end{bmatrix}\begin{bmatrix} 1 & 3 \end{bmatrix} + \begin{bmatrix} -3 \\ 1 \end{bmatrix}\begin{bmatrix} 2 & 7 \end{bmatrix}$

13. If $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, simplify: (a) $2A$ (b) $A + A^T$ (c) $A + 10I_2$

14. If $E = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, simplify: (a) A^2 (b) $A^2 - I_3$ (c) $2A - I_3$

15. Suppose A is 5×7, B is 7×4, $\vec{x}$ is in $\mathbb{R}^7$, $\vec{y}$ is in $\mathbb{R}^5$. For each of the following products, determine whether it's defined or not, and if it's defined, give the size of the result.

(a) AB (b) $(AB)^T$ (c) $A\vec{x}$
(d) $\vec{y}A$ (e) $\vec{y}^TA$ (f) $A\vec{y}$
(g) BA (h) AA^T (i) A^TA

16. Suppose F is 6×3, G is 5×3, $\vec{w}$ is in $\mathbb{R}^3$, $\vec{z}$ is in $\mathbb{R}^5$. For each of the following products, determine whether it's defined or not, and if it's defined, give the size of the result.

(a) FG (b) FG^T (c) $(GF^T)^T$
(d) $G\vec{w}$ (e) $F\vec{w}$ (f) $\vec{w}^TF^T$
(g) $\vec{z}^TG$ (h) $\vec{z}\vec{z}$ (i) $\vec{z}^T\vec{z}$

17. (T/F?) The linear transformation $T(\vec{x}) = (AB)\vec{x}$ first multiplies $\vec{x}$ by A and then multiplies the result by B.

18. (T/F?) If A is 3×2 and B is 2×3, then both AB and BA are defined.

19. (T/F?) The columns of AB are linear combinations of the columns of B.

20. (T/F?) The columns of AB are linear combinations of the columns of A.

21. (T/F?) If $AB = 0$, then either $A = 0$ or $B = 0$.

22. (T/F?) If A, B, C are matrices that makes the products defined, then $(AB)C = A(BC)$.

23. (T/F?) If A and B have equal sizes, then $(A + B)^T = (B + A)^T$.

24. (T/F?) If A, B are 2×2 matrices, and the two columns of A are equal, then the two columns of AB are equal.

25. (T/F?) If A, B are 2×2 matrices, and the two columns of B are equal, then the two columns of AB are equal.

26. Suppose $A = \begin{bmatrix} 5 & -6 \\ 4 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} -7 & k \\ -6 & 8 \end{bmatrix}$. Find k that makes $AB = BA$, if possible.

27. Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 \\ 2 & k \end{bmatrix}$. Find k that makes $AB = BA$, if possible.

28. A transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ first rotates points 30° degrees counterclockwise, then reflects them across the x_2-axis. Find the standard matrix for T two ways:

(a) By finding $T(\vec{e}_1)$ and $T(\vec{e}_2)$ directly, like in section 2.1.

(b) By finding the matrix R_{30° that rotates, and the matrix F that reflects, and multiplying them in the proper order.

29. A transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ first rotates points 90° around the y-axis, in a direction that rotates the x-axis onto the z-axis. Then, it rotates points 30° around the x axis, in a direction that moves the y-axis towards the z-axis. Find the standard matrix for T two ways:

(a) By finding $T(\vec{e}_1)$, $T(\vec{e}_2)$, and $T(\vec{e}_3)$ directly, like in section 2.1.

(b) By finding two matrices for the two rotations and multiplying them in the proper order.

30. Find R_{45°, the matrix for the transformation that rotates points 45° counterclockwise. Then find R_{-45°. Multiply them to check that $R_{45^\circ} R_{-45^\circ} = I$. [C]

31. Note that if you rotate points around the origin by an angle of β and then by an angle of α, it is as if you moved them by an angle of $\alpha + \beta$. Multiply the rotation matrices R_α and R_β to derive formulas for $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$. [C]

32. Prove theorem 2.5 part (i).

33. Prove theorem 2.5 part (iv).

34. Prove theorem 2.5 part (vi).

35. Prove theorem 2.7 part (ii).

36. Prove theorem 2.7 part (iii).

37. Prove theorem 2.8 part (i).

38. Prove theorem 2.8 part (ii).

39. Which of these is true about $(A^T A)^T$? (pick one.)

(A) It might be undefined.

(B) It's always equal to $A^T A$.

(C) It's always equal to AA^T.

40. (T/F?) If AB has a column with all zeros, then the columns of A are linearly dependent.

41. (T/F?) If AB has a column with all zeros, then the columns of B are linearly dependent.

42. Find a counterexample that shows this is false: If A, B are square matrices, and $AB = 0$, then $BA = 0$.

43. Show that if A is a scalar multiple of the $n \times n$ identity matrix, and B is any $n \times n$ matrix, then $AB = BA$.

44. A *diagonal matrix* is a square matrix where all the entries *not* on the main diagonal (a_{ij} with $i \neq j$) are zero. Show that if A and B are $n \times n$ diagonal matrices, then $AB = BA$.

45. (*challenge*)

(a) Show that each row of AB is a linear combination of the rows of B. [H]

(b) Show that if AB has a row of all zeros, then the rows of B are linearly dependent.

46. (*challenge*) Find the matrix for the $\mathbb{R}^2 \to \mathbb{R}^2$ transformation that reflects points across the line $x_2 = \frac{-3}{2} x_1$, by finding matrices that rotate the line onto the x_1-axis, reflect across the x_1-axis, and then rotate the x_1-axis back to the line, and then multiplying the matrices in the right order. [C]

[C]30. Credit: Inspired by an exercise in Sergei Treil's excellent book *Linear Algebra Done Wrong*.
[C]31. Credit: Inspired by an exercise in Sergei Treil's excellent book *Linear Algebra Done Wrong*.

[H]45. Hint: Theorem 2.9.
[C]46. Credit: Inspired by Treil.

2.3 Inverses

Exercise 1. (*Warm-up*) Let's recall basic facts of "multiplicative inverses" of real numbers.

(a) (T/F?) There exists a real number x such that $5x = x5 = 1$.

(b) (T/F?) There exists a real number y such that $0y = y0 = 1$.

Exercise 2. (*Warm-up*) Let $A = \begin{bmatrix} 3 & 2 \\ -2 & -2 \end{bmatrix}$.

(a) Solve the equations $A\vec{x}_1 = \vec{e}_1$ and $A\vec{x}_2 = \vec{e}_2$.

(b) Let $X = \begin{bmatrix} \vec{x}_1 & \vec{x}_2 \end{bmatrix}$, with your solutions from (a) in the columns, and multiply: AX. Are you surprised by the result?

(c) Multiply XA. Are you surprised by the result?

(d) (T/F?) A is row-equivalent to I, and the row operations that reduced A to I also reduced $\vec{e}_1$ to $\vec{x}_1$, and $\vec{e}_2$ to $\vec{x}_2$.

Definition. A square ($c \times c$) matrix A is called **invertible** if there exists a $c \times c$ matrix A^{-1}, called the **inverse** of A, such that:

$$A^{-1}A = AA^{-1} = I$$

If there are two matrices P and Q each satisfying the definition of the inverse of A, then

$$P = PI = P(AQ) = (PA)Q = IQ = Q$$

and so the inverse of A is unique.

Example 1. Let $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$ and $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$. Show that $E = F^{-1}$ and $F = E^{-1}$.

Example 2. Explain why $A = \begin{bmatrix} 6 & 0 \\ 7 & 0 \end{bmatrix}$ is not invertible.

[H]2. Hint: Denote $A^{-1} = \begin{bmatrix} \vec{x}_1 & \vec{x}_2 & \vec{x}_3 \end{bmatrix}$, and think about $AA^{-1} = I$.

Example 3. Let $A = \begin{bmatrix} 1 & 3 & 4 \\ -1 & -4 & -2 \\ 0 & 1 & -1 \end{bmatrix}$. Find A^{-1} if possible. H

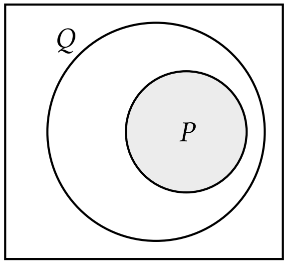
The following theorem generalizes the idea of the last example.

Theorem 2.10: If A is a $c \times c$ invertible matrix, then the $c \times 2c$ matrix $\left[A \mid I\right]$ is row-equivalent to $\left[I \mid A^{-1}\right]$

A proof is in the textbook.

Example 4. If $A = \begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}$, find A^{-1} if possible.

Recall that the *contrapositive* of a statement in the form "If P, then Q" is "If not Q, then not P." In general, a statement and its contrapositive are equivalent, as the Venn diagram suggests. The following corollary is essentially the contrapositive of theorem 2.10.



Corollary: If A is not row-equivalent to I, then A is not invertible.

In the section 2.4 we'll prove the converse of theorem 2.10: that if A is row-equivalent to I, then A is invertible; and the Venn diagram looks like just one circle.

Theorem 2.11: 2×2 inverses. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then:

(i) A is invertible if and only if $ad - bc \neq 0$.

(ii) If A is invertible, then $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Let's finish up with some nice comfortable theorems about inverses.

Theorem 2.12: Properties of inverses.
Suppose A and B are invertible matrices. Then:

(i) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(ii) A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

(iii) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

Theorem 2.13: If A is an invertible $c \times c$ matrix, then for any $\vec{y}$ in $\mathbb{R}^c$, the equation $A\vec{x} = \vec{y}$ has the unique solution $\vec{x} = A^{-1}\vec{y}$.

2.3 Exercises

3. Find the inverse of $\begin{bmatrix} 1 & 7 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

4. Find the inverse of $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -4 & 2 & 1 \end{bmatrix}$

5. Give your own example of a non-invertible 2×2 matrix A. Justify your answer.

6. Give your own example of a non-invertible 3×3 matrix A. Justify your answer.

7. Check that $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is its own inverse.

8. (T/F?) If A and B are invertible, then $(AB)^{-1} = A^{-1}B^{-1}$.

9. (T/F?) We have proven in this section that if A is row equivalent to I, then A is invertible.

10. (T/F?) If A has two identical rows, then A is not invertible.

11. (T/F?) If A is invertible, and $T(\vec{x}) = A\vec{x}$, then the range of T is equal to its codomain.

12. (T/F?) There exist matrices A, P, Q such that $AP = PA = I$, $AQ = QA = I$, and $P \neq Q$.

13. If A is an invertible $c \times c$ matrix, what is rank A?

14. If A is an invertible $c \times c$ matrix, then $\dim(\text{null } A)$?

15. Find the inverse of $A = \begin{bmatrix} -4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and check by multiplying.

16. The matrix in 15 is an example of a *diagonal* matrix, where $a_{ij} = 0$ if $i \neq j$.

(a) How can you quickly determine if a diagonal matrix is invertible or not?

(b) Explain why the inverse of an invertible diagonal matrix is also diagonal.

17. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

(a) Find A^{-1} by row-reducing $\left[A \mid I\right]$ to $\left[I \quad A^{-1}\right]$.

(b) Find A^{-1} using the formula in theorem 2.11.

(c) Use theorem 2.13 to solve $A\vec{x} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$.

(d) Use theorem 2.13 to write the solution to $A\vec{x} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ in terms of y_1 and y_2.

18. Let $A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix}$.

(a) Find A^{-1} by row-reducing $\left[A \mid I\right]$ to $\left[I \quad A^{-1}\right]$.

(b) Use theorem 2.13 to write the solution to $A\vec{x} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$ in terms of p, q and r.

19. Define a transformation $T : \mathbb{R}^5 \to \mathbb{R}^5$ by $T(x_1, x_2, x_3, x_4, x_5) = (x_5, x_1, x_2, x_3, x_4)$. This is an example of a "permutation."

(a) Find A, the standard matrix for T.

(b) Find A^{-1} and check by multiplying.

20. Suppose $B = \begin{bmatrix} \vec{b}_1 & \cdots & \vec{b}_c \end{bmatrix}$ is an invertible $c \times c$ matrix.

(a) Show that $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_c)$ is a basis of $\mathbb{R}^c$.

(b) Show that $\left[\vec{y}\right]_{\mathscr{B}} = B^{-1}\vec{y}$ for any $\vec{y}$ in $\mathbb{R}^c$.

21. If P is invertible, and $B = PAP^{-1}$, solve for A. Can you conclude that $A = B$?

22. Prove or give a counterexample: If A and B are invertible, then $A + B$ is invertible.

23. Prove or give a counterexample: If A is invertible, and $AB = BC$, then $B = C$.

24. Prove or give a counterexample: If A has two identical columns, then A is not invertible.

25. Let $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$. Describe all vectors $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ such that $A\vec{x} = \begin{bmatrix} 1 \end{bmatrix}$, aka I_1.

26. Give an example of a 2×3 matrix A and 3×2 matrix B such that $AB = I$. Is $BA = I$?

27. Suppose A is a $c \times c$ matrix, and $(\vec{b}_1, \dots, \vec{b}_c)$ is a basis of $\mathbb{R}^c$. Show that $(A\vec{b}_1, \dots, A\vec{b}_c)$ is a basis of $\mathbb{R}^c$.

H28. Hint: Show that $b \neq 0$ and $c \neq 0$.

28. Prove the following claim that was left out of the proof of theorem 2.11: If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible, and $a \neq 0$, then $ad - bc \neq 0$. H

29. Perform the two multiplications below from the proof of theorem 2.11 to check that you get I both ways.

$$\left(\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right)\begin{bmatrix} a & b \\ c & d \end{bmatrix} = I \quad \text{and} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}\left(\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) = I$$

30. Prove theorem 2.12 part (i).

31. Prove theorem 2.12 part (ii).

32. Prove that if $A^2 = 0$, then A is not invertible.

33. Show that if A is invertible and *symmetric*, which means $A^T = A$, then A^{-1} is also symmetric.

34. Prove theorem 2.14 below. H

Theorem 2.14: If A is an $r \times c$ matrix, and there exists a matrix P such that $PA = I$, then the columns of A are a linearly independent list of vectors.

35. Prove theorem 2.15 below. H

Theorem 2.15: If A is an $r \times c$ matrix, and there exists a matrix Q such that $AQ = I$, then for any $\vec{y}$ in $\mathbb{R}^r$, the equation $A\vec{x} = \vec{y}$ has a solution.

36. Suppose A is $r \times c$, F is $c \times r$, and $AF = FA = I$. The following parts outline a proof that A must be square.

(a) What does theorem 2.14 say about the pivot positions in A? What does this say about r and c?

(b) What does theorem 2.15 say about the pivot positions in A? What does this say about r and c?

H34. Hint: The goal is to prove that if $A\vec{x} = \vec{0}$, then $\vec{x} = \vec{0}$.

2.4 Elementary Matrices and the Invertible Matrix Theorem

Exercise 1. (*Warm-up*) Review theorem 7.13 (buy two, get one free) and answer these true/false questions.

(a) (T/F?) If $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ is a square matrix with linearly independent columns, then for every $\vec{y}$ in $\mathbb{R}^c$, the equation $A\vec{x} = \vec{y}$ has a solution.

(b) (T/F?) If $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$ is a square matrix such that for every $\vec{y}$ in $\mathbb{R}^c$, the equation $A\vec{x} = \vec{y}$ has a solution, then A has linearly independent columns.

(c) Is part (a) true if A is not necessarily square?

(d) Is part (b) true if A is not square?

Exercise 2. (*Warm-up*) Let $E = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$ and $F = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. Check that $EF = FE = I$. What is the row operation that transforms I to E? Can you transform E back to I with a different row operation?

Definition. An **elementary matrix** is the result of applying a *single* row operation (swapping, scaling, or multaddiplacement) to an identity matrix.

Example 1. Here are some elementary matrices:

$$\begin{bmatrix} 1 & 0 \\ \pi & 1 \end{bmatrix} \qquad \begin{bmatrix} -3 & 0 \\ 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -7 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

H35. Hint: Let $\vec{y}$ be in $\mathbb{R}^r$. What can you do with $Q\vec{y}$?

We'll soon see that multiplication by an elementary matrix *does* the corresponding row operation, but we need a little more machinery to prove it first. Recall the definition of $AB = \begin{bmatrix} A\vec{b}_1 & \cdots & A\vec{b}_c \end{bmatrix}$ shows that column j of AB is a linear combination of the columns of A, using the entries from $\vec{b}_j$ as weights. The next theorem shows an analogous result about rows.

Theorem 2.16: Rows of BA.
Suppose B is a $d \times r$ matrix, and A is an $r \times c$ matrix. Write the rows B as column vectors $\vec{\beta}_1, \ldots \vec{\beta}_d$, so that $B = \begin{bmatrix} \vec{\beta}_1^T \\ \vdots \\ \vec{\beta}_d^T \end{bmatrix}$. Then row i of BA is $\vec{\beta}_i^T A$ which is a linear combination of the rows of A using the entries in $\vec{\beta}_i$ as weights.

Example 2. Let $E = \begin{bmatrix} 1 & 0 \\ \pi & 1 \end{bmatrix}$, $F = \begin{bmatrix} -3 & 0 \\ 0 & 1 \end{bmatrix}$, and $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

(a) Multiply EA and FA using theorem 2.16. What do you notice?

(b) Find F^{-1}.

Example 3. Let $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $F = \begin{bmatrix} 1 & -7 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, and $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$.

(a) Multiply EA and FA using theorem 2.16. What do you notice?

(b) Find E^{-1}.

Note: The above examples should convince you that every elementary matrix E has an inverse, which is defined by the inverse row operation of the one that was done to I to make E.

The above examples also justify the following theorem. They're not formal proof, but there's nothing special about the particular row operations in the examples, and the concept works the same way in general.

Theorem 2.17: Elementary matrices do row operations.
If A is any $r \times c$ matrix, and E is an $r \times r$ elementary matrix that is the result of doing some row operation to I, then EA is the result of doing the same operation to A.

The next theorem combines a bunch of things we already know with one new fact to prove. Combined, it's a powerful and easy-to-use tool.

Theorem 2.18: Invertible Matrix Theorem (IMT).
Suppose A is a $c \times c$ matrix. Then the following are equivalent:

(i) A is invertible.

(ii) There exists a matrix P such that $PA = I_c$.

(iii) There exists a matrix Q such that $AQ = I_c$.

(iv) A is row-equivalent to I_c.

(v) A has c pivot positions.

(vi) The columns of A are linearly independent.

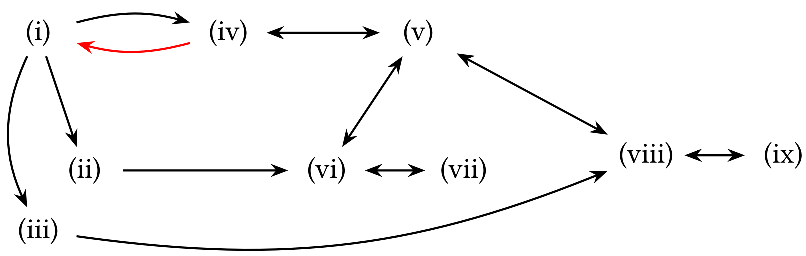
(vii) null $A = \{\vec{0}\}$.

(viii) For every $\vec{y}$ in $\mathbb{R}^c$, $A\vec{x} = \vec{y}$ has a solution.

(ix) col $A = \mathbb{R}^c$

C10. Credit: Inspired by an exercise in David Austin's excellent *Understanding Linear Algebra*.

Proof.



In the graph, a black arrow indicates an implication that we either already know, or is easy to prove. The red arrow indicates the implication we need to prove. Notice that if you know that any one of the parts is true, you can follow arrows that imply any other part is true.

[C]18. Credit: Inspired by Austin.

[H]19. Hint: Theorem 2.12 doesn't help you here, because you don't know that A and B are invertible. But the IMT might help.

2.4 Exercises

3. (T/F?) All elementary matrices are invertible.

4. (T/F?) If $\vec{a}_1, \dots, \vec{a}_c$ is a list of c linearly independent vectors in $\mathbb{R}^c$, then $[\vec{a}_1 \cdots \vec{a}_c]$ is an invertible matrix.

5. (T/F?) If A is $c \times c$, and $\dim(\text{null } A) = c$, then A is invertible.

6. (T/F?) If A is invertible, then there's a nontrivial solution to the homogeneous equation $A\vec{x} = \vec{0}$.

7. (T/F?) If the columns of $c \times c$ matrix A do not span $\mathbb{R}^c$, then A is not invertible.

8. (T/F?) If A is an invertible $c \times c$ matrix, then the columns of A are a basis of $\mathbb{R}^c$.

9. (T/F?) If A is $c \times c$, and $A\vec{x} = \vec{0}$ has only the trivial solution, then row $A = \mathbb{R}^c$.

10. (T/F?) If A is invertible, and B is row-equivalent to A, then B is invertible. [C]

11. (T/F?) An elementary matrix is any square matrix that results from applying some sequence of row operations to I.

12. (T/F?) If E is an elementary matrix, then AE is the result of applying the same row operation to A that transformed I to E.

13. Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, is invertible and let $\mathscr{A} = (\vec{a}_1, \vec{a}_2)$ be the basis of $\mathbb{R}^2$ made by the columns of A.

(a) Find $[\vec{e}_1]_{\mathscr{A}}$ and $[\vec{e}_2]_{\mathscr{A}}$.
(b) Let $\mathscr{E} = (\vec{e}_1, \vec{e}_2)$. Find $[\vec{a}_1]_{\mathscr{E}}$ and $[\vec{a}_2]_{\mathscr{E}}$.

14. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $EA = \begin{bmatrix} a & b \\ c - 2a & d - 2b \end{bmatrix}$, find E.

15. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. Since $A \sim I$, there's a sequence of row operations that reduce A to I, corresponding to elementary matrices $E_1, \dots, E_n$. Find these elementary matrices, and multiply to check that $E_n \cdots E_1 A = I$.

16. Let $A = \begin{bmatrix} 0 & 2 \\ 5 & 0 \end{bmatrix}$. Since $A \sim I$, there's a sequence of row operations that reduce A to I, corresponding to elementary matrices $E_1, \dots, E_n$. Find these elementary matrices, and multiply to check that $E_n \cdots E_1 A = I$.

[H]20. Hint: Theorem 2.12 doesn't help you here, because you don't know that A and B are invertible. But the IMT might help.

[C]21. Credit: Inspired by Austin.

17. Let $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$. Since $A \sim I$, there's a sequence of row operations that reduce A to I, corresponding to elementary matrices $E_1, \dots, E_n$. Find these elementary matrices, and multiply to check that $E_n \cdots E_1 A = I$.

18. Suppose A, B, P are square matrices such that $A = PBP^{-1}$. C

(a) Show that if A is invertible, then so is B.

(b) Show that $A^2 = PB^2P^{-1}$.

19. Show that if A is square, and AB is invertible, then A is invertible. H

20. Show that if B is square, and AB is invertible, then B is invertible. H

21. Suppose that $A^3 = AAA$ is invertible, and $(A^3)^{-1} = B$. Show that A is invertible and find A^{-1}. C

22. Let $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $F = \begin{bmatrix} 1 & -7 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, and $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Note that you can think of E as the result of adding -7 times *column* 1 of I to *column* 2 of I. Likewise F as the result of swapping *columns* 2 and 3. Multiply to find AE and AF. (Note that the result is like applying the same column operation to A; this is true in general.)

23. Suppose $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation such that $T(-3, 5) = (4, 2)$ and $T(1, -2) = (0, 3)$. Find A such that $T(\vec{x}) = A\vec{x}$ using the following steps:

(a) Fill in the ?'s: $A\begin{bmatrix} -3 & 1 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix}$.

(b) Find the inverse of $\begin{bmatrix} -3 & 1 \\ 5 & -2 \end{bmatrix}$ and use it to find A.

Chapter 3

Determinants

In this chapter, we'll see that every square matrix A is associated with a real number, called the *determinant* of A or $\det(A)$, which tells you a bunch of useful things about A:

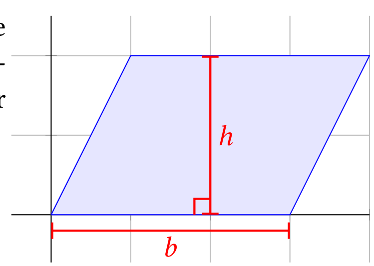
- $\det(A) \neq 0$ if and only if A is invertible.
- $\det(A)$ tells you the area of a parallelogram or volume of a parallelepiped described by A is.
- $\det(A)$ tells you how much A stretches or compresses other vectors/matrices that you multiply by A, which makes determinants important in applications of multivariable calculus.

All of these are different ways of describing the "how big" A is. Unfortunately, the formula for $\det(A)$ is pretty complicated, and it will take some work to figure it out. Fortunately, there's a lot to of interesting math to learn along the way.

This approach to covering determinants was pioneered by Sergei Treil in his excellent book *Linear Algebra Done Wrong*.

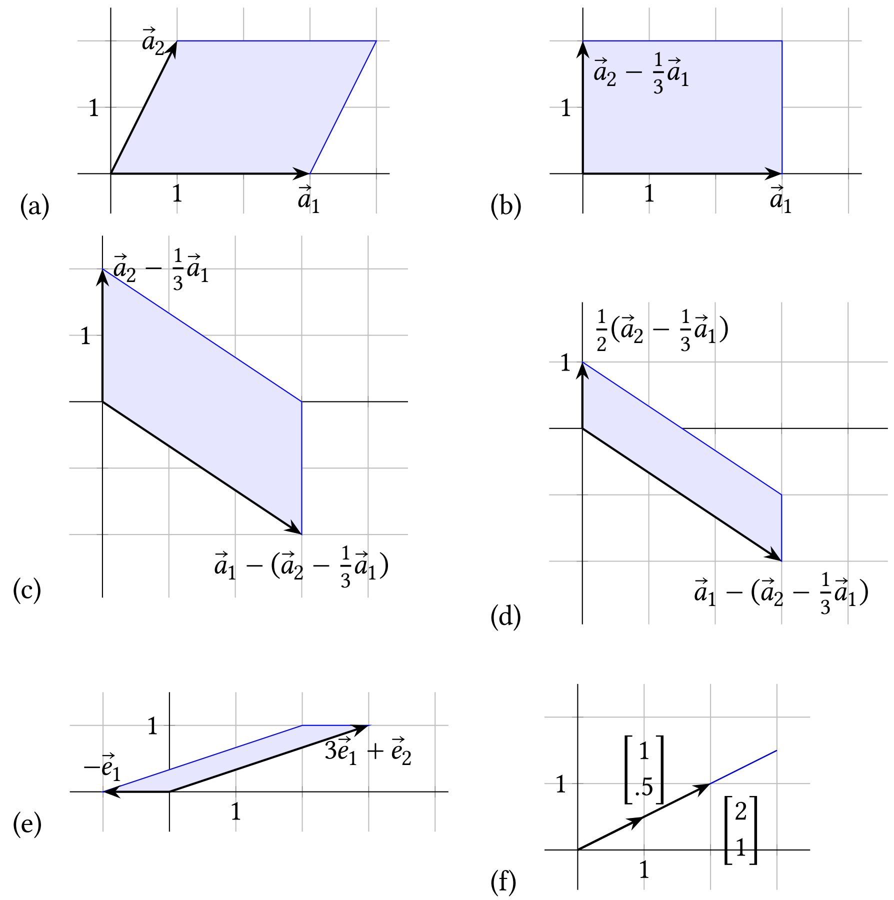
3.1 Signed Volume

Recall that the area of a parallelogram is the length of one side (called the *base*) times the distance from that side perpendicularly to the other side (the *height*).

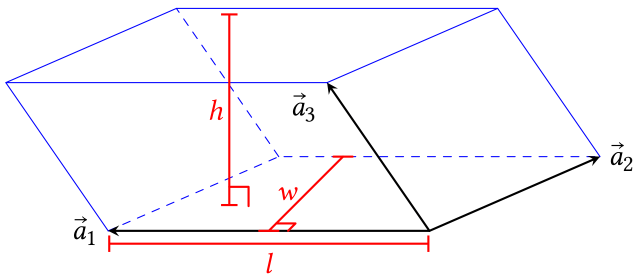


Let's describe a parallelogram in terms of vectors and matrices, which we know a lot about. If $A = [\vec{a}_1 \quad \vec{a}_2]$, we'll say the *parallelogram determined by* A, written $\varnothing A$, is the parallelogram where two sides are the arrows $\vec{a}_1$ and $\vec{a}_2$, drawn starting at $\vec{0}$. For example, the one above could be written $\varnothing\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.

Exercise 1. (*Warm-up*) Let $A = [\vec{a}_1 \quad \vec{a}_2] = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$. Find the area of each shaded parallelogram. Also, which pairs of vectors are linearly independent?



A *parallelepiped* is like a 3-dimensional parallelogram, like the one below, where the six faces are all parallelograms. Its volume is the Bh, where B is the area of one face, and h is the distance perpendicularly from that face to the opposite face. Note that $B = lw$, so its volume is lwh. We'll also describe parallelepipeds as ▱$\begin{bmatrix}\vec{a}_1 & \vec{a}_2 & \vec{a}_3\end{bmatrix}$ just like parallelograms.



If we have a 4×4 matrix $A = \begin{bmatrix}\vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \vec{a}_4\end{bmatrix}$ (or a bigger one), we can't draw a picture of ▱A, but we can talk about its "hypervolume" as its length times width times breadth times depth, where those distances are all perpendicular to each other. We'll refer to all of these things (*area* of parallelogram, *volume* of parallelepiped, etc) as the **volume** of ▱A. But, geometry in $\mathbb{R}^4$, $\mathbb{R}^{10}$, etc can be very weird. How do we even know that the volume of an object there is even well-defined? Maybe the result depends on which vector you start with to be the length and which direction is the width? What does "perpendicular" even *mean* in $\mathbb{R}^{17}$? Instead of answering all these questions, we'll define properties that *volume* should have, that are consistent with $\mathbb{R}^2$ and $\mathbb{R}^3$.

Definition. A **determinant** is a function whose domain is the set of all square matrices, and whose codomain is $\mathbb{R}$, which satisfies the properties below. Since a matrix is basically the same thing as a list of it's column vectors, we'll often write $\det(\vec{a}_1, \dots, \vec{a}_c)$ instead of $\det\left(\begin{bmatrix}\vec{a}_1 & \cdots & \vec{a}_c\end{bmatrix}\right)$.

- **Normality:** $\det(I_c) = \det(\vec{e}_1, \dots, \vec{e}_c) = 1$ for any c.
- **Linearity (scaling):** Multiplying any one column by a scalar also scales the determinant:

$$\det(\vec{a}_1, \dots, k\vec{a}_j, \dots, \vec{a}_c) = k\det(\vec{a}_1, \dots, \vec{a}_j, \dots, \vec{a}_c)$$

- **Linearity (addition):** The determinant distributes over vector addition in any one column:

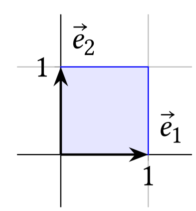
$$\det(\vec{a}_1, \dots, \vec{a}_j + \vec{b}_j, \dots, \vec{a}_c) = \det(\vec{a}_1, \dots, \vec{a}_j, \dots, \vec{a}_c) + \det(\vec{a}_1, \dots, \vec{b}_j, \dots, \vec{a}_c)$$

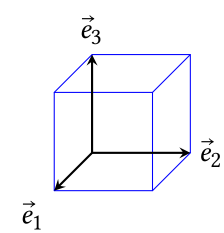
- **Preservation under column multaddiplacement:** Replacing one column with the sum of itself and a scalar multiple of a *different* column doesn't change the determinant. For $i \neq j$:

$$\det(\vec{a}_1, \dots, \vec{a}_j + k\vec{a}_i, \dots, \vec{a}_c) = \det(\vec{a}_1, \dots, \vec{a}_j, \dots, \vec{a}_c)$$

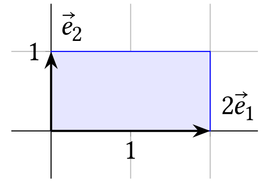
At this point in our story, maybe there doesn't actually exist a function satisfying these properties, or maybe there is more than one, so technically we should be talking about "a determinant (if it exists)." But we'll see in the next section that there exists a unique determinant function, and in the meantime, the it will be easier to talk about "the determinant" so we'll do that. Now, let's make sure the properties in the definition are consistent with volume, starting with the easiest one:

Normality. The volume of a unit square or cube is clearly 1. (Remember in $\mathbb{R}^2$, *volume* here really means area.)



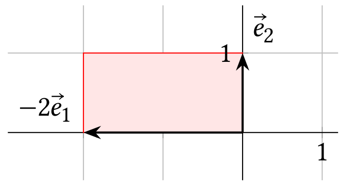


Linearity (scaling). When you scale one of the vectors by a factor of k, either the base or the height changes by that factor. For example, the volume of $\varnothing\begin{bmatrix}2\vec{e}_1 & \vec{e}_2\end{bmatrix}$ is 2 which equals $2\det(I)$.



Note this also allows for determinants to be negative. For example:

$$\det\begin{bmatrix}-2\vec{e}_1 & \vec{e}_2\end{bmatrix} = -2\det(I) = -2(1) = -2$$



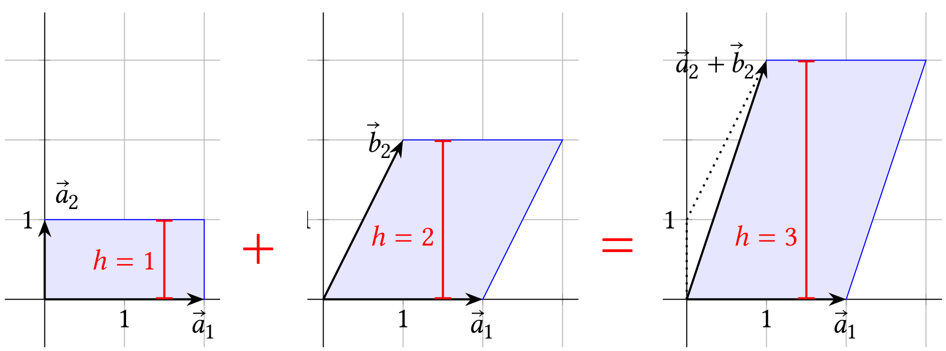
Negative "area" or "volume" might seem like a contradiction, but it lets us have linearity which makes for elegant and powerful mathematics. Sort of like how, in calculus, you allow an integral to be "signed area" between the x-axis and a curve, and it makes important rules work, such as $\int cf(x)dx = c\int f(x)dx$.

It turns out that the sign of 2×2 determinants depends on whether you turn counterclockwise or clockwise to get from the first column to the second. And there's a "right hand rule" that describes the sign of 3×3 determinants, having to do with the way the three vectors are oriented.

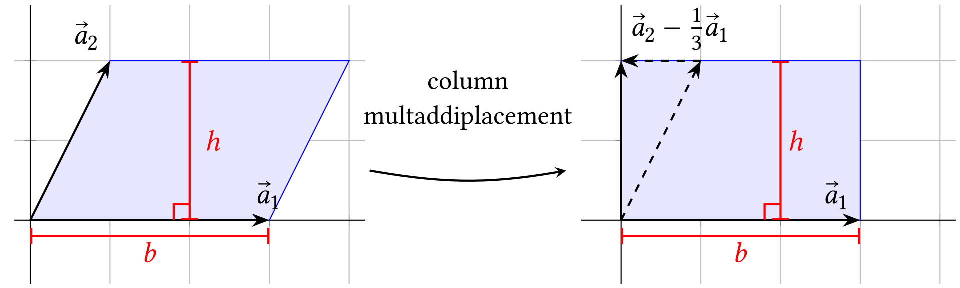
Linearity (addition). When two vectors in column j are added, we can think of the volume of the resulting parallepiped,

$$\det(\vec{a}_1, \dots, \vec{a}_j + \vec{b}_j, \dots, \vec{a}_c),$$

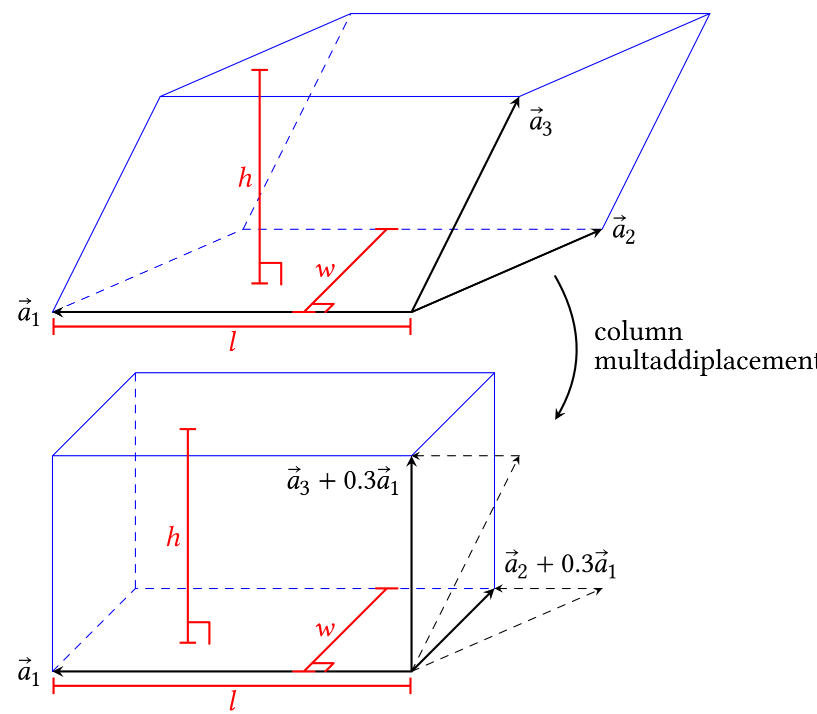
as the "height" of $\vec{a}_j + \vec{b}_j$, times the perpendicular "base," which is a lower-dimensional parallelogram in the span of the other vector(s). The height of $\vec{a}_j + \vec{b}_j$ is the sum of the heights of $\vec{a}_j$ and $\vec{b}_j$, as the picture shows.



Preservation under column multaddiplacement. As the figure below shows, replacing $\vec{a}_2$ with $\vec{a}_2 - \frac{1}{3}\vec{a}_1$ moves one side in a direction parallel to the opposite side, which doesn't change the area. Thus we want $d(\vec{a}_1, \vec{a}_2)$ to equal $d(\vec{a}_1, \vec{a}_2 - \frac{1}{3}\vec{a}_1)$.



The figure below shows a 3×3 example. To make the parallelepiped rectangular, $\vec{a}_3$ can be replaced with $\vec{a}_3 + 0.3\vec{a}_1$ without changing h, since you're moving the top face parallel to the bottom face. And $\vec{a}_2$ can be replaced with $\vec{a}_2 + 0.3\vec{a}_1$ which doesn't change w because you're moving the back edge parallel to the front edge. So the volume of $\varnothing\begin{bmatrix}\vec{a}_1 & \vec{a}_2 & \vec{a}_3\end{bmatrix}$ equals the volume of $\varnothing\begin{bmatrix}\vec{a}_1 & \vec{a}_2 + 0.3\vec{a}_1 & \vec{a}_3 + 0.6\vec{a}_2\end{bmatrix}$, so the determinants of the matrices should be equal too.



Recall the three basic row operations: scaling, swapping, multaddiplacement. The definition of the determinant how column scaling and column multaddiplacement affect the determinant. This theorem covers column swaps:

Theorem 3.1: Antisymmetry.
If you swap two column vectors, the determinant changes sign. Writing the column number underneath the vector:

$$\det(\vec{a}_1, \dots, \underset{i}{\vec{a}_i}, \dots, \underset{j}{\vec{a}_j}, \dots, \vec{a}_c) = -\det(\vec{a}_1, \dots, \underset{i}{\vec{a}_j}, \dots, \underset{j}{\vec{a}_i}, \dots, \vec{a}_c)$$

Lemma 3.2: If A is a square matrix, with $\vec{0}$ in a column, then $\det(A) = 0$.

This is intuitively consistent with the determinant representing area in $\mathbb{R}^2$: if $\vec{0}$ is one of the two vectors, then the parallelogram is just a line segment with zero area. In $\mathbb{R}^3$, if $\vec{0}$ is one of the three vectors, then the parallelepiped is a 2D parallelogram with zero volume.

Proof: Using linearity (scaling):

$$\begin{aligned}\det(\vec{a}_1, \dots, \vec{0}, \dots, \vec{a}_c) &= \det(\vec{a}_1, \dots, 0\vec{0}, \dots, \vec{a}_c)\\ &= 0\det(\vec{a}_1, \dots, \vec{0}, \dots, \vec{a}_c) = 0\end{aligned}$$

■

Definition. The **main diagonal** of a matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1c} \\ a_{21} & a_{22} & \cdots & a_{2c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r1} & a_{r2} & \cdots & a_{rc} \end{bmatrix}$ is all the entries of the form a_{ii}; that is, the row number equals the column number. A **diagonal matrix** is a square matrix where all the entries *not* on the main diagonal are zero. (For example, I is diagonal.) An **upper triangular matrix** is a square matrix where all the entries *below* the main diagonal are zero. A **lower triangular matrix** is a square matrix where all the entries *above* the main diagonal are zero. A **triangular matrix** is either upper or lower triangular.

Soon we'll find an easy shortcut for calculating the determinants of triangular matrices; but first, let's practice using the properties in the definition. The idea is to use column multaddiplacements, swaps, and scaling to transform A to I which has determinant 1.

Example 1. Find $\det(A)$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$.

Example 2. Is $A = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix}$ triangular? Find $\det(A)$.

Example 3. Find $\det(A)$ for $A = \begin{bmatrix} 3 & 2 & -6 & 3 \\ 0 & 2 & 4 & 5 \\ 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

Lemma 3.3: If A is triangular with a 0 entry in the main diagonal, then $\det(A) = 0$.

Intuitively, the 0 in the main diagonal suggests that A is not row-equivalent to I, and that the columns are linearly dependent, and like the last example, there's a sequence of column multaddiplacements that turn that column to $\vec{0}$, making $\det(A) = 0$. The proof in the book generalizes the idea of example 3.

Theorem 3.4: Determinants of triangular matrices. If A is triangular, then $\det(A)$ is the product of the entries on the main diagonal.

The proof, which you can find in the book, generalizes the idea in examples 1 and 3: The zeros on one side of the diagonal make it so column multaddiplacements transform A to a diagonal matrix, and linearity transforms that to I.

Example 4. Find $\det(A)$ if $A = \begin{bmatrix} 3 & -1 & 2 \\ 4 & 0 & 4 \\ -5 & 2 & -3 \end{bmatrix}$. Note that $\vec{a}_3 = \vec{a}_1 + \vec{a}_2$.

The next theorem gives a way to check if a matrix is invertible or not which will be important when we study eigenvalues.

Theorem 3.5: Determinants and invertibility.
Suppose A is square. Then $\det(A) \neq 0$ if and only A is invertible.

This section has been a lot. Keeping in mind that determinants are only defined for square matrices, let's summarize everything we'll need for later in one place:

Definition:

- **Normality:** $\det(I) = 1$.
- **Linearity (scaling):** Multiplying a column by a scalar also scales the determinant.
- **Linearity (addition):**

$$\det(\vec{a}_1, \ldots, \vec{a}_j + \vec{b}_j, \ldots, \vec{a}_c) = \det(\vec{a}_1, \ldots, \vec{a}_j, \ldots, \vec{a}_c) + \det(\vec{a}_1, \ldots, \vec{b}_j, \ldots, \vec{a}_c)$$

- **Preservation under column multaddiplacement:** Replacing a column with the sum of itself and a different column does not change the determinant.

Theorem 3.1 (antisymmetry): Swapping two columns changes the sign of the determinant.
Theorem 3.4: If A is triangular, then $\det(A)$ is the product of the entries on the main diagonal.
Theorem 3.5: $\det(A) \neq 0$ if and only if A is invertible.
Theorem 3.6: $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$. (to be proven in exercises.)

3.1 Exercises

For exercises 2 to 11 find $\det(A)$ by using column operations to transform A into a triangular matrix.

2. $A = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix}$

3. $A = \begin{bmatrix} 3 & 6 \\ 1 & 1 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & 1 \\ 3 & 6 \end{bmatrix}$

5. $A = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{-\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$, the matrix for a 45° rotation.

6. $A = \begin{bmatrix} 3 & 2 & 1 \\ 5 & 4 & 0 \\ 6 & 0 & 0 \end{bmatrix}$

7. $A = \begin{bmatrix} 1 & -1 & 3 \\ 2 & -2 & 3 \\ 0 & 1 & -3 \end{bmatrix}$

8. $A = \begin{bmatrix} 1 & -2 & 2 & -2 \\ 3 & -5 & 6 & -6 \\ -2 & 7 & -3 & 4 \\ 0 & -3 & 5 & 0 \end{bmatrix}$

9. $\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

10. $\begin{bmatrix} 0 & 0 & 0 & a \\ 0 & 0 & b & 0 \\ 0 & c & 0 & 0 \\ d & 0 & 0 & 0 \end{bmatrix}$

11. $\begin{bmatrix} 0 & 0 & 0 & 0 & a \\ 0 & 0 & 0 & b & 0 \\ 0 & 0 & c & 0 & 0 \\ 0 & d & 0 & 0 & 0 \\ e & 0 & 0 & 0 & 0 \end{bmatrix}$

12. (T/F?) Since $\vec{e}_1 = (1, 0)$ and $\vec{e}_2 = (0, 1)$ lie flat on the plane, $\varDelta\begin{bmatrix} \vec{e}_1 & \vec{e}_2 \end{bmatrix}$ has zero volume and so $\det(\vec{e}_1, \vec{e}_2) = 0$.

13. (T/F?) The matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \\ 6 & 0 & 0 \end{bmatrix}$ is triangular.

14. (T/F?) If A is a 3×3 matrix, $\det(2A) = 2\det(A)$. [H]

15. (T/F?) Triangular matrices can't be square matrices, because, like, triangles

[H]14. Hint: Think about a rectangular box whose length, width, and height are all doubled.

and squares are, like, different shapes.

16. (T/F?) If A is invertible, then $\det(A) = 0$.

17. (T/F?) The linearity (addition) property in the definition says that $\det(A+B) = \det(A) + \det(B)$.

18.

(a) Draw the parallelogram with vertices at (0,0), (1,2), (4,1), and (5,3). Note that this is the parallelogram determined by vectors $\begin{bmatrix}1\\2\end{bmatrix}$ and $\begin{bmatrix}4\\1\end{bmatrix}$. Find its area.

(b) Suppose you didn't draw the parallelogram first and thought it was determined by vectors $\begin{bmatrix}1\\2\end{bmatrix}$ and $\begin{bmatrix}5\\3\end{bmatrix}$. Explain how you know this mistake doesn't matter because the parallelogram has the same area as the correct one.

19. Suppose $T(\vec{x}) = A\vec{x}$ is a transformation that reflects points across some line through the origin. For some arbitrary line, draw a picture of $T(\vec{e}_1)$ and $T(\vec{e}_2)$. Thinking about area of $\varTriangle A$, what is $\det(A)$? C

20. Suppose $T(\vec{x}) = A\vec{x}$ is a transformation that projects all points to the nearest point on some line through the origin. For some arbitrary line, draw a picture of $T(\vec{e}_1)$ and $T(\vec{e}_2)$. Thinking about area of $\varTriangle A$, what is $\det(A)$? C

21. Use the antisymmetry theorem (but no other theorems/lemmas from this section) to show that if two columns of A are equal, then $\det(A) = 0$.

22. Use the column replacement property of the definition, and lemma 3.2 (but no other theorems/lemmas from this section) to show that if two columns of A are equal, then $\det(A) = 0$.

23. Use theorem 3.5 (but no other theorems/lemmas from this section) to show that if two columns of A are equal, then $\det(A) = 0$.

24. Prove theorem 3.6 below by following these steps:

(a) Show the theorem is true if $a \neq 0$. Use column multaddiplacement to turn $\begin{bmatrix}a & b\\c & d\end{bmatrix}$ into a triangular matrix.

(b) Show the theorem is true if $a = 0$. Use a swap to turn $\begin{bmatrix}a & b\\c & d\end{bmatrix}$ into a triangular matrix. (This completes the proof.)

(c) Explain how theorems 2.11, 3.5, and 3.6 agree with each other.

[C]19. Credit: Inspired by an exercise in David Austin's excellent *Understanding Linear Algebra*.
[C]20. Credit: Inspired by D. Austin.

Theorem 3.6: $\det\begin{bmatrix}a & b\\c & d\end{bmatrix} = ad - bc.$

25. Suppose (p, y_p) and (q, y_q) two points in $\mathbb{R}^2$, and you want to find b, m such that the graph of $y = b + mx$ includes the two points.

(a) Explain why finding the parabola that includes the three points is equivalent to solving $A\begin{bmatrix}b\\m\end{bmatrix} = \begin{bmatrix}y_p\\y_q\end{bmatrix}$ for (b, m), where $A = \begin{bmatrix}1 & p\\1 & q\end{bmatrix}$.

(b) Use column multaddiplacement to show $\det(A) = q - p$.

(c) What needs to be true about the points (p, y_p) and (q, y_q) to guarantee there's a unique linear function that passes through the points?

26. *Challenge:* Suppose (p, y_p), (q, y_q), and (r, y_r) are three points in $\mathbb{R}^2$, and you want to find a, b, c such that the graph of $y = c + bx + ax^2$ includes the three points. C

(a) Explain why finding the parabola that includes the three points is equivalent to solving $A\begin{bmatrix}c\\b\\a\end{bmatrix} = \begin{bmatrix}y_p\\y_q\\y_r\end{bmatrix}$ for (c, b, a), where $A = \begin{bmatrix}1 & p & p^2\\1 & q & q^2\\1 & r & r^2\end{bmatrix}$.

(b) Use column multaddiplacement to show $\det(A) = (q-p)(r-p)(r-q)$.

(c) What needs to be true about the points (p, y_p), (q, y_q), and (r, y_r) to guarantee there's a unique quadratic function that passes through all three?

27. Show that if A is lower triangular, with no zeros in the main diagonal, then $\det(A)$ is the product of the diagonal entries of A. This will finish the proof of theorem 3.4. You may use the fact that the theorem works with diagonal matrices, which was already proven.

[C]26. Credit: Inspired by D. Austin.

3.2 More Determinants and the Cofactor Expansion

Exercise 1. (*Warm-up*) For each of the following column operations, write your own example of a 3×3 elementary matrix that is the result doing the same operation to I. Find the determinant of each.

(a) Scaling a column.

(b) Column multaddiplacement.

(c) Swapping a column.

Exercise 2. (*Warm-Up*) Let $A = \begin{bmatrix} 2 & 4 & \pi \\ 0 & 3 & 7 \\ 0 & 0 & 1 \end{bmatrix}$ and $E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(a) Find AE. What do you notice?

(b) Find $\det(A)$, $\det(E)$, and $\det(AE)$. What do you notice?

3.2.1 Some Nice Theorems About Determinants

Example 1. Let $A = \begin{bmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{bmatrix}$, and $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{-e}{f} & 1 \end{bmatrix}$. Note that E is an elementary matrix, the result of doing the row operation $r_3 \leftarrow r_3 + \frac{-e}{f} r_2$ to I; and E is also the result of doing the *column* operation $c_2 \leftarrow c_2 + \frac{-e}{f} c_3$.

(a) Find AE. What do you notice?

(b) Find $\det(A)$, $\det(E)$, and $\det(AE)$. What do you notice?

Lemma 3.7: If A is any $r \times c$ matrix, and E is an $c \times c$ elementary matrix that is the result of doing some *column* operation to I, then AE is the result of doing the same *column* operation to A.

Note that this lemma is the same as theorem 2.17, but about AE instead of EA, and column instead of row operations.

Lemma 3.8:
If A is square and E is elementary (of the same size) then:

$$\det(AE) = \det(A)\det(E)$$

The next theorem should be surprising, because the vectors that make ▱A^T are very different from the vectors that make ▱A.

Theorem 3.9: If A is square, then $\det(A) = \det(A^T)$.

Theorem 3.10:
If A and B are $c \times c$ matrices, then $\det(AB) = \det(A)\det(B)$.

Remarks: Existence and Uniqueness of Determinant

The definition of the determinant in section 3.1 gives four properties that a function should have to be called a determinant. But we haven't shown that there actually exists a function that satisfies all those properties. And, if there does exist a determinant function, we haven't shown that it's unique. What we have done throughout this chapter is *implicitly assume* that we have a function, det(), that satisfies the definition properties, and showed that some other things about det() must be true. For example the wording for theorem 3.5 should have been:

Theorem 3.5 (precise version): Suppose det() is a function which satisfies the definition of a determinant, and A is square. Then $\det(A) \neq 0$ if and only if A is invertible.

But instead we went with the following instead:

Theorem 3.5 (readable version): Suppose A is square. Then $\det(A) \neq 0$ if and only if A is invertible.

Note that we did come up with a way of calculating $\det(A)$, by column-reducing A to a triangular matrix, keeping track of how the operations change the determinant, and then multiplying those factors by the diagonal entries. But it's not obvious this is a well-defined function because there are many triangular matrices equivalent to A. Fortunately, all is well and good, as the next two theorems show.

You can find the proof of the theorem below in Sergei Treil's *Linear Algebra Done Wrong* section 3.4 (the book that inspired this book's whole approach to determinants and is available free online). It's quite interesting and illuminating, but is long and uses some math that is slightly beyond the scope of this course..

Theorem 3.11: A function that satisfies the definition of the determinant exists.

In the next subsection we show the determinant is unique and find a formula for it.

3.2.2 The Cofactor Expansion

First some notation we'll use in the big theorem: A_{ij} is the result of removing row i and column j from A. For example, if $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ then $A_{12} = \begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix}$.

Theorem 3.12: Cofactor Expansion.
Suppose det is a function satisfying the definition of a determinant. Define a function $d : \{c \times c \text{ matrices}\} \to \mathbb{R}$ by the formula:

$$\begin{aligned} d(A) &= (-1)^{1+1} a_{11} \det(A_{11}) + (-1)^{2+1} a_{21} \det(A_{21}) + \\ &\qquad + \cdots + (-1)^{c+1} a_{c1} \det(A_{c1}) \\ &= \sum_{k=1}^{c} (-1)^{k+1} a_{k1} d(A_{k1}) \end{aligned}$$

(i) $\det(A) = d(A)$. (This also implies the determinant is unique.)

(ii) Instead of row 1, $d(A)$ can be calculated along any row or column, and the result is the same:

$$\begin{aligned} d(A) &= \sum_{k=1}^{c} (-1)^{i+k} a_{ik} d(A_{ik}) \quad \text{(along row } i\text{)} \\ &= \sum_{k=1}^{c} (-1)^{k+j} a_{kj} d(A_{kj}) \quad \text{(along column } j\text{)} \end{aligned}$$

First let's think about the $(-1)^{i+j}$. In row 1, column 1, we have $(-1)^{1+1} = +1$. And every time you increase the row or column index by one, the sign of $(-1)^{i+j}$ changes. This leads to the pattern on the right.

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

The reason for the term "cofactor expansion" is that $C_{ij} = (-1)^{i+j} \det(A_{ij})$ is called the (i,j)-**cofactor** of A. Cofactors can describe some nice formulas for A^{-1} as well as solutions of $A\vec{x} = \vec{y}$ that will probably be included in a future version of this book.

Example 2. Let for $A = \begin{bmatrix} 2 & 4 & \sqrt{7} \\ 0 & 3 & -8 \\ 0 & 0 & 1 \end{bmatrix}$. Use the cofactor expansions along column 1 and row 2 to calculate $\det(A)$.

In practice, calculating determinants (either by hand or computer) with the cofactor expansion becomes extremely inefficient with matrices bigger than 3×3. A $c \times c$ matrix requires more than $c!$ multiplications to compute the cofactor expansion, which means that doing column or row operations until you get a triangular matrix is much more efficient for larger matrices.

3.2 Exercises

3.2.1 Some Nice Theorems About Determinants

3. (T/F?) If A is 3×3 and $A\begin{bmatrix}1\\2\\3\end{bmatrix} = \vec{0}$, then $\det(A) \neq 0$.

4. (T/F?) If A and B are row-equivalent, then $\det(A) = \det(B)$.

5. (T/F?) If A and B are row-equivalent, then either $\det(A)$ and $\det(B)$ both equal zero, or neither of them are zero.

6. (T/F?) The volume of the parallelogram/parallelepiped determined by the columns of A equals the volume of the parallelogram/parallelepiped determined by the *rows* of A.

7. (T/F?) If A is $c \times c$ and $\det A \neq 0$, then the columns of A are a basis of $\mathbb{R}^c$.

8. (T/F?) If A is square, then $\det(A^T A) \geq 0$.

9. (T/F?) If A is square, then $\det(-A) = -\det(A)$.

10. Suppose $\vec{b}_1$ and $\vec{b}_2$ are vectors in $\mathbb{R}^2$, and that A is a 2×2 matrix. We know that the signed area of $\varTheta\begin{bmatrix}\vec{b}_1 & \vec{b}_2\end{bmatrix}$ is $\det(B)$. How does multiplication by A affect this area? In other words, what is the area of $\varTheta\begin{bmatrix}A\vec{b}_1 & A\vec{b}_2\end{bmatrix}$?

11. In chapter 4, we'll say that square matrices A and B are *similar* if there exists an invertible P such that $A = PBP^{-1}$. Show that if A and B are similar, then $\det(A) = \det(B)$.

12. In chapter 5, we'll look at *orthogonal* matrices, which are square matrices that turn out to have the property $A^T A = I$. What are possible values of $\det(A)$ if A is orthogonal? [C]

13. Prove that if A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

14. A^n means $AA \cdots A$ multiplied with n copies of A. Show that $\det(A^n) = \det(A)^n$.

15. Let $A_1, A_2, \ldots, A_n$ be $c \times c$ matrices. Show that the product $A_1 A_2 \cdots A_n$ is invertible if and only if each matrix A_i is invertible. [C]

16. Explain why row operations on A affect $\det(A)$ the same way the corresponding column operations do. [H]

17. Prove that $\det(AB) = \det(BA)$.

18. Suppose $A = \begin{bmatrix}\vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \vec{a}_4\end{bmatrix}$ and B are 4×4 matrices, with $\det(A) = 2$ and $\det(B) = -3$. Find:

(a) $\det(3\vec{a}_1, \vec{a}_1 + \vec{a}_2, \vec{a}_4, \vec{a}_3)$
(b) $\det(\vec{a}_2, \vec{a}_3, \vec{a}_4, \vec{a}_1)$
(c) $\det(2A)$
(d) $\det(AB)$
(e) $\det(B^2)$
(f) $\det(BA^{-1})$

3.2.2 The Cofactor Expansion

19. (T/F?) To compute a 6×6 determinant with a cofactor expansion requires computing 6 different 5×5 determinants.

For exercises 20 through 23, calculate the determinant with two cofactor expansions, along different rows or columns, and check you get the same result.

20. $\det\begin{bmatrix}2 & -1 & 4\\3 & 0 & 1\\-2 & 5 & 3\end{bmatrix}$

21. $\det\begin{bmatrix}2 & 1 & 0\\-3 & 4 & 7\\-1 & 5 & 7\end{bmatrix}$

22. $\det\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\\7 & 8 & 9\end{bmatrix}$

23. $\det\begin{bmatrix}1 & 3 & 0 & -1\\0 & 2 & 3 & 0\\1 & 0 & 0 & 5\\2 & 1 & 0 & 3\end{bmatrix}$

24. Suppose you don't yet know that $\det\begin{bmatrix}a & b\\c & d\end{bmatrix} = ad - bc$, but you do know how to calculate 1×1 determinants: $\det[a] = a$. Use this and the cofactor expansion to find $\det\begin{bmatrix}a & b\\c & d\end{bmatrix}$.

25. Do a column 1 cofactor expansion to calculate each of the following. Do you notice a pattern?

(a) $\det\begin{bmatrix}1 & 1\\-1 & 1\end{bmatrix}$

[C]12. Credit: Inspired by an exercise in Ken Kuttler's excellent *A First Course in Linear Algebra.*

[C]15. Credit: Inspired by an example in Dan Margalit and Joseph Rabinoff's excellent *Interactive Linear Algebra.*

[H]16. Hint: Theorem 3.9.

(b) $\det\begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

(c) $\det\begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix}$

26. A simple computer program calculates determinants using the cofactor expansion. The program doesn't do row operations, or any tricks to look for zero factors; it just applies the formula until it gets to a 2×2 matrix where it does $ad - bc$. How many multiplications does it compute for:

(a) 2×2 determinants?

(b) 3×3 determinants?

(c) 4×4 determinants?

(d) 5×5 determinants?

Chapter 4

Diagonalization and Eigenstuff

In this chapter, we'll study techniques for analyzing sequences of vectors of the form

$$\vec{y}_{n+1} = A\vec{y}_n$$

This simple-looking notation hides lots of depth. As we'll see, this kind of formula can model many situations where there are many quantities that change at a rate that depends on the other quantities–making the theory useful in differential equations, dynamical systems, probability theory, and many real-life applications of these.

4.1 Review and Diagonalization

A solid understanding of the concepts of bases, coordinates, and dimension, and how they relate to matrices, will make this chapter much easier. So we'll start with an extra long set of warm-up exercises to review these concepts, as well as introduce a little new stuff.

Exercise 1. *(Warm-up)* Review the following definitions and theorems (unless you're confident you already have them mastered).

(a) The definition of a basis in section 1.6.

(b) The definition of $\mathscr{B}$-coordinates in section 1.6.

(c) The definition of dimension in section 1.7.

(d) Theorem 7.13 (buy two, get one free) in section 1.7.

If you struggle with the rest of these warm-ups, the examples and discussions in sections 1.6 and 1.7 might help.

Exercise 2. *(Warm-up)* (T/F?) The list of vectors $\begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ is a basis of $\mathbb{R}^3$.

Exercise 3. *(Warm-up)* (T/F?) The list of vectors $\begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ is a basis of $\operatorname{span}\left(\begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right)$, which is a 2-dimensional subspace of $\mathbb{R}^3$.

Exercise 4. *(Warm-up)* (T/F?) The list of vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ is a basis of $\mathbb{R}^2$.

Exercise 5. *(Warm-up)* (T/F?) The list of vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ is a basis of $\operatorname{span}\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \end{bmatrix}\right)$ which is a 3-dimensional subspace of $\mathbb{R}^2$.

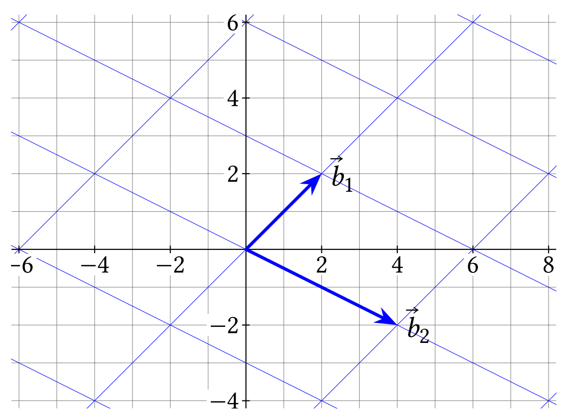
Exercise 6. *(Warm-up)* The grid below shows $\mathscr{B} = (\vec{b}_1, \vec{b}_2)$, a basis of $\mathbb{R}^2$. Find:

(a) $\begin{bmatrix} 6 \\ 0 \end{bmatrix}_{\mathscr{B}}$

(b) $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$

(c) $\begin{bmatrix} -2 \\ 4 \end{bmatrix}_{\mathscr{B}}$

(d) $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$



Exercise 7. *(Warm-up)* With $\vec{b}_1 = \begin{bmatrix}1\\1\end{bmatrix}, \vec{b}_2 = \begin{bmatrix}2\\-1\end{bmatrix}$ as in the last exercise, let $P = \begin{bmatrix}\vec{b}_1 & \vec{b}_2\end{bmatrix}$.

(a) Explain how you know P is invertible.

(b) In the last exercise you found $\begin{bmatrix}6\\0\end{bmatrix}_{\mathscr{B}}$. Check that $P\begin{bmatrix}6\\0\end{bmatrix}_{\mathscr{B}} = \begin{bmatrix}6\\0\end{bmatrix}$.

(c) Find P^{-1}.

(d) In the last exercise you found $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = \begin{bmatrix}6\\0\end{bmatrix}$. Check that $[\vec{y}]_{\mathscr{B}} = P^{-1}\vec{y}$.

Exercise 8. *(Warm-up)* Using the IMT, convince yourself that the following statement is true: A list of vectors $\vec{b}_1, \dots, \vec{b}_c$ in $\mathbb{R}^c$ is a basis of $\mathbb{R}^c$ if and only if the matrix $\begin{bmatrix}\vec{b}_1 & \cdots & \vec{b}_c\end{bmatrix}$ is invertible. No need to write anything; just ask for help if you don't see why it's true. (Warm-ups end here.)

4.1.1 Difference Equations and Diagonalizability

Now let's look at a certain type of application problem, and we'll soon see how to apply bases and coordinates to them.

Example 1. Spock accidentally transports a population of 100 baby tribbles (cute, furry, little creatures) to an alien planet where there is plenty of food, but also predators. Every week, 80% of the baby tribbles are eaten by predators, but the other 20% grow to become adult tribbles. And, every week, every adult tribble gives birth to 5 baby tribbles, and at the end of the week, 60% of the adults are eaten. Let $\vec{x}_w = (b_w, a_w)$ represent the number of baby and adult tribbles w weeks later; starting with $\vec{x}_0 = (100, 0)$.

(a) Find $\vec{x}_1, \vec{x}_2$ and $\vec{x}_3$.

(b) Find a matrix A so that $\vec{x}_{w+1} = A\vec{x}_w$.

(c) Use a computer to see what happens to the tribble population in the long run (as $w \to \infty$).

Of course, in the real world, the population of tribbles wouldn't grow to infinity for many reasons. For one, the *real* Spock doesn't make mistakes. Some other factors to consider:

- The tribbles' food supply on the planet isn't infinite. Assuming it's a living thing, the rate it can replenish itself depends on how much of it there is, and how many tribbles are eating it.
- The well-fed population of predators would also increase, and the more predators there are, the smaller a percentage of tribbles survive to reproduce.
- Over thousands of years, the predators, the tribbles, and the tribbles' food will all evolve by natural selection, gradually changing their abilities to eat and reproduce.

Many of the mathematical models that take these factors into account are based on more or less complicated versions of $\vec{y}_{n+1} = A\vec{y}_n$. As we saw this is equivalent to $\vec{y}_n = A^n\vec{y}_0$; either way it's called a **linear difference equation**.

Example 2. A new virus emerges in a population of 1000 susceptible people. Each day, 8% of the susceptible people become infected, and 40% of the infected people recover. Also, 1% of the recovered population become reinfected. Let $\vec{p}_n = (s_n, i_n, r_n)$ represent the number of susceptible, infected, and recovered people, n days later.

(a) Find $\vec{p}_1$, $\vec{p}_2$ and $\vec{p}_3$.

(b) Find A so that $\vec{p}_{n+1} = A\vec{p}_n$.

(c) Use a computer to see what happens in the long run.

(d) Find $A\begin{bmatrix} 0 \\ 24.39 \\ 975.61 \end{bmatrix}$.

Notice that in the last example the entries in each column of A add up to 1, because each column describes what happens to 100% of each group from the previous day. This kind of matrix is very useful in statistics, and is called a *stochastic matrix*, and when A is stochastic, the sequence $\vec{x}_n = A^n\vec{x}_0$ is called a *Markov chain*.

When calculating A^n in many real world applications, the matrix A might be enormous. For example, instead of just baby and adult tribbles, you may have entries in $\vec{y}_w$ for thousands of species in the ecosystem whose populations affect each other in various ways. And Google's PageRank algorithm, for assessing the quality of web pages, uses a powers of a square matrix with a row and column for *every page on the web*, and there are many trillions of those. In general, if A is $c \times c$, calculating A^n requires nc^3 multiplications. We'll soon see that the concept of diagonalization makes calculation much easier, as well as helping with theoretical analysis.

Definition. A square matrix A is **similar** to B if there exists an invertible matrix P such that $A = PBP^{-1}$. And A is **diagonalizable** if A is similar to a diagonal matrix D.

Example 3. Let $A = \begin{bmatrix} -7 & 9 \\ -6 & 8 \end{bmatrix}$. It can be shown that if $P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, and $D = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$, then $A = PDP^{-1}$, with $P^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$.

(a) Calculate A^2 and A^3 by hand (to see that it's tedious).

(b) Calculate D^2 and D^3 by hand, and find a general formula for D^n.

(c) Find a formula for A^n in terms of P and D.

In general, if D is any diagonal matrix,

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_c \end{bmatrix}$$

it is easy to calculate powers of D. Using the row-column rule, any entry in D^2 not on the main diagonal (say, row i, column j, with $i \neq j$) will be of the form:

And any entry of D^2 on the main diagonal will be:

This gives the following formula for D^2. Repeated multiplication by D likewise gives the formula for D^n:

Like in the last example, if P is invertible, then $(PDP^{-1})^n = PD^nP^{-1}$. The next theorem states a nice version of this in terms of coordinates.

4.1.2 Diagonal Coordinates

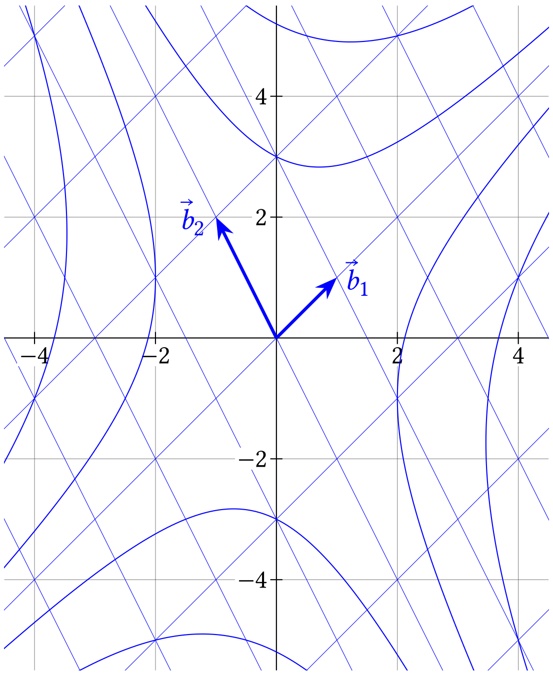
Theorem 4.1: Diagonal coordinates.
Suppose $A = PDP^{-1}$ with D diagonal, and $P = \begin{bmatrix} \vec{b}_1 & \cdots & \vec{b}_c \end{bmatrix}$. Let $\mathscr{B} = (\vec{b}_1, \ldots, \vec{b}_c)$ be the basis made of the columns of P. Then for any positive integer n and any $\vec{y}$ in $\mathbb{R}^c$:

$$[A^n\vec{y}]_{\mathscr{B}} = D^n[\vec{y}]_{\mathscr{B}}$$

Example 4. Let $A = \begin{bmatrix} 1.5 & 0.5 \\ 1 & 1 \end{bmatrix}$. It can be shown that if $P = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 \end{bmatrix}$, and $D = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}$, then $A = PDP^{-1}$.

(a) Suppose $\vec{x}_n = A^n \vec{x}_0$, with $\vec{x}_0 = 1\vec{b}_1 + 1\vec{b}_2$. Use theorem 4.1 to plot $\vec{x}_1$ and $\vec{x}_2$.

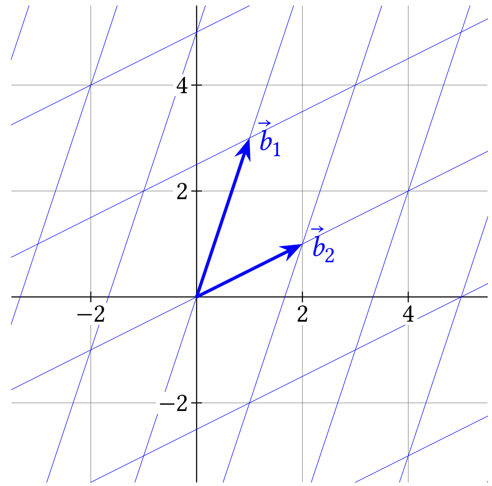
(b) Repeat for some other choices of $\vec{x}_0$.



Example 5. Let $A = \begin{bmatrix} 0.4 & 0.2 \\ -0.3 & 1.1 \end{bmatrix}$. It can be shown that if $P = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 \end{bmatrix}$, and $D = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix}$, then $A = PDP^{-1}$.

(a) Suppose $\vec{x}_n = A^n \vec{x}_0$, with $\vec{x}_0 = 1\vec{b}_1 + 2\vec{b}_2$. Use theorem 4.1 to plot $\vec{x}_1$ and $\vec{x}_2$.

(b) Repeat for some other choices of $\vec{x}_0$.

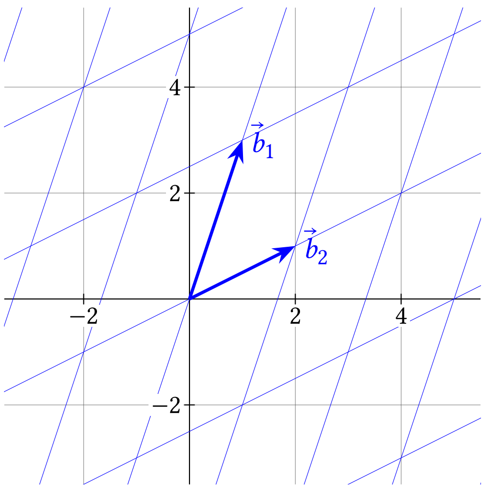


Example 6. Let $A = \begin{bmatrix} -0.2 & 0.4 \\ -0.6 & 1.2 \end{bmatrix}$. It can be shown that if $P = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 \end{bmatrix}$ is the same as the last example, and $D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (note the 0.5 was replaced with 0), then $A = PDP^{-1}$.

(a) Suppose $\vec{x}_n = A^n\vec{x}_0$, with $\vec{x}_0 = 1\vec{b}_1 + 2\vec{b}_2$. Use theorem 4.1 to plot $\vec{x}_1$ and $\vec{x}_2$.

(b) Repeat for some other choices of $\vec{x}_0$.

(c) Which of the matrices A from the last three examples are invertible?



4.1 Exercises

4.1.1 Difference Equations and Diagonalizability

9. Let $D = \begin{bmatrix} 0.8 & 0 \\ 0 & 1 \end{bmatrix}$.

(a) Find $\lim_{n\to\infty} D^n \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

(b) Find $\lim_{n\to\infty} D^n \begin{bmatrix} p \\ q \end{bmatrix}$ in terms of p and q.

(c) Find $\vec{x}_0$ such that $\lim_{n\to\infty} D^n\vec{x}_0 = \vec{0}$.

10. Let $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 0.7 \end{bmatrix}$.

(a) Find $\lim_{n\to\infty} D^n\vec{x}_0$ for $\vec{x}_0 = (-5, 0, 10)$.

(b) Find $\lim_{n\to\infty} D^n\vec{x}_0$ for $\vec{x}_0 = (1, 1, 1)$.

(c) Find $\vec{x}_0$ such that $\lim_{n\to\infty} D^n\vec{x}_0 = \vec{0}$.

11. A population of a species of plants is made up of either young plants or mature plants. Every year, 10% of the young plants survive to maturity, and the other 90% dies. And every mature plant produces seeds which become two mature plants every year. Also, 10% of the mature plants die every year. Find A so that $\begin{bmatrix} y_{n+1} \\ m_{n+1} \end{bmatrix} = A\begin{bmatrix} y_n \\ m_n \end{bmatrix}$ describes how the population changes over n years.

12. A Blockbuster^{TM} video rental store has three locations, x, y, and z. Every week, 90% of the videos at location x remain there, and 5% are rented from x and returned to y, with the other 5% returned to z. And 80% of the videos at location y remain there; the other 20% are rented from y and returned to z. At location z, 99% of the videos remain there every week, and the other 1% is rented from z and returned to x. Find A so that $\begin{bmatrix} x_{n+1} \\ y_{n+1} \\ z_{n+1} \end{bmatrix} = A\begin{bmatrix} x_n \\ y_n \\ z_n \end{bmatrix}$ describes how the videos move around, where n is the number of weeks.

13. (T/F?) Define a sequence of vectors starting with $\vec{x}_0$, and continuing with $\vec{x}_{n+1} = A\vec{x}_n$. Then $\vec{x}_n = A^n\vec{x}_0$.

14. (T/F?) If $A = PDP^{-1}$, then $A = D$ since the P and P^{-1} cancel out.

C21. Credit: Inspired by an exercise in Robert Beezer's excellent *A First Course in Linear Algebra.*

C22. Credit: Inspired by R. Beezer.

15. (T/F?) Every matrix that is row-equivalent to a diagonal matrix is diagonalizable.

16. (T/F?) If A is diagonalizable, it is invertible.

17. Show that if A is similar to B, then B is similar to A.

18. Show that if A is diagonalizable, then A^2 is diagonalizable.

19. Show that if A^2 is diagonalizable, then A is diagonalizable.

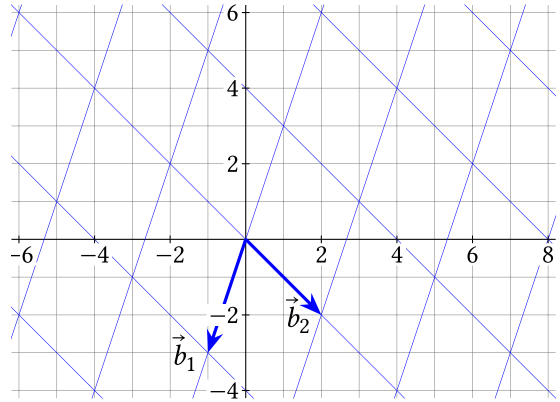
20. Show that if A and B are similar matrices, then A^2 is similar to B^2.

21. Show that if A and B are similar matrices and A is invertible, then B is invertible, and A^{-1} is similar to B^{-1}. C

22. Show that if A is invertible, then AB is similar to BA. C

4.1.2 Diagonal Coordinates

23. The grid below shows $\mathscr{B} = (\vec{b}_1, \vec{b}_2)$, a basis of $\mathbb{R}^2$. Find:



(a) $\begin{bmatrix} 6 \\ 2 \end{bmatrix}_{\mathscr{B}}$

(b) $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

(c) $\begin{bmatrix} -1 \\ 5 \end{bmatrix}_{\mathscr{B}}$

(d) $\vec{y}$ if $[\vec{y}]_{\mathscr{B}} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$

(e) Let $P = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 \end{bmatrix}$ and multiply to check that $P\begin{bmatrix} 6 \\ 2 \end{bmatrix}$ gives your answer to (b), and that P times your answer to (a) gives $\begin{bmatrix} 6 \\ 2 \end{bmatrix}$.

(f) Use the grid to find $P^{-1}\begin{bmatrix} 8 \\ 0 \end{bmatrix}$.

(g) Let $D = \begin{bmatrix} 0.5 & 0 \\ 0 & 2 \end{bmatrix}$, $A = PDP^{-1}$, and $\vec{x}_n = A^n\vec{x}_0$, with $\vec{x}_0 = (5, -1)$. Use the method in examples 4 through 6 to find $\vec{x}_1$ and $\vec{x}_2$.

24. An electric scooter rental company has two locations where people can pick up or drop off scooters: the park, and the quarry. Let $\vec{y}_n = (p_n, q_n)$ represent the number of scooters at each location after n days. They start with 100 scooters at each location, so $\vec{y}_0 = (100, 100)$. Each day, 70% of scooters remain at the park, and the others are rented and return to the quarry. And 90% of the scooters at the quarry remain at the quarry, and the rest are rented and returned to the park.

(a) Find $\vec{y}_1$ and $\vec{y}_2$.

(b) Find A so that $\vec{y}_{n+1} = A\vec{y}_n$.

(c) Before continuing, make a guess at how many scooters end up at the quarry in the long run. Do you think $\lim_{n\to\infty} q_n$ will be... 0? 200? Between 0 and 100? Between 100 and 200?

(d) Check by multiplying that $A = PDP^{-1}$, if $P = \begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 0.6 & 0 \\ 0 & 1 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} -0.75 & 0.25 \\ 0.25 & 0.25 \end{bmatrix}$.

(e) Let $\mathscr{B} = (\vec{b}_1, \vec{b}_2)$ be the basis made of the columns of P. Find $[y_0]_{\mathscr{B}}$.

(f) Using the diagonal coordinates theorem, find $[A\vec{y}_0]_{\mathscr{B}}$ and $[A^2\vec{y}_0]_{\mathscr{B}}$. Multiply these by P to get $A\vec{y}_0$ and $A^2\vec{y}_0$. Your answers should agree with $\vec{y}_1$ and $\vec{y}_2$ from part (a).

(g) Find $\lim_{n\to\infty} A^n\vec{y}_0$.

25. In a certain national park there are hawks that eat rabbits. Let $\vec{y}_m = (h_m, r_m)$ be the number of hawks and rabbits after m months, starting with $\vec{y}_0 = (100, 500)$. The following simplified model describes how their populations affect each other:

$$\begin{bmatrix} h_{m+1} \\ r_{m+1} \end{bmatrix} = \begin{bmatrix} .4 & .03 \\ -.07 & 1.06 \end{bmatrix} \begin{bmatrix} h_m \\ r_m \end{bmatrix}$$

(a) Find $\vec{y}_1$.

(b) Suppose the rabbits evolve to grow their own spiked armor that protects them from hawks. Which of the four coefficient(s) do you think would change?

(c) Suppose the hawks evolve to be able to eat berries and acorns in addition to rabbits, so they can more of them can survive even if the rabbit population is low. Which of the four coefficient(s) do you think would change?

(d) It can be shown that $A = PDP^{-1}$, with $P = \begin{bmatrix} 9.383 & 0.046 \\ 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 0.403 & 0 \\ 0 & 1.057 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} .107 & -.0049 \\ -.107 & 1.0049 \end{bmatrix}$. If $\mathscr{B}$ is the basis made from the columns of P, find $[\vec{y}_0]_{\mathscr{B}}$.

(e) Find $\lim_{m\to\infty} A^m \vec{y}_0$, and describe what happens to the populations in the long run.

26. Suppose $A = PDP^{-1}$ is $c \times c$, with D diagonal. Explain why the diagonal coordinates theorem implies that for any $\vec{y}$ in $\mathbb{R}^c$, $A\vec{y} = PD[\vec{y}]_{\mathscr{B}}$ where $\mathscr{B}$ is the basis made of columns of P.

4.2 Eigenstuff

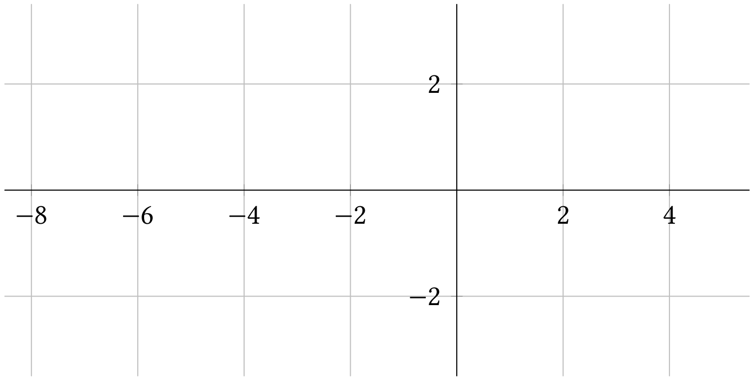
Exercise 1. (*Warm-up*) Let $A = \begin{bmatrix} -3.5 & 6 \\ -0.75 & 1 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and $\vec{v} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$.

(a) Find $A\vec{x}$ and $A\vec{v}$.

(b) Sketch a graph of $\vec{x}$, $\vec{v}$, $A\vec{x}$, and $A\vec{v}$.

(c) T/F? $A\vec{x}$ is a scalar multiple of x.

(d) T/F? $A\vec{v}$ is a scalar multiple of v.



Exercise 2. (*Warm-up*) Suppose $\mathscr{B} = (\vec{b}_1, \ldots, \vec{b}_c)$ is a basis of $\mathbb{R}^c$, and $P = \begin{bmatrix} \vec{b}_1 & \cdots & \vec{b}_c \end{bmatrix}$.

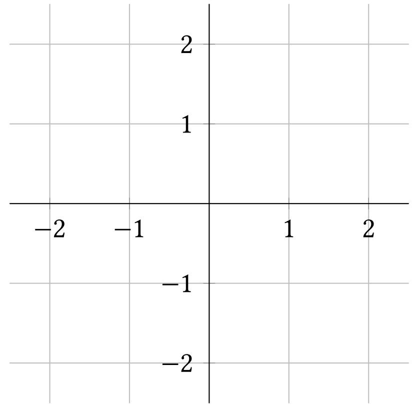
(a) Which of the following are equal to $\left[\vec{b}_2\right]_{\mathscr{B}}$? (Hint: Two are correct.)
$\vec{b}_2 \quad P\vec{b}_2 \quad P^{-1}\vec{b}_2 \quad \vec{e}_2$

(b) Which of the following are equal to $\left[\vec{e}_3\right]_{\mathscr{B}}$? (Hint: Only one is correct.)
$\vec{e}_3 \quad P\vec{e}_3 \quad P^{-1}\vec{e}_3 \quad \vec{b}_3$

(c) Which of the following are equal to $\vec{b}_3$? (Hint: Only one is correct.)
$\vec{e}_3 \quad P\vec{e}_3 \quad P^{-1}\vec{e}_3 \quad P^{-1}\vec{b}_3$

> **Definition.** Suppose A is a $c \times c$ matrix. We say $\vec{x}$ is an **eigenvector** corresponding with an **eigenvalue** λ (a scalar) of A, if:
>
> $$A\vec{x} = \lambda\vec{x} \quad \text{and} \quad \vec{x} \neq \vec{0}$$

Example 1. Let $A = \begin{bmatrix} \frac{1}{2} & \frac{-\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$. Explain why A has no real eigenvalues.



Example 2. If $A = \begin{bmatrix} -3.5 & 6 \\ -0.75 & 1 \end{bmatrix}$, $P = \begin{bmatrix} 2 & -2 \\ 0.5 & -1 \end{bmatrix} = \begin{bmatrix} \vec{b}_1 & \vec{b}_2 \end{bmatrix}$, and $D = \begin{bmatrix} -2 & 0 \\ 0 & -0.5 \end{bmatrix}$, it can be shown that $A = PDP^{-1}$. With $\mathscr{B} = (\vec{b}_1, \vec{b}_2)$, show that $\vec{b}_2$ is an eigenvector of A two ways:

(a) By multiplying $A\vec{b}_2$ directly.

(b) Using the diagonal coordinates theorem (4.1).

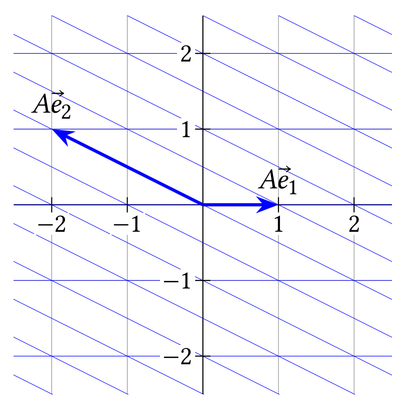
Part (b) of the last example illustrates a connection between diagonalization and eigenstuff. The next theorem fleshes it out some more.

Theorem 4.2: Diagonalization Theorem.
A $c \times c$ matrix A is diagonalizable if and only if A has c linearly independent eigenvalues.
Specifically, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are c linearly independent eigenvectors of A, which correspond with eigenvalues in the diagonal of D (in matching left-to-right order).

Example 3. Suppose A is 3×3, with two eigenvalues, $\lambda_1 = 4$ and $\lambda_2 = 5$. Every eigenvector corresponding with $\lambda_1 = 4$ is a multiple of $\vec{b}_1 = (1, 2, 3)$ and every eigenvector corresponding with $\lambda_2 = 5$ is a multiple of $\vec{b}_2 = (4, 5, 6)$. Is A diagonalizable?

Example 4. Suppose A is 3×3, with two eigenvalues, $\lambda_1 = 4$ and $\lambda_2 = 5$. Every eigenvector corresponding with $\lambda_1 = 4$ is a multiple of $\vec{b}_1 = (1, 2, 3)$ and the set of eigenvectors associated with $\lambda_2 = 5$ is $\text{span}(\vec{b}_2, \vec{b}_3)$, with $\vec{b}_2 = (4, 5, 6)$ and $\vec{b}_3 = (0, 3, 0)$. Is A diagonalizable?

Example 5. Let $A = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$, which is the matrix for a horizontal shear transformation as shown. Is A diagonalizable?



Example 6. Show that a square matrix A is invertible if and only iff 0 is an eigenvalue of A.

This shows that part (xi) can be added to the IMT. Part (x) was theorem 3.5.

Theorem 4.3: IMT v2.
Suppose A is a $c \times c$ matrix. Then the following are equivalent:

(i) A is invertible.

(ii) There exists a matrix P such that $PA = I_c$.

(iii) There exists a matrix Q such that $AQ = I_c$.

(iv) A is row-equivalent to I_c.

(v) A has c pivot positions.

(vi) The columns of A are linearly independent.

(vii) $\operatorname{null} A = \{\vec{0}\}$.

(viii) For every $\vec{y}$ in $\mathbb{R}^c$, $A\vec{x} = \vec{y}$ has a solution.

(ix) $\operatorname{col} A = \mathbb{R}^c$

(x) $\det(A) \neq 0$

(xi) 0 is not an eigenvalue of A.

Example 7. Suppose A is both diagonalizable and invertible. Show that A^{-1} is diagonalizable, and find its eigenstuff.

[H]11. Hint: Drawing a picture might help you find eigenstuff.
[H]12. Hint: Drawing a picture might help you find eigenstuff.
[H]13. Hint: Drawing a picture will help finding eigenstuff.
[H]14. Hint: Drawing a picture will help finding eigenstuff.
[C]18. Credit: Inspired by an exercise in Ken Kuttler's *A First Course in Linear Algebra.*

4.2 Exercises

3. Construct a matrix A with eigenvectors $\begin{bmatrix} 3 \\ -5 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -4 \end{bmatrix}$ which correspond respectively with eigenvalues 2 and 3.

4. Let $A = \begin{bmatrix} -1.5 & 5 \\ -1 & 3 \end{bmatrix}, \vec{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \vec{v} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$.

(a) Show that $\vec{u}$ and $\vec{v}$ are eigenvectors of A and find corresponding eigenvalues.

(b) Diagonalize A (Find P and a diagonal matrix D so that $A = PDP^{-1}$.)

(c) Find $A^{1000000}$.

TODO: Problem with vector grid and several powers of $A\vec{x}_0$ with different starting $\vec{x}_0$

TODO: More basic T/F problems.

5. (T/F?) If A is 2×2 with eigenvectors $\vec{u}$ and $\vec{v}$ that correspond respectively to eigenvalues μ and λ, then $A^n(x_1\vec{u} + x_2\vec{v}) = x_1\mu^n\vec{u} + x_2\lambda^n\vec{v}$.

6. (T/F?) If A is diagonalizable, then the diagonalization is unique. In other words, if $A = P_1D_1P_1^{-1} = P_2D_2P_2^{-1}$, then $P_1 = P_2$ and $D_1 = D_2$.

7. (T/F?) If 0 is an eigenvalue of A, then A is invertible.

8. (T/F?) If A is square, then $\vec{0}$ is an eigenvector of A corresponding with eigenvalue $\frac{-1}{12}$ because $A\vec{0} = \frac{-1}{12}\vec{0}$.

9. (T/F?) If $AP = PD$ with D diagonal, then the columns of P are eigenvectors of A.

10. (T/F?) If $AP = PD$ with D diagonal, then P is invertible.

11. Let A be the matrix that makes the transformation $T(\vec{x}) = A\vec{x}$ a reflection across the x_2-axis. Diagonalize A. H

12. Let A be the matrix that makes the transformation $T(\vec{x}) = A\vec{x}$ a reflection across the line $x_2 = x_1$. Diagonalize A. H

13. Let A be the 2×2 matrix so that $T(\vec{x}) = A\vec{x}$ projects points onto the nearest point on the x_2-axis. For example, $A\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$. Diagonalize A. H

14. Let A be the 2×2 matrix so that $T(\vec{x}) = A\vec{x}$ projects points onto the nearest point on the line $x_2 = 3x_1$. Find A by finding the eigenstuff of A and multiplying PDP^{-1}. H

15. Let A be the 3×3 matrix such that $T(\vec{x}) = A\vec{x}$ rotates points 30° around the x_2-axis. Find an eigenvector and its eigenvalue. Is A diagonalizable?

16. Let R_θ be the 2×2 matrix such that $T(\vec{x}) = R_\theta\vec{x}$ rotates points an angle θ around the origin. What values of θ make R_θ diagonalizable?

17. Let $A = 0$ be a square matrix with all zero entries. Is A diagonalizable?

18. Let A, B be square matrices such that $AB = BA$ (ie, they *commute*). Suppose $\vec{x}$ is an eigenvector of B. Show that $A\vec{x}$ is also an eigenvector of B. C

19. Suppose A has eigenvalue λ corresponding with eigenvector $\vec{x}$. Find an eigenvalue and corresponding eigenvector of kA, where k is any nonzero constant.

20. Give an example of a matrix that is diagonalizable but not invertible.

21. Give an example of a matrix that is invertible but not diagonalizable.

22. Prove or find a counterexample: If A is diagonalizable, then A^T is diagonalizable.

23. Prove or find a counterexample: If A is diagonalizable, then A^T has the same eigenvalues as A.

24. Prove or find a counterexample: If A is diagonalizable, then A^T has the same eigenvectors as A.

25. Suppose A is 3×3, and every vector in $\mathbb{R}^3$ is an eigenvector corresponding with eigenvalue 7. Show that $A = 7I$.

26. Show that if A is $c \times c$ and has c distinct eigenvalues, then A is diagonalizable.

27. Let A be a square matrix such that $A^n = A$ for some positive integer $n > 1$. Show that if λ is an eigenvalue of A, then $|\lambda|$ equals either $-1, 0$, or 1. C

28. Suppose that A is a $c \times c$ matrix such that every row of A adds up to the same number k. Show that k is an eigenvalue of A by finding a corresponding eigenvector.

[C] 27. Credit: Inspired by K. Kuttler.

4.3 Finding Eigenstuff

Exercise 1. (*Warm-up*) If A is an invertible $c \times c$ matrix...

(a) How many pivot positions does A have?
(b) How many solutions does $A\vec{x} = \vec{0}$ have?
(c) What do you know about the determinant of A?

Exercise 2. (*Warm-up*) Write the null space of each matrix as the span of some vectors. (Review section 1.4 if necessary.) (a) $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & -1 & -\pi \\ 0 & 0 & 13 \end{bmatrix}$

(b) $M = \begin{bmatrix} 2 & 4 & 8 \\ 0 & -4 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

We learned a lot about the usefulness and properties of eigenstuff in the last two sections, but we haven't learned how to find eigenvectors and eigenvalues if you don't already know them. Let's start by finding an eigenvector, given an eigenvalue.

Example 1. Show that 2 is an eigenvalue of $A = \begin{bmatrix} -2 & 3 \\ -8 & 8 \end{bmatrix}$ by finding a corresponding eigenvector.

This idea works in general; the following equations are equivalent:

$$\begin{aligned} A\vec{x} &= \lambda\vec{x} \\ A\vec{x} - \lambda\vec{x} &= \vec{0} \\ A\vec{x} - \lambda I\vec{x} &= \vec{0} \\ (A - \lambda I)\vec{x} &= \vec{0} \end{aligned}$$

This proves:

Theorem 4.4: Alternative definition of eigenstuff.
Suppose A is a $c \times c$ matrix. Then $\vec{x}$ is an eigenvector corresponding with an eigenvalue λ (a scalar) of A, if and only if:

$$(A - \lambda I)\vec{x} = \vec{0} \quad \text{and} \quad \vec{x} \neq \vec{0}$$

This means that, for any eigenvalue λ of A, the set of eigenvectors corresponding with λ equals the null space of $A - \lambda I$ (*except* $\vec{0}$ *which is in* $\mathrm{null}(A - \lambda I)$ *but isn't an eigenvector*). So, the vector space $\mathrm{null}(A - \lambda I)$ is called the **eigenspace** of A corresponding with λ.

Now we know how to find eigenvectors given eigenvalues. But how do you find eigenvalues of A, if you don't know any eigenstuff at all? Fill in the blanks to make equivalent statements, using the IMT:

- A is *not* invertible.
- $\mathrm{null}(A)$ $\qquad$ $\{\vec{0}\}$.
- $\det(A)$ $\qquad$ 0.

Applying this to $A - \lambda I$, the following are equivalent:

- λ is an eigenvalue of A.
- $\mathrm{null}(A - \lambda I)$ $\qquad$ $\{\vec{0}\}$.
- $\det(A - \lambda I)$ $\qquad$ 0.

This shows that solutions (values of λ) that are solutions of the *characteristic equation* are eigenvalues of A:

Definition. The **characteristic equation** of A is $\det(A - \lambda I) = 0$.

Example 2. Find the eigenstuff of the stochastic matrix $A = \begin{bmatrix} 0.9 & 0.05 \\ 0.1 & 0.95 \end{bmatrix}$.

Example 3. Using the characteristic equation, find the eigenvalues of the stochastic matrix from the infectious disease example from section 4.1. Then find an eigenvector corresponding with $\lambda = 1$:

$$A = \begin{bmatrix} 0.92 & 0 & 0 \\ 0.08 & 0.6 & 0.01 \\ 0 & 0.4 & 0.99 \end{bmatrix}$$

Example 4. In section 4.2 example 1 we saw that the matrix $A = \begin{bmatrix} \frac{1}{2} & \frac{-\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$ has no *real* eigenvalues because multiplying by A rotates all vectors (except $\vec{0}$) away from their span by 60°. Diagonalize A, by finding the *complex* eigenvalues of A.

In general, multiplication by complex numbers is an elegant and powerful way to represent rotation. In fact, it can be shown that $e^{at}(cos(bt) + i\sin(bt)) = e^{(a+bi)t}$, and functions of this form are solutions of many differential equations. And an important class of differential equations, called *linear* differential equations, can be written in the form $\vec{y}'(t) = A\vec{y}(t)$, which are analogous to our linear difference equations, $\vec{y}_{n+1} = A\vec{y}_n$.

For this book, the above example (and a single exercise) is the only time we'll talk about complex numbers. But in a slightly more advanced linear algebra course, you'll learn that *every* $c \times c$ matrix has c complex eigenvalues (counting repeated factors in the characteristic equation); and there's a similar result for the analogous *differential* equations.

The next theorem describes a nice way that eigenspaces are distinct from each other. It will be useful in chapter 6.

Theorem 4.5: If A is a square matrix with eigenvectors $\vec{b}_1, \ldots, \vec{b}_n$ corresponding to *distinct* eigenvalues $\lambda_1, \ldots, \lambda_n$, then $\vec{b}_1, \ldots, \vec{b}_n$ is a linearly independent list.

4.3 Exercises

3. Find eigenvalues and corresponding eigenvectors of $A = \begin{bmatrix} 5 & -2 \\ 3 & -2 \end{bmatrix}$.

4. Find eigenvalues and corresponding eigenvectors of $A = \begin{bmatrix} 6 & 3 \\ -4 & -1 \end{bmatrix}$.

5. Find eigenvalues and corresponding eigenvectors of $A = \begin{bmatrix} 2 & 5 \\ -7 & -10 \end{bmatrix}$.

6. (*Challenge*) Find eigenvalues and corresponding eigenvectors, which will involve complex numbers, of $A = \begin{bmatrix} 2 & -5 \\ 1 & -2 \end{bmatrix}$.

7. Let $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ be the rotation matrix that rotates points an angle θ around the origin. Find the characteristic polynomial. For what values of θ does R_θ have real eigenvalues? (If you did section 4.2 exercise 16, check that your answers agree.)

8. Find eigenvalues and corresponding eigenvectors of $A = \begin{bmatrix} 0 & 2 & -2 \\ 0 & 3 & 2 \\ 0 & 2 & -2 \end{bmatrix}$.

9. Write the characteristic equation of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. No need to solve or even simplify. The point of this problem is to show you that for most matrices bigger than 2×2, the characteristic equation is an inefficient way to find eigenstuff.

10. (T/F?) If λ is an eigenvalue of A, then every vector in $\operatorname{null}(A - \lambda I)$ is an eigenvector of A.

11. (T/F?) If A has eigenvalue 3, then $A - 3I$ is not invertible.

12. (T/F?) If $\det A = 0$, then 0 is an eigenvalue of A.

13. (T/F?) If A has eigenvalues 3 and 6, corresponding respectively with eigenvectors $\vec{x}$ and $\vec{y}$, then $\vec{y} = 2\vec{x}$.

14. (T/F?) If A has eigenvalues 3 and 6, corresponding respectively with eigenvectors $\vec{x}$ and $\vec{y}$, then $\vec{x}$ and $\vec{y}$ are linearly independent.

15. Suppose A is a 5×5 matrix with two eigenvalues. One eigenspace is 4-dimensional. Can you conclude if A is diagonalizable or not?

16. Suppose B is a 6×6 matrix with 4 eigenvalues. One eigenspace is 2-dimensional. Can you conclude if B is diagonalizable or not?

17. Prove theorem 4.6 below.

> **Theorem 4.6:** If A is triangular, then the eigenvalues of A are the entries on the main diagonal of A.

18. Find P and D so that $A = PDP^{-1}$ for $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ -7 & -1 & -2 \end{bmatrix}$.

19. Show that A is not diagonalizable: $A = \begin{bmatrix} 3 & 0 & 0 \\ 7 & 3 & 0 \\ -7 & -1 & -2 \end{bmatrix}$.

20. Find a matrix R so that $R^2 = A = \begin{bmatrix} 4 & \frac{3}{2} \\ 0 & 1 \end{bmatrix}$. [H] [C]

21. In section 2.2 exercise 44 you showed that if D_1 and D_2 are diagonal matrices, then $D_1 D_2 = D_2 D_1$. Show that if A and B are $c \times c$ matrices that share a set of c linearly independent eigenvectors, then $AB = BA$.

22. Show that λ is an eigenvalue of A if and only if λ is an eigenvalue of A^T. (Note A might not be diagonalizable. [H]

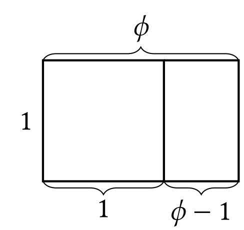
23. Use the results of exercise 28 of section 4.2 and exercise 22 above to show that every stochastic matrix (whose columns all add up to 1) has 1 as an eigenvalue.

24. Show that if A is similar to B, then A and B have the same eigenvalues. [H]

25. Show that every 3×3 matrix A has at least one real eigenvalue.

26. Show that if A is a non-zero diagonalizable matrix, then $A^2 \neq 0$.

27. The rectangle illustrates ϕ, the *golden ratio*, which makes the ratio of the length to the width of the big rectangle equal to the ratio of length to width of the small rectangle. Find ϕ.



[H]20. Hint: Diagonalize A. It's easy to find M so that $M^2 = D$. You can leave your answer as a product.

[C]20. Credit: Inspired by an exercise in Sergei Treil's excellent *Linear Algebra Done Wrong*.

[H]22. Hint: How are $A - \lambda I$ and $A^T - \lambda I$ related?

[H]24. Hint: Show they have the same characteristic equation.

28. (*Challenge*) The Fibonacci sequence is defined as follows: $F_0 = 0, F_1 = 1$ and $F_{n+2} = F_{n-1} + F_n$ for $n \geq 2$. This gives the sequence:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$$

(a) Find A so that $\begin{bmatrix} F_{n+2} \\ F_{n+1} \end{bmatrix} = A \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix}$.

(b) Use the characteristic equation to find the eigenvalues of A. (Notice that one of them is the golden ratio ϕ and the other is $1 - \phi$. The rest of this problem will be easier if you write them this way.)

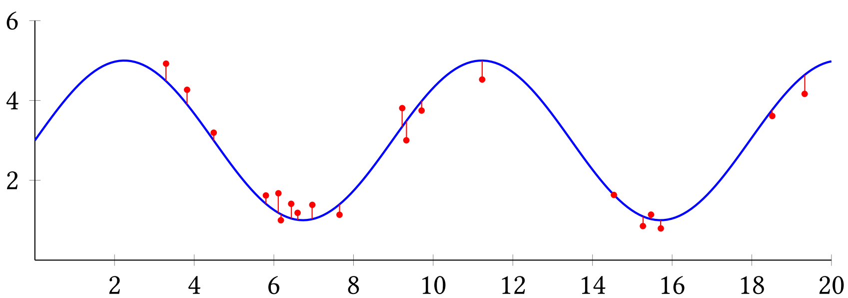
(c) Diagonalize A.

(d) Use the diagonalization and the fact that $\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = A^n \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = A^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to find a formula for F_n. (You'll have to find inverses and multiply some matrices. It gets pretty messy and there are many correct ways to write the formula. Check that your answer works with, say, $F_4 = 3$.)

(e) Show $\lim_{n \to \infty} \frac{F_{n+1}}{F_n} = \phi$.

Chapter 5

Orthogonality and Projections



The above plot shows 20 data points $(x_1, y_1), \dots, (x_{20}, y_{20})$, taken from some real-world measurements, in red. You, a scientist, think there is a periodic or sinusoidal phenomenon here, so you want to find a function of the form $f(x) = \beta_0 + \beta_1 \sin(0.7x)$ to model the phenomenon. Or maybe you want this function because you're an electrical engineer doing signal processing. In any case, the goal is find β_0, β_1 values that make all of the differences $y_i - f(x_i)$ as small as possible. You can think of this as the distance, in $\mathbb{R}^{20}$, between the two points $(y_1, \dots, y_{20})$ and $(f(x_1), \dots, f(x_{20}))$. If you can picture it, the smallest distance from a point in a room to the nearest wall is in a direction *perpendicular* to the wall. The same principle applies in $\mathbb{R}^{20}$, but the point in the room is $(y_1, \dots, y_{20})$, and the "wall" is the set of all points of the form $(f(x_1), \dots, f(x_{20}))$ where f of the form $f(x) = \beta_0 + \beta_1 \sin(0.7x)$. In this chapter, we'll learn how to find a perpendicular line between them.

This concept of finding a *model* that gets as close as possible to *data* is fundamentally what statistics is about, making it useful for all kinds of science and engineering where data is involved. It's also the simplest kind of machine learning model.

5.1 Dot Products and Orthogonality

Exercise 1. (*Warm-up*) Let $\vec{x} = (1, 0, -3) = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$ and $\vec{y} = (4, 7, 0) = \begin{bmatrix} 4 \\ 7 \\ 0 \end{bmatrix}$. Multiply the matrices:

(a) $\vec{x}^T \vec{y}$ (Note: a 1x1 matrix is basically just a number, so you don't need to write the brackets.)

(b) $\vec{x} \vec{y}^T$

Exercise 2. (*Warm-up*) Let $\vec{x} = (x_1, x_2)$ be any vector in $\mathbb{R}^2$.

(a) Multiply $\vec{x}^T \vec{x}$.

(b) (T/F?) $\vec{x}^T \vec{x} \geq 0$.

(c) (T/F?) $\sqrt{\vec{x}^T \vec{x}}$ is the distance from $(0, 0)$ to (x_1, x_2).

(d) If $\vec{x}^T \vec{x} = 0$, what can you say about $\vec{x}$?

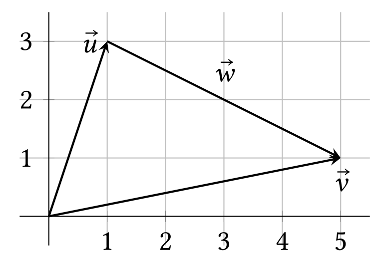
Exercise 3. (*Warm-up*) The graph shows $\vec{u} = (1, 3)$ and $\vec{v} = (5, 1)$.

(a) Find $\vec{w}$ so that $\vec{u} + \vec{w} = \vec{v}$.

(b) (T/F?) The distance from $(1, 3)$ to $(5, 1)$ is the length of the arrow $\vec{w}$.

(c) (T/F?) $\vec{u} + (\vec{v} - \vec{u}) = \vec{v}$.

(d) (T/F?) $\vec{v} - \vec{u} = \vec{w}$.



5.1.1 Dot Products and Orthogonality: Basics

Definition. If $\vec{u}$ and $\vec{v}$ are vectors in $\mathbb{R}^c$, then their **dot product** is:

$$\vec{u} \cdot \vec{v} = \vec{u}^T \vec{v} = u_1 v_1 + u_2 v_2 + \cdots + u_c v_c$$

You may have studied dot products in another class, like multivariable calculus or physics. Everything you learned there still applies here. But since we can write it as a product of matrices, we can use everything we've learned about matrix multiplication study dot products, and then use what we learn about dot products to understand some types of matrix multiplication.

Theorem 5.1: Properties of the dot product. If $\vec{u}$, $\vec{v}$, and $\vec{w}$ are vectors in $\mathbb{R}^c$ and k is a scalar, then:

(i) $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$

(ii) $(k\vec{u}) \cdot \vec{v} = k(\vec{u} \cdot \vec{v})$

(iii) $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$

(iv) $\vec{u} \cdot \vec{u} \geq 0$

The proofs are left as exercises.

Definition. The **length** (aka *norm* or *magnitude*) of a vector is:

$$\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}} = \sqrt{u_1^2 + \cdots + u_c^2}$$

A **unit vector** has length 1.

Like you saw in warm-up 2, in $\mathbb{R}^2$ this definition agrees with pythagorean theorem/distance formula for the length of $\vec{u}$ as an arrow. It's not too hard to show it agrees in $\mathbb{R}^3$ as well; see exercise 20.

Theorem 5.2: Properties of length. If k is a scalar and $\vec{u}$ is a vector, then:

(i) $\|\vec{u}\|^2 = \vec{u} \cdot \vec{u}$

(ii) $\|\vec{u}\| = 0$ if and only if $\vec{u} = 0$

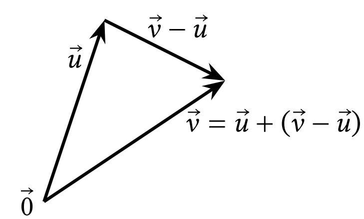
(iii) $\|k\vec{u}\| = |k|\|\vec{u}\|$

Part (i) obviously follows from the definition, but is worth writing down here because we'll use it so much. The other proofs are straightforward exercises.

Example 1. Find two unit vectors parallel to $\vec{x} = (1, -2, 2)$

Definition.

The **distance from $\vec{u}$ to $\vec{v}$** is $\|\vec{v} - \vec{u}\|$.



Example 2. Find the distance from $\vec{u} = (1, 0, 0, 0, 0)$ to $\vec{v} = (0, 0, 0, 0, -1)$.

Definition. Two vectors $\vec{u}$ and $\vec{v}$ are called **orthogonal** if $\vec{u} \cdot \vec{v} = 0$.

The English word *orthogonal* basically means *perpendicular*. Let's look at an example to get an idea what perpendicularity has to do with $\vec{u} \cdot \vec{v} = 0$ and then we'll prove it.

Example 3. Draw each pair of vectors. Are they geometrically perpendicular? Are they orthogonal?

(a) $\vec{u} = (3, 2, 0), \vec{v} = (0, 0, 1)$

(b) $\vec{x} = (1, 0), \vec{y} = (2, 1)$

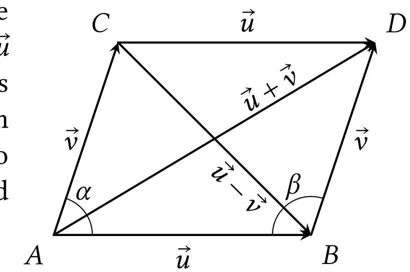
(c) $\vec{w} = (1, 3), \vec{z} = (-3, 1)$

Example 4. Suppose $\vec{u}$ and $\vec{x}$ are any vectors in $\mathbb{R}^c$. Show that $\vec{u}$ and $\vec{x} - \frac{\vec{x}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\vec{u}$ are orthogonal.

Theorem 5.3: Orthogonal means perpendicular.
Suppose $\vec{u}$ and $\vec{v}$ are non-zero vectors in $\mathbb{R}^c$. The angle between them is 90° if and only if $\vec{u} \cdot \vec{v} = 0$.

Note that if $\vec{u} = \vec{0}$ or $\vec{v} = \vec{0}$, then $\vec{u} \cdot \vec{v} = 0$ but the "angle between them" doesn't really make sense.

Proof. Exercise 25 asks you to fill in the gaps below. The graph on the right shows $\vec{u}$ and $\vec{v}$ starting at the origin, labeled A here, as well as copies of them to make a parallelogram $\varparallelogram ABCD$. Note that this doesn't just apply to $\mathbb{R}^2$, because even in higher dimensions, $\vec{u}, \vec{v}$, and $\vec{u} + \vec{v}$ all lie on the same plane.



First, we'll show that:

$$\left\|\vec{u} + \vec{v}\right\|^2 - \left\|\vec{u} - \vec{v}\right\|^2 = 4\vec{u} \cdot \vec{v} \tag{5.1}$$

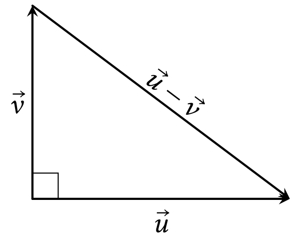
The first step is done; fill in the missing steps:

$$\begin{aligned} \left\|\vec{u} + \vec{v}\right\|^2 - \left\|\vec{u} - \vec{v}\right\|^2 &= (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) - (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) \\ &= \dots \\ &\vdots \\ &= 4\vec{u} \cdot \vec{v} \end{aligned}$$

Equation 5.1 shows that $\|\vec{u} + \vec{v}\| = \|\vec{u} - \vec{v}\|$ if and only if ________ $= 0$. Now we can prove the theorem:

($\longrightarrow$) Suppose $\alpha = 90°$. Since α and β are in the same side of a parallelogram, $\alpha + \beta = 180°$ and so $\beta =$ ____. This means the triangle $\triangle CAB$ is congruent to (same angles and side lengths) ________, since they have two equal sides ($\|\vec{u}\| = \|\vec{u}\|$ and $\|\vec{v}\| = \|\vec{v}\|$) and equal angles between those sides ($\alpha = \beta$). This means their hypotenuses are equal: $\|\vec{u} + \vec{v}\| = \|\vec{u} - \vec{v}\|$, and equation 5.1 implies $\vec{u} \cdot \vec{v} =$ ____.

($\longleftarrow$) Now suppose $\vec{u} \cdot \vec{v} = 0$. Equation 5.1 implies that $\|\vec{u} + \vec{v}\| =$ __________. The triangle $\triangle CAB$ is congruent to ________, since all three pairs of corresponding sides are equal ($\|\vec{u}\| = \|\vec{u}\|$, $\|\vec{v}\| = \|\vec{v}\|$, and the hypotenuses, as we just showed). Thus $\alpha = \beta$ and since they're on the same side of a parallelogram, $\alpha = \beta = 90°$. ■

> **Theorem 5.4: The Pythagorean theorem.**
>
> Two vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^c$ are orthogonal if and only if $\|\vec{u}\|^2 + \|\vec{v}\|^2 = \|\vec{u} - \vec{v}\|^2$
>
> 
>

Note a simple proof would be to apply theorem 5.3 and use the Pythagorean theorem as you learned it years ago. But we're here to do Linear Algebra so that's what we're gonna do. The gaps in the proof are for you to fill in for exercise 26.

Proof. Suppose $\vec{u}$ and $\vec{v}$ are any vectors in $\mathbb{R}^c$. Fill in the missing steps:

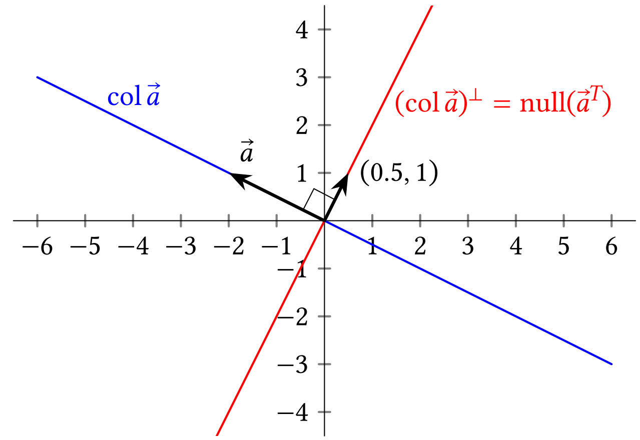
$$\begin{aligned}
\|\vec{u} - \vec{v}\|^2 &= (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) \\
&= \dots \\
&\vdots \\
\|\vec{u} - \vec{v}\|^2 &= \|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2
\end{aligned}$$

Thus $\|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$ if and only if $2\vec{u} \cdot \vec{v} =$ ____ which is true if and only if $\vec{u}$ and $\vec{v}$ are ________________. ■

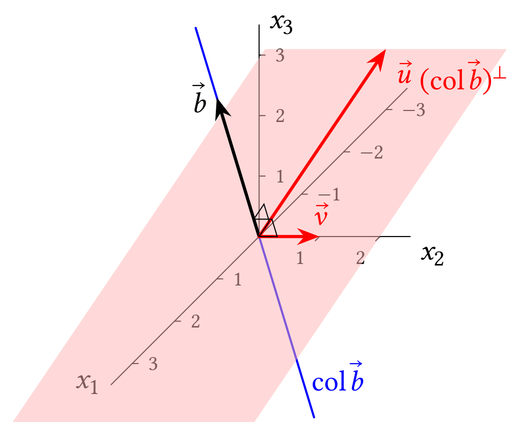
5.1.2 Orthogonal Complements

> **Definition.** Suppose W is a subspace of $\mathbb{R}^c$. The **orthogonal complement** of W, written $W^\perp$, is the set of all vectors that are perpendicular to every vector in W. In symbols:
>
> $$W^\perp = \{\vec{u} \text{ in } \mathbb{R}^c \ : \ \vec{u} \cdot \vec{w} = 0 \text{ for all } \vec{w} \text{ in } W\}$$

Example 5. Let $\vec{a} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$. Note that $\vec{a}$ is a matrix with one column so we can write $\operatorname{col}\vec{a} = \operatorname{span}(\vec{a})$. Describe $(\operatorname{col}\vec{a})^\perp$ and make sure your answer agrees with the graph below.



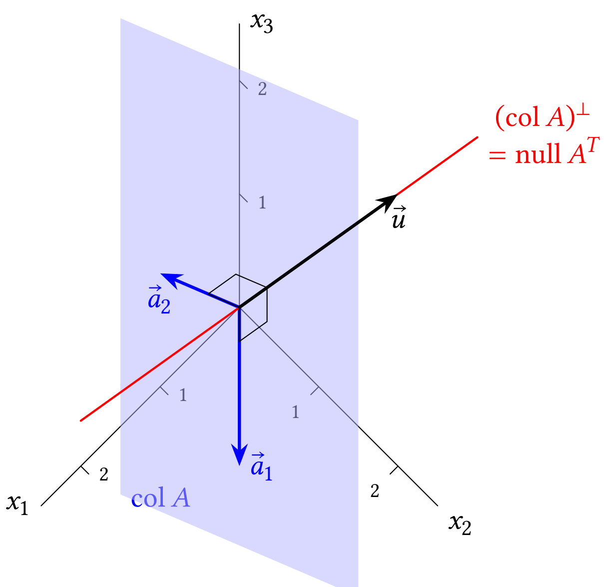
Example 6. Let $\vec{b} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$ and note $\operatorname{col}\vec{b} = \operatorname{span}(\vec{b})$. Describe $(\operatorname{col}\vec{b})^\perp$. Hopefully the picture agrees with the answer.



The next theorem should be intuitive, if you're able to decipher the graphs in the previous examples and the next. The proof is left as an exercise.

Theorem 5.5: A vector $\vec{x}$ is in $\operatorname{span}(\vec{a}_1, \dots, \vec{a}_c)^\perp$ if and only if $\vec{x}$ is orthogonal to each vector $\vec{a}_1, \dots, \vec{a}_c$.

Example 7. Let $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. Describe $(\operatorname{col} A)^\perp$.



These examples illustrate the next theorem, which is sometimes called THE FUNDAMENTAL THEOREM OF LINEAR ALGEBRA.

Theorem 5.6: Four related subspaces. If A is an $r \times c$ matrix, then:

(i) $(\operatorname{col} A)^\perp = \operatorname{null} A^T$ and

(ii) $(\operatorname{row} A)^\perp = \operatorname{null} A$.

Theorem 5.7: Properties of orthogonal complements.
If W is a subspace of $\mathbb{R}^n$, then:

(i) $W^\perp$ is also a subspace of $\mathbb{R}^n$.

(ii) $\dim W + \dim W^\perp = n$.

(iii) $(W^\perp)^\perp = W$.

The proofs of (i) and (ii) are left as exercises. The proof of (iii) is ironically the most difficult of them, and is in the textbook.

Example 8. Suppose $\text{rank}(A) = 1$ and $A\begin{bmatrix} 1 \\ 0 \\ -2 \\ 3 \end{bmatrix} = \vec{0}$. Attempt to find $\dim(\text{null } A)$ and $\dim(\text{null } A^T)$.

Example 9. Find $W^\perp$ if $W = \{(w_1, w_2, w_3) : 7w_1 - w_2 = 2w_3\}$.

[H]20. Hint: The points $(0, 0, 0), (u_1, u_2, 0)$ and (u_1, u_2, u_3) form a right triangle. You can use either the version of the Pythagorean theorem that you learned years ago, or the linear algebra version, thm 5.4.

5.1 Exercises

5.1.1 Dot Products and Orthogonality Basics

4. Find the length of each vector:

$$\vec{x} = \begin{bmatrix} 7 \\ -7 \\ \sqrt{2} \end{bmatrix} \quad \vec{y} = \begin{bmatrix} 0 \\ -5 \\ 0 \end{bmatrix} \quad \vec{x} - \vec{y}$$

5. Which pairs of the following vectors are orthogonal to each other? Which are unit vectors?

$$\vec{a} = \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} -0.8 \\ 0.6 \\ 0 \end{bmatrix} \quad \vec{p} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \quad \vec{q} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

6. If $\vec{u} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, find $\vec{u} \cdot \vec{v}$, $\vec{v} \cdot \vec{v}$, and $\frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}\vec{v}$.

7. Prove theorem 5.1 part (i).

8. Prove theorem 5.1 part (ii).

9. Prove theorem 5.1 part (iii).

10. Prove theorem 5.1 part (iv).

11. Prove theorem 5.2 part (ii).

12. Prove theorem 5.2 part (iii).

13. Find two unit vectors in $\mathbb{R}^2$ which are each parallel to $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$.

14. Find two unit vectors in $\mathbb{R}^2$ which are each orthogonal to $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$.

15. A triangle has three sides $(2, 2), (5, 6)$, and $(11.6, -5.2)$. Find the lengths of each side. Is it a right triangle?

16. (T/F?) $(\vec{u} \cdot \vec{u})^2 = \|\vec{u}\|$.

17. (T/F?) For any scalar k, $\|k\vec{u}\| = k\|\vec{u}\|$.

18. (T/F?) The zero vector in $\mathbb{R}^c$ is orthogonal to every other vector in $\mathbb{R}^c$.

[C]21. Credit: Inspired by an exercise in Gilbert Strang's excellent *Linear Algebra and its Applications.*

[C]28. Credit: Inspired by Strang.

19. (T/F?) If $\vec{u}\cdot\vec{u}=0$, then $\vec{u}=\vec{0}$.

20. Use the Pythagorean theorem to find the distance from $(0,0,0)$ to (u_1,u_2,u_3), and check that it agrees with the definition of $\|(u_1,u_2,u_3)\|$. H

21. Let $\vec{a}=(a_1,a_2)$ and $\vec{b}=(b_1,b_2)$ be two non-zero $\mathbb{R}^2$ vectors. Note that the slopes of the vectors are $\frac{a_2}{a_1}$ and $\frac{b_2}{b_1}$. You probably learned in an earlier class that two lines in $\mathbb{R}^2$ are perpendicular when their slopes are opposite reciprocals: $m_2=\frac{-1}{m_1}$. Show this is equivalent to $\vec{a}\cdot\vec{b}=0$. C

22. Give an example of two vectors that are linearly independent but not orthogonal.

23. Give an example of two vectors that are orthogonal but not linearly independent.

24. Suppose A is an invertible matrix. Explain how you know row i of A is orthogonal to column j of A^{-1}, if $i\neq j$.

25. Fill in the gaps in the proof of theorem 5.3.

26. Fill in the gaps in the proof of theorem 5.4.

27. (T/F?) If $\vec{u}\cdot\vec{v}=0$, then $\|\vec{u}\|^2+\|\vec{v}\|^2=\|\vec{u}+\vec{v}\|^2$. (Hint: draw a picture.)

28. Show that $\vec{u}-\vec{v}$ is orthogonal to $\vec{u}+\vec{v}$ if and only if $\|u\|=\|v\|$. C

5.1.2 Orthogonal Complements

29. Show that if $\vec{x}$ is in W and in $W^\perp$, then $\vec{x}=\vec{0}$.

30. Recall that $\{\vec{0}\}$, the set containing only the zero vector, is always a subspace of $\mathbb{R}^c$. What is $\{\vec{0}\}^\perp$?

31. Consider $\vec{a}=\begin{bmatrix}-2\\1\end{bmatrix}$, the 2×1 matrix from example 5. Describe row(A) and row$(\vec{a})^\perp$.

32. Consider $A=\begin{bmatrix}1&1\\1&0\\0&1\end{bmatrix}$, the matrix from example 7. Describe row A and (row $A)^\perp$.

33. Let $A=\begin{bmatrix}10&5\\-2&-1\end{bmatrix}$. Find one vector in each of:

(a) $(\text{col}(A))^\perp$

(b) $(\text{null } A)^\perp$

(c) $(\text{row } A)^\perp$

34. Let $A=\begin{bmatrix}1&2&3\\4&5&6\\3&3&3\end{bmatrix}$. Note the second row minus the first row equals the third row. And the third column is the opposite of the first column plus twice the second column. Find one vector in each of:

(a) $(\text{col}(A))^\perp$

(b) $(\text{null}(A))^\perp$

(c) $(\text{row}(A))^\perp$

35. Suppose A is a 10×15 matrix with rank$(A)=6$. Find dim(null(A)) and dim(null(A^T)).

36. (T/F?) If A is an $r\times c$ matrix, then col(A) is a subspace of $\mathbb{R}^r$ and row(A) is a subspace of $\mathbb{R}^c$.

37. (T/F?) If $A\vec{x}=\vec{0}$, then $\vec{x}$ is orthogonal to every column of A.

38. (T/F?) The vectors in col(A^T) are orthogonal to the vectors in null(A).

39. (T/F?) There exists a matrix A such that $(1,2,3)$ is in null(A), and $(0,2,0)$ is in row(A).

40. (T/F?) The solution set of $\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}\vec{x}=\vec{0}$ is orthogonal to the span of $(1,2,3)$ and $(4,5,6)$.

41. Find a counterexample to the false statement: If $\vec{x}\cdot\vec{y}=0$, then span$(\vec{x})^\perp=$ span$(\vec{y})$.

42. Let $A=\begin{bmatrix}1&2&3\\-1&-2&-3\end{bmatrix}$. Find a basis for each subspace: col(A), row(A), null(A), and null(A^T).

43. Let W be the plane in $\mathbb{R}^3$ with equation $2x_1-x_2+4x_3=0$. Find a matrix A such that $W=$ null(A), and a matrix B such that $W=$ row(B). C

44. Prove theorem 5.5.

45. Prove part (ii) of theorem 5.6 directly (that is, without using part (i)).

46. Prove theorem 5.7 part (i).

47. (*Challenge*) Prove theorem 5.7 part (ii). H

C43. Credit: Inspired by Strang.

H47. Hint: One way to do it is with 1.17, 1.16, 1.22, and 5.6.

5.2 Projections and Least-Squares Solutions

Exercise 1. (*Warm-up*) Suppose you are lost and thirsty in the desert. For a moment you get some cell reception and your maps app tells you the Derecho River is in the distance, which, oddly enough, is in the shape of a perfectly straight line, all the way to the ocean.

(a) If you walk in a straight line to the river, in a direction that makes your walk as short as possible, at what angle will your path meet the river?

(b) How many points are there on the river that your straight line path could meet at that angle?

Exercise 2. (*Warm-up*) If $\vec{v}$ is a unit vector, then $\vec{v} \cdot \vec{v} = 1$.

Exercise 3. (*Warm-up*) Let $\vec{u} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

(a) Is $\vec{u}$ orthogonal to $\vec{v}$?

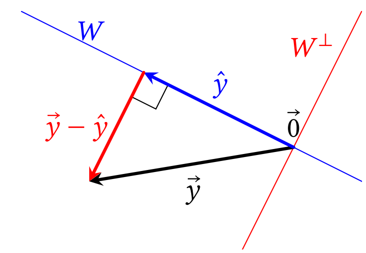
(b) Find $\vec{v} - \frac{\vec{u}\cdot\vec{v}}{\vec{u}\cdot\vec{u}}\vec{u}$. Is it orthogonal to $\vec{v}$?

5.2.1 Projections: Basic Idea

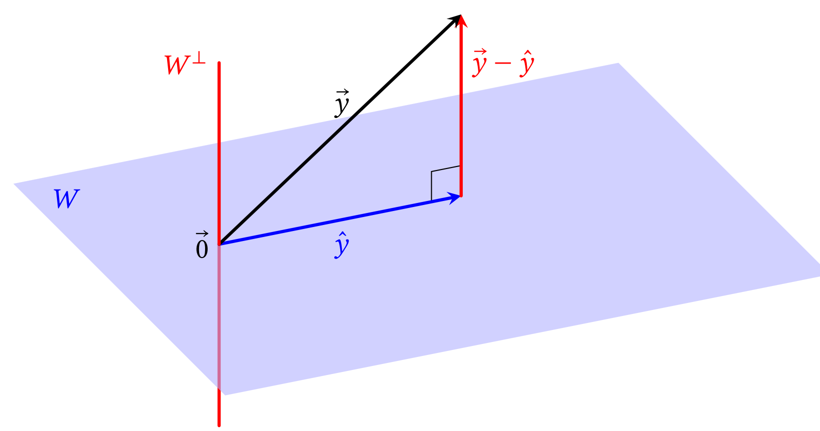
First, let's define some notation that describes the idea of exercise 1, and show that your intuition there was correct in general.

Theorem 5.8: Existence and uniqueness of projections.
If W is a subspace of $\mathbb{R}^r$ and $\vec{y}$ is a vector in $\mathbb{R}^r$, then there exists a unique $\hat{y}$ in W such that $\vec{y} - \hat{y}$ is in $W^\perp$.
Definition: The $\hat{y}$ that makes this work is called the **projection of $\vec{y}$ onto W** and is written $\text{proj}_W \vec{y}$.

In the case of $\mathbb{R}^2$ or $\mathbb{R}^3$, the figures on the right and below, plus a hand-wavy appeal to intuition, might be better than the proof (which you can find in the book if you're interested). Remember any subspace W of $\mathbb{R}^2$ or $\mathbb{R}^3$ is either a single point $\{\vec{0}\}$, a line, or a plane, or all of $\mathbb{R}^3$. The theorem is saying that for any point $\vec{y}$, there exists exactly one point $\hat{y}$ in W that makes the line from $\hat{y}$ to $\vec{y}$ perpendicular to W.



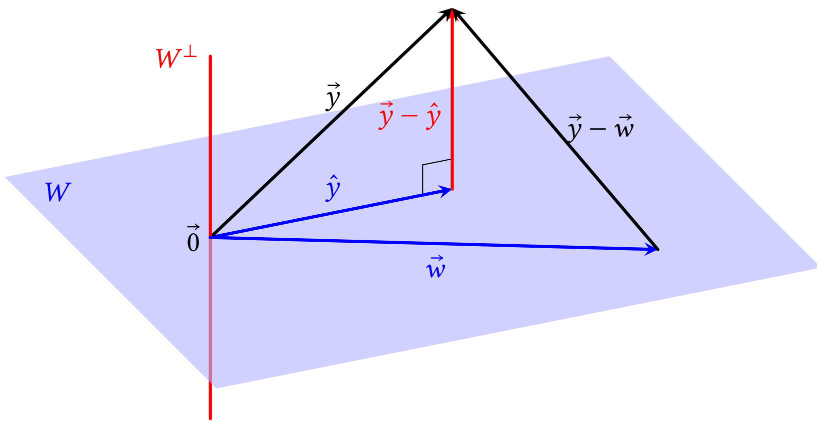
Note that the *point* $\vec{y} - \hat{y}$ is in $W^\perp$, which you can see if you draw the arrow starting at $\vec{0}$, but the pictures show a copy of $\vec{y} - \hat{y}$ drawn tip-to-tail to illustrate $\vec{y} = \hat{y} + \vec{y} - \hat{y}$.



Theorem 5.9: Best approximation theorem. If W is a subspace of $\mathbb{R}^r$, $\vec{y}$ is a vector in $\mathbb{R}^r$, and $\hat{y} = \text{proj}_W \vec{y}$, then for any vector $\vec{w}$ in W with $\vec{w} \neq \hat{y}$:

$$\|\vec{y} - \hat{y}\| < \|\vec{y} - \vec{w}\|$$

As the graph below illustrates, this theorem says that the any line segment from $\vec{y}$ to W that meets W non-orthogonally is longer than the line segment that meets it orthogonally. The proof shows that $\vec{y} - \vec{w}$ is the hypotenuse of a right triangle with $\vec{y} - \hat{y}$ as a leg.



Example 1. Let $\vec{y} = (4, 5, 6)$, $W = \text{col}[\vec{e}_1 \quad \vec{e}_2]$, and $\vec{v} = \frac{\vec{e}_1 \cdot \vec{y}}{\vec{e}_1 \cdot \vec{e}_1}\vec{e}_1 + \frac{\vec{e}_2 \cdot \vec{y}}{\vec{e}_2 \cdot \vec{e}_2}\vec{e}_2$.

(a) Find $\vec{v}$ and $\vec{y} - \vec{v}$.

(b) Show that $\text{proj}_W \vec{y} = \vec{v}$.

We'll soon see that the formula for $\vec{v} = \text{proj}_W \vec{y}$ in example 1 turns out to work in general, when you have a basis of W made of vectors that are orthogonal to each other.

5.2.2 Projections with Orthogonal Lists

Definition. A list of non-zero vectors $(\vec{u}_1, \dots, \vec{u}_c)$ is called an **orthogonal list** if every vector is orthogonal to every other vector: $\vec{u}_i \cdot \vec{u}_j = 0$ for $i \neq j$. If all the vectors have length one, it's an **orthonormal list**.

Note that if a list contains only one vector, it's vacuously orthogonal to every other vector in the list, making the list orthogonal.

Example 2. The standard basis vectors of $\mathbb{R}^3$, $\vec{e}_1, \vec{e}_2, \vec{e}_3$, are an orthonormal list.

Example 3. Show that the columns of U make an orthonormal list of vectors: $U = \begin{bmatrix}\vec{u}_1 & \vec{u}_2 & \vec{u}_3\end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 \end{bmatrix}$.

Example 4. With the same U from example 3, find UU^T and U^TU.

Theorem 5.10: If U is a matrix with orthonormal columns, then $U^TU = I$.

The proof of the case where U has three columns was done in the last example. If U has more columns, the same principle applies: All the diagonal entries of U^TU are $\vec{u}_i \cdot \vec{u}_i = \|\vec{u}_i\|^2 = 1$ because they're unit vectors, and the non-diagonal entries, with $i \neq j$, are $\vec{u}_i \cdot \vec{u}_j = 0$ because they're orthogonal.

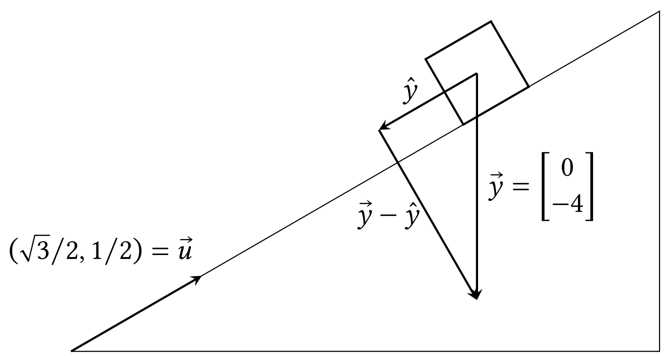
Theorem 5.11: Orthogonal projection formula.
If $\vec{u}_1, \ldots, \vec{u}_c$ is an orthogonal basis of a subspace W of $\mathbb{R}^r$, and $\vec{y}$ is in $\mathbb{R}^n$, then:

$$\operatorname{proj}_W \vec{y} = \frac{\vec{u}_1 \cdot \vec{y}}{\vec{u}_1 \cdot \vec{u}_1}\vec{u}_1 + \cdots + \frac{\vec{u}_c \cdot \vec{y}}{\vec{u}_c \cdot \vec{u}_c}\vec{u}_c$$

Note that if $\vec{u}_1, \ldots, \vec{u}_c$ is an ortho*normal* basis of W, then all the denominators of the coefficients equal one, making the formula simpler.

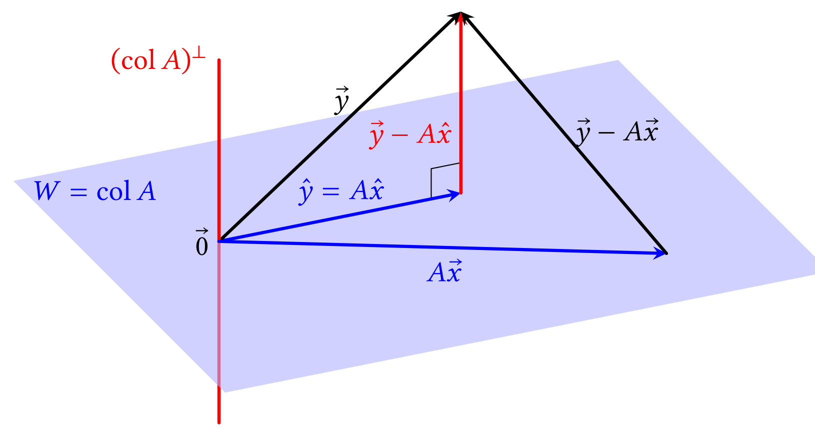
Example 5. Let $W = \operatorname{span}(1,1,1)$ and $\vec{y} = (4,5,6)$. Find the nearest point in W to $\vec{y}$, and find the distance from $\vec{y}$ to that point.

Example 6. A block weighing 4 pounds is resting on an inclined plane with slope $\frac{1}{\sqrt{3}}$. Find components of its weight that are parallel and perpendicular to the slope.



5.2.3 Least-Squares Solutions

Now, what if you want to find $\operatorname{proj}_W \vec{y}$, but you don't have an orthogonal basis of W? What if you don't even have a basis of W? Let's say that at least you have a list $\vec{a}_1, \dots, \vec{a}_c$ (maybe linearly dependent) that spans W, so that $W = \operatorname{col} A$ with $A = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_c \end{bmatrix}$. In this case, finding $\operatorname{proj}_{\operatorname{col} A} \vec{y}$ amounts to finding a $\hat{y}$ that can be written $\hat{y} = A\hat{x}$ for some $\hat{x}$, so that $A\hat{x}$ is *as close as possible* to $\vec{y}$. Here's the illustration of theorem 5.9 in the situation $W = \operatorname{col} A$, where $A\vec{x}$ is some other vector in $\operatorname{col} A$:



Theorem 5.8 guarantees that such a $\hat{y}$ exists and is unique. The trick to finding $\hat{x}$ and $\hat{y}$ is to notice that theorem 5.6 says that $(\operatorname{col} A)^\perp = \operatorname{null}(A^T)$. So, we're trying to find $\hat{x}$ such that $\vec{y} - A\hat{x}$ is in $\operatorname{null}(A^T)$ which means:

$$\begin{aligned} A^T(\vec{y} - A\hat{x}) &= \vec{0} \\ A^T\vec{y} - A^T A\hat{x} &= \vec{0} \\ A^T A\hat{x} &= A^T\vec{y} \qquad (5.2) \end{aligned}$$

Note that theorem 5.8 guarantees that $\hat{y} = A\hat{x}$ exists and is unique, but that doesn't mean that $\hat{x}$ is unique; there may be more that one $\hat{x}$ that satisfies equation 5.2. In any case, this proves:

Theorem 5.12: Least-squares solutions.
Let A be an $r \times c$ matrix, and $\vec{y}$ be a vector in $\mathbb{R}^r$. Then $A^T A\hat{x} = A^T\vec{y}$ if and only if $A\hat{x} = \operatorname{proj}_{\operatorname{col} A} \vec{y}$, which is the closest point in $\operatorname{col} A$ to $\vec{y}$ (that is, $\|\vec{y} - A\hat{x}\| < \|\vec{y} - A\vec{x}\|$ for any $\vec{x} \neq \hat{x}$).
Definition. The $\hat{x}$ described in the theorem is called the **least-squares solution** of the equation $A\vec{x} = \vec{b}$, because it minimizes $\|\vec{y} - A\hat{x}\|$, whose formula involves a sum of squares.

Example 7. In example 1 we found that with $A = \begin{bmatrix}\vec{e}_1 & \vec{e}_2\end{bmatrix}$ and $\vec{y} = (4, 5, 6)$, $\text{proj}_{\text{col}\,A}\,\vec{y} = (4, 5, 0)$. Check that theorem 5.12 gives the same result.

Example 8. Describe least-squares solutions to $A\vec{x} = \vec{y}$, with $A = \begin{bmatrix} -1 & 0 & 2 \\ 2 & 0 & -4 \\ 1 & 2 & 3 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$.

5.2 Exercises

5.2.1 Projections: Basic Idea

4. If $\vec{y}$ is in W, what is $\text{proj}_W\,\vec{y}$? What is $\vec{y} - \text{proj}_w\,\vec{y}$?

5. If $W = \{\vec{0}\}$, and $\vec{y}$ is not in W, what is $\text{proj}_W\,\vec{y}$? What is $\vec{y} - \text{proj}_w\,\vec{y}$?

6. Let $\vec{y} = (\pi, \sqrt{2})$ and $W = \text{col}\begin{bmatrix}\vec{e}_1\end{bmatrix}$ aka the x_1-axis. Find $\text{proj}_W\,\vec{y}$ and $\vec{y} - \text{proj}_W\,\vec{y}$.

7. Let $\vec{y} = (-4, -2, 7)$ and $W = \text{col}\begin{bmatrix}\vec{e}_1 & \vec{e}_3\end{bmatrix}$, aka the x_1x_3-plane. Find $\text{proj}_W\,\vec{y}$ and $\vec{y} - \text{proj}_W\,\vec{y}$.

8. (T/F?) For some subspaces W of $\mathbb{R}^r$, and some $\vec{y}$ in $\mathbb{R}^r$, there does not exist a $\hat{y}$ in W such that $\vec{y} - \hat{y}$ is in $W^\perp$.

9. (T/F?) For some subspaces W of $\mathbb{R}^r$, and some $\vec{y}$ in $\mathbb{R}^r$, there exists more than one vector $\hat{y}$ in W such that $\vec{y} - \hat{y}$ is in $W^\perp$.

10. (T/F?) If W is a subspace of $\mathbb{R}^r$ and $\vec{y}$ is in $\mathbb{R}^r$, then $\text{proj}_W\,\vec{y}$ is closer to $\vec{y}$ than any other point in W.

11. (T/F?) If W is a subspace of $\mathbb{R}^r$ and $\vec{y}$ is in $\mathbb{R}^r$, then $\|\text{proj}_W\,\vec{y}\|$ is the distance from $\vec{y}$ to the closest point in W.

12. (T/F?) Let $\vec{y}$ be in $\mathbb{R}^r$, and A be $r \times c$. If $W = \text{col}\,A$, then the equation $A\vec{x} = \text{proj}_W\,\vec{y}$ is consistent.

13. If $\vec{y}$ is in $W^\perp$, what is $\text{proj}_W\,\vec{y}$?

5.2.2 Projections with Orthogonal Lists

14. Let $U = \begin{bmatrix}\vec{u}_1 & \vec{u}_2\end{bmatrix} = \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$. Is the list $\vec{u}_1, \vec{u}_2$ orthogonal? Is it orthonormal? Find U^TU.

15. Let $U = \begin{bmatrix}\vec{u}_1 & \vec{u}_2\end{bmatrix} = \begin{bmatrix} 0.6 & 0.8 \\ 0.8 & 0.6 \end{bmatrix}$. Is the list $\vec{u}_1, \vec{u}_2$ orthogonal? Is it orthonormal? Find U^TU.

16. Let $U = \begin{bmatrix}\vec{u}_1 & \vec{u}_2 & \vec{u}_3\end{bmatrix} = \begin{bmatrix} -1/3 & 2/3 \\ 2/3 & -1/3 \\ 2/3 & 2/3 \end{bmatrix}$. Is the list $\vec{u}_1, \vec{u}_2$ orthogonal? Is it orthonormal? Find U^TU and UU^T.

17. With the same $\vec{u}_1, \vec{u}_2$ as in exercise 16, find $\vec{u}_3$ so that $\vec{u}_1, \vec{u}_2, \vec{u}_3$ is an orthonor-

[H]17. Hint: $(\text{col}\,U)^\perp = \text{null}(U^T)$.
[H]21. Hint: One way to do this is with theorem 5.10. Another way is with theorems 5.8 and 5.11.
[H]22. Hint: Theorem 5.13.
[H]24. Hint: $U(U^T\vec{y})$.

mal list. [H]

18. Let $A = \begin{bmatrix} 2 & 10 & -5 \\ 2 & -11 & -2 \\ 1 & h & k \end{bmatrix}$.

(a) Find h and k so that the columns of A form an orthogonal basis of $\mathbb{R}^3$.

(b) Make the columns of A into an orthonormal list by dividing each vector by its length.

19. (T/F?) If $\vec{u}_i$ is one vector from an orthogonal list of vectors, then $\vec{u}_i \cdot \vec{u}_i = 0$.

20. (T/F?) If $\vec{b}_1, \ldots, \vec{b}_c$ is a basis of W, then $\operatorname{proj}_W \vec{y} = \frac{\vec{b}_1 \cdot \vec{y}}{\vec{b}_1 \cdot \vec{b}_1}\vec{b}_1 + \cdots + \frac{\vec{b}_c \cdot \vec{y}}{\vec{b}_c \cdot \vec{b}_c}\vec{b}_c$.

21. Prove theorem 5.13 below. [H]

Theorem 5.13: Orthogonal lists are independent.
If $\vec{u}_1, \ldots, \vec{u}_n$ is an orthogonal list of vectors, then it's linearly independent too.

22. Suppose U is an $r \times c$ matrix with orthogonal columns.

(a) Show that $r \geq c$. [H]

(b) What are the sizes (rows × columns) of UU^T and U^TU?

23. Suppose $\vec{u} = k\vec{v}$ with $k \neq 0$, and let $W = \operatorname{span}(\vec{v})$. We know the vectors are parallel, so $W = \operatorname{span}(\vec{v}) = \operatorname{span}(\vec{u})$. Hopefully the formula in theorem 5.11 gives the same result for $\operatorname{proj}_W \vec{y}$ whether you use $\vec{u}$ or $\vec{v}$ as a basis of W. Check that this is indeed the case.

24. Suppose $U = \begin{bmatrix} \vec{u}_1 & \cdots & \vec{u}_c \end{bmatrix}$ and is a matrix with orthonormal columns, and let $W = \operatorname{col} U$. Show that the formula in theorem 5.11 can be written $\operatorname{proj}_W \vec{y} = UU^T\vec{y}$. [H]

25. Let $U = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \end{bmatrix} = \begin{bmatrix} -1/3 & 2/3 \\ 2/3 & -1/3 \\ 2/3 & 2/3 \end{bmatrix}$ and $\vec{y} = (9, 0, 0)$. Find $\operatorname{proj}_{\operatorname{col} U} \vec{y}$. (You showed in exercise 16 that $(\vec{u}_1, \vec{u}_2)$ is an orthonormal list.)

26. Let $U = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \end{bmatrix} = \begin{bmatrix} -12 & -4 \\ -4 & 3 \\ 3 & -12 \end{bmatrix}$. It can be shown that $(\vec{u}_1, \vec{u}_2)$ is an orthogonal list. Find the closest point in $\operatorname{col} U$ to $\vec{y} = (13, 13, 13)$ and the distance from $\vec{y}$ to that point.

27. It can be shown that the columns of $U = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \\ 1/2 & -1/2 \\ 1/2 & -1/2 \end{bmatrix}$ are an orthonormal list of vectors. Find the nearest point in $\operatorname{col} U$ to $\vec{y} = (4, 0, 0, 0)$ and the distance from $\vec{y}$ to that point.

28. Suppose U is an $r \times c$ matrix with orthonormal columns, and let $\vec{x}, \vec{y}$ be $\mathbb{R}^c$ vectors. Show that $(U\vec{x}) \cdot (U\vec{y}) = \vec{x} \cdot \vec{y}$. [H]

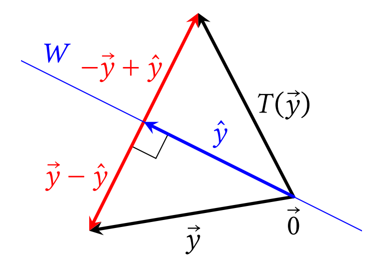
29. Suppose U is an $r \times c$ matrix with orthonormal columns, and let $\vec{x}$ be in $\mathbb{R}^c$. Show that $\|U\vec{x}\| = \|\vec{x}\|$.

30. Suppose $\vec{u}$ is a unit vector in $\mathbb{R}^c$, and let $W = \operatorname{span}(\vec{u})$. Define a function $T(\vec{y})$ that reflects $\vec{y}$ to the point on the opposite side of W from $\vec{y}$, as shown in the graph below.

(a) Show that $T(\vec{y}) = 2\operatorname{proj}_W \vec{y} - \vec{y}$.

(b) Use the result of exercise 24 to find A so that $T(\vec{y}) = A\vec{y}$.

(c) Use the result of (b) to find A so that $T(\vec{y})$ reflects $\vec{y}$ across the line $x_2 = 3x_1$.



5.2.3 Least-Squares Solutions

For problems 31 to 34, Assume A is an $r \times c$ matrix and $\vec{y}$ is in $\mathbb{R}^r$.

31. (T/F?) If $\vec{x}$ is a solution of $A\vec{x} = \vec{y}$, then $\vec{x}$ is also a least-squares solution of $A\vec{x} = \vec{y}$.

32. (T/F?) The equation $A^TA\hat{x} = A^T\vec{y}$ always has a solution for $\hat{x}$.

33. (T/F?) The equation $A^TA\hat{x} = A^T\vec{y}$ has a *unique* solution for $\hat{x}$.

34. (T/F?) If $A^TA\hat{x} = A^T\vec{y}$ has two solutions $\hat{x} = \vec{p}$ and $\hat{x} = \vec{q}$, then $A\vec{p} = A\vec{q}$.

35. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $\vec{y} = (-3, 2, 1)$. It can be shown that $\hat{x} = (3, -2, -1/3)$ is a least-squares solution of $A\hat{x} = \vec{y}$.

(a) Find $\operatorname{proj}_{\operatorname{col} A} \vec{y}$.

[H]28. Hint: The definition of $\vec{u} \cdot \vec{v}$ is $\vec{u}^T\vec{v}$.

(b) It turns out that $\hat{x} = (10/3, -8/3, 0)$ is also a least-squares solution. Do you expect this to give a different $\text{proj}_{\text{col}\,A}\,\vec{y}$ than your answer to (a) or the same? Explain.

36. Let $\vec{u} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$.

(a) Use the projection formula in 5.11 to find $\text{proj}_{\text{col}\,\vec{u}}\,\vec{y}$.

(b) Solve $\vec{u}^T\vec{u}x = \vec{u}^T\vec{y}$ to find a least-squares solution to $\vec{u}x = \vec{y}$, and find $\text{proj}_{\text{col}\,\vec{u}}\,\vec{y}$. Note the similarities to part (a).

37. Let $U = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$, and $\vec{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. Find $\text{proj}_{\text{col}\,U}\,\vec{y}$ two ways:

(a) Notice that the columns of U are orthogonal. Use the projection formula in 5.11.

(b) Solve $U^TU\hat{x} = U^T\vec{y}$ to find a least-squares solution to $U\vec{x} = \vec{y}$, and multiply it by U. Note the similarities to part (a).

38. Let $A = \begin{bmatrix} -1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} 0 \\ -11 \\ 7 \end{bmatrix}$. Find the least-squares solution of $A\vec{x} = \vec{y}$.

39. Let $A = \begin{bmatrix} -1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}$.

(a) Find $\hat{x}$, the least-squares solution of $A\vec{x} = \vec{y}$.

(b) How far is $A\hat{x}$ from $\vec{y}$? What does this tell you about $A\vec{x} = \vec{y}$?

40. Let $A = \begin{bmatrix} 8 & 3 & 5 \\ 1 & 2 & -1 \\ 8 & 3 & 5 \end{bmatrix}$, and $\vec{y} = (14, 0, 12)$. Find the least-squares solution of $A\vec{x} = \vec{y}$ and $\text{proj}_{\text{col}\,A}\,\vec{y}$.

41. Prove the (i $\leftrightarrow$ii) part of theorem 5.14 below. Here's a guide to the proof:

(a) ($\longleftarrow$) Assume A^TA is invertible. To show that the columns of A are independent, suppose $A\vec{z} = \vec{0}$, and show this implies $\vec{z} = \vec{0}$.

(b) ($\longrightarrow$) Assume the columns of A are linearly independent. Explain why, to show that A^TA is invertible, it suffices to show the columns of A^TA are linearly independent. To show the columns of A^TA are independent, suppose $A^TA\vec{z} = \vec{0}$, and show this implies $\vec{z} = \vec{0}$. [H]

42. Prove the (ii $\leftrightarrow$iii) part of theorem 5.14 below.

Theorem 5.14: Uniqueness of least-squares solutions.
Suppose A is an $r \times c$ matrix. The following are equivalent:

(i) The columns of A are linearly independent.

(ii) A^TA is invertible.

(iii) For every $\vec{y}$ in $\mathbb{R}^r$, the equation $A\vec{x} = \vec{y}$ has a *unique* least-squares solution $\hat{x}$.

43. If the columns of A are linearly independent, the matrix $(A^TA)^{-1}A^T$ is called the *pseudo-inverse* of A. What does it have in common with actual inverses and how is it different? (This question is more open-ended than usual; there are many correct answers.)

[H]41. Hint: One step is to show $\vec{z}^TA^TA\vec{z} = 0$; what does this tell you about $A\vec{x}$?

5.3 Curve Fitting

Exercise 1. (*Warm-up*) Suppose the size of A is 3×5 and $\vec{y}$ is in $\mathbb{R}^3$. Determine the size of each product, if it's defined:

(a) AA^T (b) A^TA (c) $A\vec{y}$ (d) $A^T\vec{y}$

Exercise 2. (*Warm-up*) Suppose A is $r \times c$ and $\vec{y}$ is in $\mathbb{R}^r$. Does $A^TA\hat{x} = A^T\vec{y}$ always have a solution for $\hat{x}$?

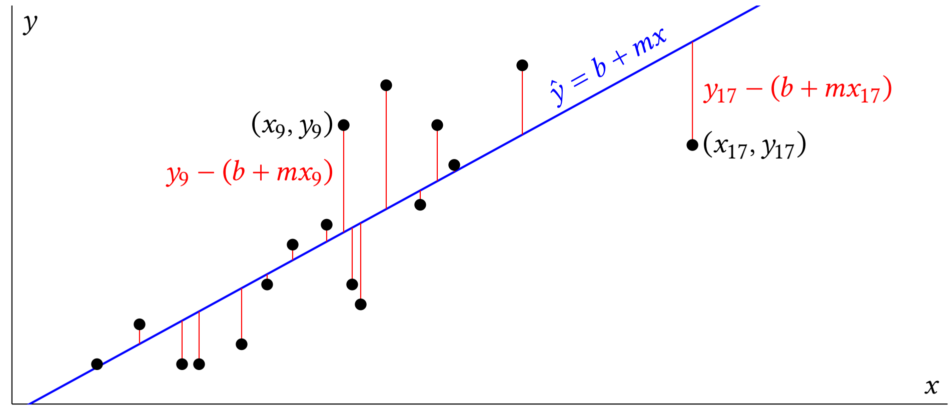
A big part of any kind of science, from astronomy to biology to economics, is analyzing data, which is what statistics is all about. And a big part of data analysis is coming up with a *model*, which is a way of summarizing the data in a simpler way. Some examples of models:

- A *least-squares regression* model, which is an equation of a line (or some other curve) that gets as close as possible to a bunch of data points. For a single-variable linear model, you can describe it with only two *parameters*: the slope and intercept. This is the focus of this section.
- A *weather forecasting* model, which is a mathematical representation of the atmosphere. This kind of model may store millions of parameters, for the temperature, velocity, and humidity of millions of sections of air, but it is still much simpler than the atmosphere itself, with, like, forty gazillion molecules.
- A *large language* model, which contains billions of parameters that represent the likelihood of certain phrases coming next in a piece of text, depending on all the phrases that came before. The data this kind of model is summarizing the vast amount of text on the internet that it was trained on.

All of these models can *describe* the data they summarize, and also try to *predict* new data. Either way, the model should get as close as possible to the data; the parameters should be chosen so that the model *best fits* the data. Roughly speaking, we want the *ERROR* to be as small as possible:

$$\textit{ERROR} = \textit{DATA} - \textit{MODEL}$$

To get more specific, consider a list of data points $(x_1, y_1), (x_2, y_2), \ldots, (x_c, y_c)$. Suppose we want a model to describe/predict y based on values of x: $\hat{y} = b + mx$. So, we want b, m that makes all of the different errors $y_i - (b + mx_i)$, for each i between 1 and c, as small as possible.



Fortunately, we have a good way describe how big/small a big list of numbers is–the length of a vector! The goal is to minimize:

$$\left\| \begin{bmatrix} y_1 - (b + mx_1) \\ \vdots \\ y_c - (b + mx_c) \end{bmatrix} \right\| = \left\| \begin{bmatrix} y_1 \\ \vdots \\ y_c \end{bmatrix} - \begin{bmatrix} b + mx_1 \\ \vdots \\ b + mx_c \end{bmatrix} \right\|$$

We can use what we know about matrix-vector multiplication to help with this if we denote:

$$\vec{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_c \end{bmatrix}, \quad A = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_c \end{bmatrix}, \quad \hat{\beta} = \begin{bmatrix} b \\ m \end{bmatrix}$$

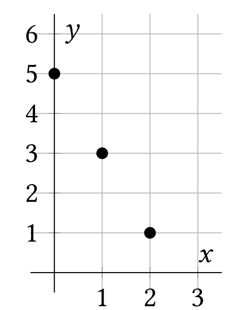
Here A is called the **design matrix**. So, we want to find $\hat{\beta}$ that minimizes the error:

$$\left\| \vec{y} - A\hat{\beta} \right\|$$

In other words, to make $\hat{y} = b + mx$ best fit the data, we want to find $\hat{\beta}$, the *least-squares solution* of $A\hat{\beta} = \vec{y}$. Theorem 5.12 tells us $\hat{\beta}$ is a solution of:

$$\boxed{A^TA\hat{\beta} = A^T\vec{y}} \tag{5.3}$$

Example 1. Use equation 5.3 to find the best-fit line for the data points $(0, 5), (1, 3)$, and $(2, 1)$. Note that the points are collinear, so it should be easy to check the answer.



The next example shows that we can use the same method for finding best-fit functions for multi-variable data.

Example 2. Use equation 5.3 To find the best-fit linear function of two variables, of the form $\hat{z} = b + mx + ny$ for some constants b, m, n, for the following data points. Also find the *least-squares error*, $\|\vec{z} - \hat{z}\|$.

$$\begin{aligned}(x_1, y_1, z_1) &= (-2, -2, -4)\\ (x_2, y_2, z_2) &= (-2, 2, 2)\\ (x_3, y_3, z_3) &= (2, 2, 8)\\ (x_4, y_4, z_4) &= (2, -2, 6)\end{aligned}$$

In fact, the method of equation 5.3 works for finding many kinds (but not all kinds) of best-fit functions.

Example 3. Describe the design matrix you would use to find the best-fit function of the form $\hat{y} = a\cos(x) + b\sin(x)$ for the data points $(x_1, y_1), (x_2, y_2), \ldots, (x_c, y_c)$.

The above methods are pretty flexible and powerful, but they don't cover every kind of curve fitting problem. For example, if you want to find a best-fit function of the form $\hat{y} = \cos(ax + b)$, there is no way to construct a design matrix that works:

$$\begin{bmatrix}\cos(ax_1 + b)\\ \vdots\\ \cos(ax_c + b)\end{bmatrix} = \begin{bmatrix}?? & ??\\ \vdots & \vdots\\ ?? & ??\end{bmatrix}\begin{bmatrix}a\\ b\end{bmatrix}$$

This kind of *nonlinear regression* problem is beyond the scope of this course, but you might learn about algorithms for it in a numerical methods class.

5.3 Exercises

3. Suppose you have data points $(x_1, y_1), \ldots, (x_{376}, y_{376})$, and denote $\vec{x} = (x_1, \ldots, x_{376})$, $\vec{y} = (y_1, \ldots, y_{376})$. You want to find the best fit line $\hat{y} = b + mx$ for the data. Fill in the blanks with numbers: The design matrix A is ____ $\times$ ____. The column space of A is a ____-dimensional subspace of $\mathbb{R}^{\_\_\_\_}$.

4. Find the equation in the form $\hat{y} = b + mx$ that best fits the (x, y) data points $(-1, 2), (0, 4), (1, -2), (2, -6)$.

5. Find the equation in the form $\hat{y} = b + mx$ that best fits the (x, y) data points $(2, 3), (4, 2), (6, 7)$.

6. Find the equation in the form $\hat{y} = b + mx$ that best fits the (x, y) data points $(2, 7), (4, 7), (6, 4), (8, 0)$. Also find the least-squares error $\|\vec{y} - \hat{y}\|$.

7. (T/F?) If $\hat{y} = 2x + 3$ is the best-fit line for data points $(x_1, y_1), \ldots, (x_c, y_c)$, then $\sum(y_i - (2x_i + 3))^2 \leq \sum(y_i - (1.9x_i + 2.9))^2$.

8. (T/F?) The methods in this section can be used to find a best-fit function of the form $\hat{y} = e^{ax} + \sin(x + b)$.

9. Describe the design matrix you would use to find the best-fit function of the form $\hat{y} = a_0 + a_1x + a_2x^2 + a_3x^3$ for data points $(x_1, y_1), (x_2, y_2), \ldots, (x_c, y_c)$. (See example 3.)

10. Describe the design matrix you would use to find the best-fit function of the form $\hat{z} = ae^{x+y} - bx + cy^2$ for data points $(x_1, y_1, z_1), (x_2, y_2, z_2), \ldots, (x_c, y_c, z_c)$. (See example 3.)

11. Describe the design matrix you would use to find the best-fit function of the form $\hat{z} = a_0x + a_1x^2 + a_2y + a_3y^2 + a_4xy$ for data points $(x_1, y_1, z_1), (x_2, y_2, z_2), \ldots, (x_c, y_c, z_c)$. (See example 3.)

12. Suppose you are creating a robot outfielder to play robot baseball. When a fly ball is hit, a system of cameras estimates the ball's height at several times, giving approximate data in the form $(t_1, h_1), (t_2, h_2), \ldots, (t_c, h_c)$. The robot must find want to find h_0, v_0, and a for a best-fit function of the form $\hat{h} = h_0 + v_0t - \frac{1}{2}at^2$. Describe the design matrix needed to do so.

13. (*Challenge*) Suppose you are still working on your robot outfielder. You realize that the ball is always hit from an initial height of $h_0 = 1$ meter, and gravity has a constant acceleration $a = 9.8\frac{\text{m}}{\text{s}^2}$, so you should only be estimating the initial velocity v_0. Do the methods in this section work to find a best-fit function of the form $\hat{h} = 1 + v_0t - \frac{9.8}{2}t^2$? Is there a way to modify the methods to make it work?

14. Cesium-137 has a half-life of 30.04 years and strontium-90 has a half-life of 28.91 years. Both are harmful components of nuclear waste. If you had a sample of nuclear waste (where these only radioactive isotopes), and measured its mass at several times, you'd have data points of the form $(t_1, m_1), (t_2, m_2), \ldots, (t_c, m_c)$. To estimate the initial amounts s and c of the two isotopes, you could find the best-fit function of the form $\hat{m} = b + c(0.5)^{t/30.04} + s(0.5)^{t/28.91}$. Describe the design matrix you'd need. (Note: If you could take perfectly precise measurements, you'd be able to find the exact values of s and c with three data points. But in real life, measurements are not perfectly precise and small errors can make a big difference in situations like this.)

15. This question refers to the four data points in example 2. One naive way to find a plane that gets close to the four points might be the plane that goes through three of the points, and hope it gets close enough to the fourth point. It turns out the plane that goes through $(x_1, y_1, z_1), (x_2, y_2, z_2)$, and (x_3, y_3, z_3) has equation $z = 2 + 1.5x + 1.5y$. If $\vec{\gamma} = (2, 1.5, 1.5)$, find the error for this plane, $\|\vec{y} - A\vec{\gamma}\|$. Which plane has smaller error: $\hat{z} = 3 + 2x + y$ or $z = 2 + 1.5x + 1.5y$? Is this what you expected?

16. Suppose you want to find a best-fit function $\hat{y} = b + mx$ for the (x, y) data points $(5, 1), (5, 3)$. Show there isn't a unique best-fit function by finding at least two that have the same least-squares error $\|\vec{y} - \hat{y}\|$.

17. What must be true about data points $(x_1, y_1), \ldots, (x_c, y_c)$ to guarantee that the best-fit line $\hat{y} = b + mx$ is *unique*? Explain.[H]

18. Find the best-fit function of the form $\hat{y} = c + bx + ax^2$ for the (x, y) data points $(-2, 7), (-1, -2), (1, 2)$, and $(2, 5)$. Also find the least-squares error $\|\vec{y} - \hat{y}\|$ for your function.

19. Suppose you're finding a best-fit line $\hat{y} = b + mx$ for data points $(x_1, y_1), \ldots, (x_c, y_c)$.

(a) If you calculate the mean $\bar{x}$ of the data's x-values, and write a new variable with values $x_i^* = x_i - \bar{x}$, we say x^* is written in *mean-deviation form*, and we'll see how this saves some computation. Write the data in example 1 in mean-deviation form and find the best-fit line for it.

(b) It can be shown that $\sum(x_i - \bar{x}) = 0$ for any list of numbers $x_1, \ldots, x_c$. Use this to show that if x^* is in mean-deviation form, with corresponding design matrix A, then $A^T A$ is a diagonal matrix.

[H]17. Hint: See theorem 5.14.

Chapter 6

Symmetric Matrices and Quadratic Forms

TODO: Write this chapter

Chapter 7

Abstract Vector Spaces

Many of the concepts of linear algebra apply to mathematical objects other than $\mathbb{R}^c$ vectors. For example, polynomial addition and scalar multiplication works the same way:

$$5(z^2 - z + 3) + (2z - 1) = 5z^2 + 3z + 14$$

looks a lot like:

$$5\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 14 \end{bmatrix}$$

Another example is that derivatives and integrals have a linearity property, which means we can think of them as *linear transformations*:

$$\frac{d}{dz}(kf(z) + cg(z)) = k\frac{d}{dz}f(z) + c\frac{d}{dz}g(z)$$
$$\int (kf(z) + cg(z))dz = k\int f(z)dz + c\int g(z)dz$$

These look a lot like:

$$T(k\vec{x} + c\vec{z}) = kT(\vec{x}) + cT(\vec{z})$$

When we call objects like polynomials and other functions *vectors*, we can bring the powerful theorems of linear algebra to bear when studying them. But of course, it's not just *calling* objects vectors, it's the fact that the objects have certain properties in common with $\mathbb{R}^c$ vectors that allows us to apply the theorems. First we'll see what those properties are, and later we'll see exactly how the theorems of linear algebra apply to objects like polynomials.

7.1 Vector Spaces other than $\mathbb{R}^c$

Exercise 1. (*Warm-up*) Let $\mathbb{P}_2$ be the set of all functions of the form $f(z) = az^2 + bz + c$, for any real numbers a, b, and c. (We'll use z as the variable because x and t are already used for other things.) So $\mathbb{P}_2$ contains functions like $3z^2 - 17.5z + \pi$, or $2z$, or just 3. Suppose that $f(z) = az^2 + bz + c$ is in $\mathbb{P}_2$.

(a) If k is a real number, is $kf(z) = kaz^2 + kbz + kc$ in $\mathbb{P}_2$?

(b) If $g(z) = pz^2 + qz + r$ is in $\mathbb{P}_2$, is $f(z) + g(z)$ in $\mathbb{P}_2$?

(c) Is $f(z) + g(z) = g(z) + f(z)$?

(d) Is $f(z) + 0 = f(z)$?

Definition. A set of objects V is called a **vector space**, and the objects are called **vectors**, if it comes with addition and scalar multiplication operations for which the following properties hold:

(i) **$\vec{0}$ exists and works:** There exists an object, called $\vec{0}$, in V such that $\vec{0} + \vec{u} = \vec{u}$ for all $\vec{u}$ in V.

(ii) **Closure under vector addition:** $\vec{u} + \vec{v}$ is in V for all $\vec{u}, \vec{v}$ in V.

(iii) **Closure under scalar multiplication:** $r\vec{u}$ is in V for all r in $\mathbb{R}$ and $\vec{u}$ in V.

(iv) **Vector addition is commutative:** $\vec{u} + \vec{v} = \vec{v} + \vec{u}$, for all $\vec{u}, \vec{v}$ in V.

(v) **Vector addition is associative:** $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$ for all $\vec{u}, \vec{v}, \vec{w}$ in V.

(vi) **Scalar multiplication is associative:** $r(t\vec{u}) = (rt)\vec{u}$ for all r, t in $\mathbb{R}$ and all $\vec{u}$ in V.

(vii) **Additive inverses exist:** For every $\vec{u}$ in V, there exists a vector $\vec{v}$ such that $\vec{u} + \vec{v} = \vec{0}$.

(viii) **1 works:** $1\vec{u} = \vec{u}$ for all $\vec{u}$ in V.

(ix) **Distributive rules work:** $r(\vec{u} + \vec{v}) = r\vec{u} + r\vec{v}$ and $(r + t)\vec{u} = r\vec{u} + t\vec{v}$ for all r, t in $\mathbb{R}$ and all $\vec{u}, \vec{v}$ in V.

Example 1. $\mathbb{R}^c$ is a vector space, for any whole number c. Each of the properties of the definition is clearly familiar in $\mathbb{R}^c$. ■

Example 2. We denote $\mathbb{P}$ to be the set of all polynomials, which are functions that can be written in the form $\vec{p}(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ for some non-negative integer n and real numbers $a_0, \dots, a_n$. Is $\mathbb{P}$ a vector space?

Example 3. Are the integers, $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$, a vector space?

The following properties are familiar from $\mathbb{R}^c$, but it takes a little work to show that they are true for other vector spaces.

Theorem 7.1: Basic vector space properties.
If V is a vector space, then:

(i) There is only one vector $\vec{0}$ that satisfies property (i) of the definition.

(ii) $0\vec{u} = \vec{0}$ for all $\vec{u}$ in V.

(iii) $(-1)\vec{u}$ is the additive inverse of $\vec{u}$ for each $\vec{u}$ in V.

(iv) $r\vec{0} = \vec{0}$ for all r in $\mathbb{R}$.

Let's prove (i) and (ii) and leave the others for exercises.

Example 4. The following can each be shown to be vector spaces:

- The set of all $\mathbb{R} \to \mathbb{R}$ functions is a vector space.
- The set of all $\mathbb{R} \to \mathbb{R}$ *differentiable* functions is a vector space.
- Given any homogeneous differential equation, the set of solutions to it is a vector space.
- The set of all 3×2 matrices is a vector space.
- The set of all $\mathbb{R}^2 \to \mathbb{R}^3$ linear transformations is a vector space. (This is equivalent to the previous one).

Many definitions we learned in chapter 1 can be generalized for vector spaces. Let's include them here for completeness.

Definition. If $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_c$ is a list of vectors in a vector space V, and $x_1, x_2, \dots, x_c$ are in $\mathbb{R}$, then $x_1\vec{u}_1 + x_2\vec{u}_2 + \cdots + x_c\vec{u}_c$ is called a **linear combination** of $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_c$. The set of all linear combinations of $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_c$ is called $\operatorname{span}(\vec{u}_1, \vec{u}_2, \dots, \vec{u}_c)$.

Example 5. In the vector space of all $\mathbb{R} \to \mathbb{R}$ functions, describe $\operatorname{span}(\cos(3z), \sin(2z))$.

Definition. A linear system of equations that can be written in the form

$$x_1\vec{u}_1 + \cdots + x_c\vec{u}_c = \vec{0}$$

is called **homogeneous**. Homogeneous systems are always consistent since $\vec{x} = (x_1, \dots, x_c) = \vec{0}$ is always a solution, called the **trivial solution.** A **non-trivial** solution is any solution $\vec{x} \neq \vec{0}$.
When a non-trivial solution exists, we say the list $\vec{u}_1, \dots, \vec{u}_c$ is **linearly dependent**. When the only solution is the trivial solution, we say the list is **linearly independent.** The empty list () is defined to be linearly independent as well.

The next theorem may be so intuitive that it seems obvious, but the proof is a little tricky. You can find it in the textbook.

Theorem 7.2: Zero polynomial. If the polynomial $\vec{p}(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ equals the $\vec{0}(z)$ function (that is, if $\vec{p}(z) = 0$ for all real numbers z), then all the coefficients are zero: $a_0 = 0, \dots, a_n = 0$.

Example 6. Is each list of polynomials linearly independent in $\mathbb{P}$?

(a) $3 + z^2, 30 + 10z^2$

(b) $1, z, z^2$

(c) $1 - 2z^2, 2 - z - 5z^2, 3z + 3z^2$

Note that in the last example, there's a striking similarity between the equation with polynomials in $\mathbb{P}_2$

$$x_1(1 - 2z^2) + x_2(2 - z - 5z^2) + x_3(3z + 3z^2) = 0 + 0z + 0z^2$$

and the $\mathbb{R}^3$ vector equation:

$$x_1 \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -1 \\ -5 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The following theorems are analogous to theorems 1.4, 1.5, and 1.6.

Theorem 7.3: If a list contains only a single vector and it's not $\vec{0}$, then the list is linearly independent.

Theorem 7.4: If a list of vectors contains $\vec{0}$, then it's linearly dependent.

Theorem 7.5: Characterization of linear dependence.
A list of vectors $W = \vec{a}_1, \dots, \vec{a}_c$ is linearly dependent iff one of the vectors can be written as a linear combination of the others.
Specifically, if W is linearly dependent, and $\vec{a}_1 \neq \vec{0}$, then some $\vec{a}_i$ with $i > 1$ is a linear combination of the preceding vectors:

$$x_1\vec{a}_1 + \cdots + x_{i-1}\vec{a}_{i-1} = \vec{a}_i$$

The proof of theorem 1.6 did not rely on any special properties of $\mathbb{R}^n$; each of the steps only relies on the properties of the definition of a vector space. So it doubles as a proof of theorem 7.5. In chapter 1, the proof of the first two theorems 1.4 and 1.5 were assigned as exercises; your answers probably likewise would stand in as proofs of theorems 7.3, 7.4.

Definition. A **subspace** U of a vector space V is a subset of V for which:

(i) The zero vector of V is in U.

(ii) If $\vec{p}$ and $\vec{q}$ are in U, then $\vec{p}+\vec{q}$ is in U. (U is **closed under vector addition**.)

(iii) If $\vec{p}$ is in U and c is in $\mathbb{R}$, then $c\vec{p}$ is in U. (U is **closed under scalar multiplication**.)

Example 7. Just like in chapter 1, the smallest subspace of a vector space V is $\{\vec{0}\}$, and the largest subspace of V is V itself. ■

Example 8. Is the set $U = \{a + bz^2 \;:\; a \geq 0, b \geq 0\}$ a subspace of $\mathbb{P}_2$?

Note that properties (i), (ii), and (iii) in the definition of a subspace are the same as properties (i), (ii), and (iii) of a vector space. And, if U is a subspace of V, then any vectors $\vec{u}, \vec{v}, \vec{w}$ in U are also vectors in V, which makes properties the rest of the properties, (iv) through (ix) in the definition of a vector space, also true for U. This proves:

Theorem 7.6: Subspaces are vector spaces. If U is a subspace of a vector space V, then U itself is a vector space.

The following theorems are just like theorems 1.12 and 1.13, but with $\mathbb{R}^c$ replaced with an abstract vector space V.

Theorem 7.7: Spans are subspaces.
If $\vec{a}_1, \dots, \vec{a}_c$ is a list of vectors in a vector space V, then $\text{span}(\vec{a}_1, \dots, \vec{a}_c)$ is a subspace of V.

Like before, if you read the proof of theorem 1.12, you'll find that it only relies on properties that are true about vector spaces in general, not just $\mathbb{R}^c$. So theorem 7.7 is already proven. The next definitions are also the same as in chapter 1, but with V instead of $\mathbb{R}^n$.

Definition. Let U be a subspace of a vector space V. (Note that since V is a subspace of itself, this definition also applies to $U = V$.) A list of vectors $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_c)$ is called a **basis** for U if:

(i) $\mathscr{B}$ is linearly independent.

(ii) $\text{span}\,\mathscr{B} = U$.

Definition. Suppose $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_c)$ is a basis for a subspace U of a vector space V, and that $\vec{y}$ is a vector in V. If

$$x_1\vec{b}_1 + \cdots + x_c\vec{b}_c = \vec{y}$$

then the vector $(x_1, \dots, x_c)$ is called the **$\mathscr{B}$-coordinate vector of $\vec{y}$** and we write $(x_1, \dots, x_c) = [\vec{y}]_{\mathscr{B}}$.

Example 9. The *standard basis of* $\mathbb{P}_3$ is $\mathscr{B} = (1, z, z^2, z^3)$.

(a) Find $\left[\pi - 7z^2\right]_{\mathscr{B}}$.

(b) If $\left[\vec{y}\right]_{\mathscr{B}} = (5, 4, 3, 2)$, find $\vec{y}$.

Here are some more theorems that have $\mathbb{R}^c$ replaced with an arbitrary vector space V, that were really proved in chapter 1:

> **Theorem 7.8: Unique representation theorem.**
> Let $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_c)$ be a basis for a subspace U of $\mathbb{R}^n$. Then for each $\vec{y}$ in U, there is only one $\left[\vec{y}\right]_{\mathscr{B}}$.

> **Theorem 7.9: Properties of coordinates.**
> Suppose $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_n)$ is a basis of a vector space V. Then, for any $\vec{u}, \vec{v}$ in V and any real number k:
>
> (i) $\left[k\vec{u}\right]_{\mathscr{B}} = k\left[\vec{u}\right]_{\mathscr{B}}$.
>
> (ii) $\left[\vec{u} + \vec{v}\right]_{\mathscr{B}} = \left[\vec{u}\right]_{\mathscr{B}} + \left[\vec{v}\right]_{\mathscr{B}}$.
>
> (iii) If $\left[\vec{u}\right]_{\mathscr{B}} = \left[\vec{v}\right]_{\mathscr{B}}$, then $\vec{u} = \vec{v}$. (We say $f(\vec{u}) = \left[\vec{u}\right]_{\mathscr{B}}$ is an *injective* or *one-to-one* function.)
>
> (iv) If $\vec{x}$ is any vector in $\mathbb{R}^n$, then there is a $\vec{u}$ in V so that $\left[\vec{u}\right]_{\mathscr{B}} = \vec{x}$. (We say $f(\vec{u}) = \left[\vec{u}\right]_{\mathscr{B}}$ is a *surjective* or *onto* function.)

To sum up theorem 7.9: vectors $\vec{u}, \vec{v}$ in an n-dimensional vector space V really behave just like their coordinate vectors $[\vec{u}]_{\mathscr{B}}, [\vec{v}]_{\mathscr{B}}$ in $\mathbb{R}^n$, at least as far as scalar multiplication and vector addition are concerned; and there's a one-to-one correspondence between them. Next we'll show that this extends to the span, linear independence, solutions of vector equations. In the next section we'll extend this to linear transformations.

> **Theorem 7.10: More properties of coordinates.**
> Let $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_n)$ be a basis for a vector space V.
>
> (i) $x_1\vec{a}_1 + \cdots + x_p\vec{a}_p = \vec{y}$ is true and only if $x_1[\vec{a}_1]_{\mathscr{B}} + \cdots + x_p[\vec{a}_p]_{\mathscr{B}} = [\vec{y}]_{\mathscr{B}}$ is true.
>
> (ii) A list $(\vec{a}_1, \dots, \vec{a}_p)$ in V is linearly dependent if and only if $([\vec{a}_1]_{\mathscr{B}}, \dots, [\vec{a}_p]_{\mathscr{B}})$ is linearly dependent in $\mathbb{R}^n$.
>
> (iii) A list $(\vec{a}_1, \dots, \vec{a}_p)$ in V spans V if and only $([\vec{a}_1]_{\mathscr{B}}, \dots, [\vec{a}_p]_{\mathscr{B}})$ spans $\mathbb{R}^n$.

Let's prove (iii) and leave (i) and (ii) for exercises.

The next two theorems were proved in chapter 1.

> **Theorem 7.11: More vectors than a basis are dependent.** If a subspace U of a vector space V has a basis with n vectors in it, then any list with more than n vectors in U is linearly dependent.

> **Theorem 7.12:** If U is a subspace of a vector space V, then every basis of U has the same number of vectors.

Definition. The **dimension** of a subspace U of a vector space V, written $\dim U$, is the number of vectors in a basis of U. If $\dim U = n$, we say U is n-**dimensional**.
And something new: If U does not have a finite basis, we say that U is **infinite-dimensional**.

Example 10. Show that $\mathbb{P}$, the vector space of all polynomials, is infinite dimensional.

Theorem 7.13: Buy two, get one free.
Suppose $\vec{v}_1, \dots, \vec{v}_n$ is a list of n vectors in a subspace U of $\mathbb{R}^c$. If any two of the following are true, then so is the third:

(i) $\dim U = n$.

(ii) $\vec{v}_1, \dots, \vec{v}_n$ is linearly independent.

(iii) $\vec{v}_1, \dots, \vec{v}_n$ spans U.

[H]13. Hint: You may use part (ii) of the theorem.
[H]14. Hint: $\vec{0} = \vec{0} + \vec{0}$.

Example 11. If $(\vec{p}_1, \vec{p}_2, \vec{p}_3) = (3 - z + 6z^2, -2 + z - 2z^2, -4 + 2z - 5z^2)$, find the dimension of $\operatorname{span}(\vec{p}_1, \vec{p}_2, \vec{p}_3)$.

7.1 Exercises

2. (T/F?) Every vector space is a subspace of itself.

3. (T/F?) The list of polynomials $z, 5z^2$ is linearly dependent because $-5z$ and 1 are scalars that make $-z(z) + 1(5z^2) = \vec{0}$.

4. (T/F?) In $\mathbb{P}_2$, the function $\vec{f}(z) = 3z^2$ is the zero vector because $\vec{f}(0) = 0$.

5. (T/F?) In the vector space of all $\mathbb{R} \to \mathbb{R}$ functions, $\operatorname{span}(\cos(z), \sin(z))$ is the interval $[-1, 1]$.

6. (T/F?) In the vector space of all $\mathbb{R} \to \mathbb{R}$ functions, $\operatorname{span}(\cos(z), \sin(z))$ is the set of all functions that can be written $a\cos(z) + b\sin(z)$ for some real numbers a, b.

7. Determine if $3z$ is in $\operatorname{span}(1 + z, 1 - z)$.

8. Is the list of polynomials $(\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4) = (1+2z-z^2, -5-9z+5z^2, 4+11z-3z^2, 2 + 4z + 2z^2)$ linearly independent in $\mathbb{P}_2$? Explain.

9. Consider the list of polynomials $(\vec{p}_1, \vec{p}_2, \vec{p}_3) = (1 + 2z + 3z^2,\ 4 + 5z + 6z^2,\ 7 + 8z + 9z^2)$.

(a) Is the list linearly independent? Explain.

(b) Does the list span $\mathbb{P}_2$? Explain.

[C]17. Credit: Inspired by an exercise in Axler's excellent book *Linear Algebra Done Right*.
[C]18. Credit: Inspired by Axler.

10. It can be shown that $\mathscr{B} = (2+z, 3+z)$ is a basis of $\mathbb{P}_1$.

(a) Find $[-4+z]_{\mathscr{B}}$.

(b) If $[\vec{p}] = (3,5)$, find $\vec{p}$.

11. It can be shown that $\mathscr{B} = (2+z+z^2, 4+4z+3z^2, -5-z-2z^2)$ is a basis of $\mathbb{P}_2$.

(a) Find $[z]_{\mathscr{B}}$.

(b) If $[\vec{p}] = (1,2,3)$, find $\vec{p}$.

12. Explain why every basis of $\mathbb{P}_3$ contains at least one polynomial with degree 3.

13. Prove theorem 7.1 part (iii). H

14. Prove theorem 7.1 part (iv). H

15. Prove theorem 7.10 part (i).

16. Prove theorem 7.10 part (ii).

17. Suppose $\vec{u}_1, \dots, \vec{u}_n$ is linearly independent in a vector space V and $\vec{w}$ is in V. Prove that if $u_1 + \vec{w}, \dots, \vec{u}_n + \vec{w}$ is linearly dependent, then $\vec{w}$ is in span$(\vec{u}_1, \dots, \vec{u}_n)$. C

18. Let $U = \{\vec{p} \text{ in } \mathbb{P}_4 : \vec{p}(6) = 0\}$.C

(a) Show that U is a subspace of $\mathbb{P}_4$.

(b) Find a basis for U.

19. Let $U = \left\{\vec{p} \text{ in } \mathbb{P}_5 : \int_{-1}^{1} \vec{p}(z)dz = 0\right\}$. Find a basis for U.C

(a) Show that U is a subspace of $\mathbb{P}_5$.

(b) Find a basis for U.

20. Let V be the set of all functions of the form $\vec{f}(z) = a\cos(5z) + b\sin(5z)$ for some a, b in $\mathbb{R}$. Is V a subspace of the vector space of all $\mathbb{R} \to \mathbb{R}$ functions? If so, what is its dimension?

21. Let W be the set of all functions of the form $\vec{f}(z) = a\cos(kz) + b\sin(kz)$ for some a, b, k in $\mathbb{R}$. Is W a subspace of the vector space of all $\mathbb{R} \to \mathbb{R}$ functions? If so, what is its dimension?

22. Let E be the set of all even $\mathbb{R} \to \mathbb{R}$ functions. That is, the set of functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(-z) = f(z)$ for all z in $\mathbb{R}$. Is E a subspace of the vector space all $\mathbb{R} \to \mathbb{R}$ functions? Explain.

23. Let U be the set of polynomials $\vec{p}$ in $\mathbb{P}_2$ such that $\vec{p}(3) = 4$. Is U a subspace of $\mathbb{P}_2$? Explain.

24. Let D be the set $\{\vec{p}(z) = az^2 + bz + c \text{ in } \mathbb{P}_2 : b^2 - 4ac = 0\}$. Is D a subspace of $\mathbb{P}_2$? Explain.

25. Let $U = \{\vec{p} \text{ in } \mathbb{P}_4 : \vec{p}''(7) = 0\}$. Find a basis for U. C

26. Suppose r is in $\mathbb{R}$ and $\vec{u}$ is in a vector space V. Show that if $r\vec{u} = \vec{0}$, then either $r = 0$ or $\vec{u} = \vec{0}$.

27. (*Challenge*) Prove or give a counterexample: If $(\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4)$ is a basis of a vector space V, and U is a subspace of V, with $\vec{b}_1, \vec{b}_2$ in U, but $\vec{b}_3$ and $\vec{b}_4$ are not in U, then $(\vec{b}_1, \vec{b}_2)$ is a basis of U. C

28. (*Challenge*) Suppose that $\vec{p}_0, \vec{p}_1, \dots, \vec{p}_m$ are polynomials in $\mathbb{P}_m$, such that $\vec{p}_j(2) = 0$ for each j. Show that $\vec{p}_0, \vec{p}_1, \dots, \vec{p}_m$ is linearly dependent in $\mathbb{P}_m$. C

29. Many curious students ask questions like, "if we can expand the real numbers to include complex numbers like i, with $i^2 = -1$, could we also expand real numbers to include ∞ and $-\infty$?" Let's denote $\mathbb{I}$ to be the set that includes all real numbers, and also two distinct objects ∞ and $-\infty$, and define multiplication and addition in a reasonable way. Specifically, for any real number t:

$$t + \infty = \infty + t = \infty \qquad t + (-\infty) = (-\infty) + t = \infty,$$
$$\infty + \infty = \infty \qquad (-\infty) + (-\infty) = (-\infty)$$
$$\infty + (-\infty) = 0$$
$$t\infty = \begin{cases} -\infty \text{ if } t < 0 \\ 0 \text{ if } t = 0 \\ \infty \text{ if } t > 0 \end{cases} \qquad t(-\infty) = \begin{cases} \infty \text{ if } t < 0 \\ 0 \text{ if } t = 0 \\ -\infty \text{ if } t > 0 \end{cases}$$

Is $\mathbb{I}$ a vector space? Justify your answer. C

30. Suppose $\vec{p}_1, \vec{p}_2, \dots, \vec{p}_n, \vec{p}_{n+1}$ is a list of $n+1$ polynomials in $\mathbb{P}_n$, all with the property that $\vec{p}_i(3) = 0$. Show that the list is linearly dependent.

C19. Credit: Inspired by Axler.
C25. Credit: Inspired by Axler.
C27. Credit: Inspired by Axler.
C28. Credit: Inspired by Axler.
C29. Credit: Credit: Inspired by Axler.

7.2 Abstract Linear Transformations

In this section, we'll expand the notion of linear transformations to abstract vector spaces, and see how they are essentially the same as the $\mathbb{R}^c \to \mathbb{R}^r$ linear transformations introduced in section 2.1 and studied throughout the book. First let's warm up with a review.

Exercise 1. (*Warm-up*) (T/F?) If A is a $c \times r$ matrix, then $T(\vec{x}) = A\vec{x}$ is an $\mathbb{R}^c \to \mathbb{R}^r$ linear transformation.

Exercise 2. (*Warm-up*) (T/F?) If $T : \mathbb{R}^c \to \mathbb{R}^r$ is a linear transformation, then there is a $c \times r$ matrix A such that $T(\vec{x}) = A\vec{x}$.

Exercise 3. (*Warm-up*) (T/F?) Suppose A is a matrix. Then the range of the transformation $T(\vec{x}) = A\vec{x}$ is the column space of A.

Exercise 4. (*Warm-up*) Let $T(\vec{x}) = A\vec{x}$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. Let $\mathscr{B} = (\vec{e}_1, \vec{e}_2, \vec{e}_3)$ and $\mathscr{D} = (\vec{e}_1, \vec{e}_2)$ be the standard bases of $\mathbb{R}^3$ and $\mathbb{R}^2$, respectively.

(a) Find $T(\vec{e}_3)$.

(b) (T/F?) $A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) & T(\vec{e}_3) \end{bmatrix}$.

(c) Find $\begin{bmatrix} T(\vec{e}_3) \end{bmatrix}_{\mathscr{D}}$.

(d) (T/F?) For any $\vec{y} = (y_1, y_2)$ in $\mathbb{R}^2$, we have $\begin{bmatrix} \vec{y} \end{bmatrix}_{\mathscr{D}} = (y_1, y_2)$.

Definition. Suppose V and W are vector spaces. A function $T : V \to W$ is a **linear transformation** if for all real numbers k and all vectors $\vec{u}, \vec{v}$ in V:

(i) $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$

(ii) $T(k\vec{u}) = kT(\vec{u})$

Example 1. Let $T : \mathbb{P} \to \mathbb{P}$ be differentiation; that is, $T(\vec{p}(z)) = \frac{d}{dz}\vec{p}(z)$. Is T a linear transformation?

Example 2. Is the function $T : \mathbb{P} \to \mathbb{P}$ defined by $T(\vec{p}(z)) = \vec{p}(z) + 5z$ a linear transformation?

Example 3. Suppose $T : V \to W$ is a linear transformation. Denote the zero vector of V as $\vec{0}_V$ and the zero vector of W as $\vec{0}_W$. Show that $T(\vec{0}_V) = \vec{0}_W$.

Recall that every $\mathbb{R}^c \to \mathbb{R}^r$ linear transformation can be written $T(\vec{x}) = A\vec{x}$, with $A = \begin{bmatrix} T(\vec{e}_1) & \cdots & T(\vec{e}_c) \end{bmatrix}$, called the *standard matrix* of T (theorem 2.4). The next example and theorem show how this concept can be generalized to abstract vector spaces. This will reinforce the idea that, as far as linear algebra is concerned, finite-dimensional vector spaces are essentially the same as $\mathbb{R}^c$.

Example 4. Let $T : \mathbb{P}_2 \to \mathbb{P}_1$ be the differentiation transformation, $T(\vec{p}(z)) = \frac{d}{dz}\vec{p}(z)$, and denote the standard bases of $\mathbb{P}_2$ and $\mathbb{P}_1$ as $\mathscr{B} = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\} = \{1, z, z^2\}$ and $\mathscr{D} = \{\vec{d}_1, \vec{d}_2\} = \{1, z\}$.

(a) Find each: $T(\vec{b}_1), T(\vec{b}_2)$, and $T(\vec{b}_3)$.

(b) Find each: $[T(\vec{b}_1)]_{\mathscr{D}}, [T(\vec{b}_2)]_{\mathscr{D}}$, and $[T(\vec{b}_3)]_{\mathscr{D}}$.

(c) Find $T(10 - 9z + 7z^2)$.

(d) Find $[10 - 9z + 7z^2]_{\mathscr{B}}$.

(e) Find $[T(10 - 9z + 7z^2)]_{\mathscr{D}}$.

(f) Find $A[10 - 9z + 7z^2]_{\mathscr{B}}$, with $A = \begin{bmatrix} [T(\vec{b}_1)]_{\mathscr{D}} & [T(\vec{b}_2)]_{\mathscr{D}} & [T(\vec{b}_3)]_{\mathscr{D}} \end{bmatrix}$.

(g) Explain why it's not a coincidence that $[T(10 - 9z + 7z^2)]_{\mathscr{D}}$ equals $A[10 - 9z + 7z^2]_{\mathscr{B}}$.

The justification for part (g) in the above example didn't use any special properties of the differentiation transformation or $\mathbb{P}_1$ or $\mathbb{P}_2$; the proof of the theorem below is basically the same. You can find it in the textbook.

Theorem 7.14: Matrix of a transformation.
Suppose V and W are vector spaces with bases $\mathscr{B} = \{\vec{b}_1, \ldots, \vec{b}_c\}$ and $\mathscr{D}$, respectively, and $T : V \to W$ is a linear transformation. Then for all $\vec{v}$ in V,

$$[T(\vec{v})]_{\mathscr{D}} = A[\vec{v}]_{\mathscr{B}}$$

where $A = \begin{bmatrix} [T(\vec{b}_1)]_{\mathscr{D}} & \cdots & [T(\vec{b}_c)]_{\mathscr{D}} \end{bmatrix}$, which is called the **matrix of T relative to $\mathscr{B}$ and $\mathscr{D}$.**

Example 5. Let $T : \mathbb{P}_2 \to \mathbb{P}_1$ be the differentiation transformation. It can be shown that $\mathscr{B} = (\vec{b}_1, \vec{b}_2, \vec{b}_3) = (z - 1, z + 1, z^2 - 4)$ and $\mathscr{D} = (\vec{d}_1, \vec{d}_2) = (\pi, 5z)$ are bases of $\mathbb{P}_2$ and $\mathbb{P}_1$. Find the matrix of T relative to $\mathscr{B}$ and $\mathscr{D}$ and check that it works with $T(2z)$.

Definition. Suppose $T : V \to W$ is a linear transformation. The **null space** of T is the set of vectors $\vec{v}$ in V such that $T(\vec{v}) = \vec{0}$. The **range** of T is the set of vectors in W that can be written $T(\vec{v})$ for some $\vec{v}$ in V. In symbols:

- $\text{null}(T) = \left\{\vec{v} \text{ in } V \,:\, T(\vec{v}) = \vec{0}\right\}$.
- $\text{range}(T) = \left\{T(\vec{v}) \,:\, \vec{v} \text{ is in } V\right\}$.

If T is an $\mathbb{R}^c \to \mathbb{R}^r$ linear transformation, we know $T(\vec{x})$ can be written $A\vec{x}$, and the null space of A is the same thing as the null space of T. And since any linear combination of the columns of A can be written $A\vec{x}$, the column space of A is the range of T.

Theorem 7.15: Suppose $T : V \to W$ is a linear transformation. Then $\text{null}(T)$ is a subspace of V and $\text{range}(T)$ is a subspace of W.

The proof is left as an exercise.

Example 6. Let $T : \mathbb{P}_3 \to \mathbb{P}_3$ be the second derivative, that is, $T(\vec{p}(z)) = \frac{d^2}{dz^2}\vec{p}(z)$, and let $\mathscr{B}$ and $\mathscr{D}$ both equal $(1, z, z^2, z^3)$, the standard basis of $\mathbb{P}_3$.

(a) Find A, the matrix of T relative to $\mathscr{B}$ and $\mathscr{D}$.

(b) Find basis for the null space and column space of A.

(c) Find bases for the null space and range of T.

The next theorem generalizes the above example.

Theorem 7.16: Suppose $T : V \to W$ is a linear transformation, with $\mathscr{B} = \left\{\vec{b}_1, \dots, \vec{b}_c\right\}$ and $\mathscr{D}$ bases for V, W, respectively. Let A be the basis of T relative to $\mathscr{B}$ and $\mathscr{D}$. Then

(i) $\dim(\text{range}(T)) = \dim(\text{col}(A))$ and

(ii) $\dim(\text{null}(T)) = \dim(\text{null}(A))$.

The proof is relegated to the textbook, because it's really tedious, but here's the gist:

7.2 Exercises

For questions 5 through 15, determine if the given transformation is linear or not. Justify your answer.

5. $T : \mathbb{P}_2 \to \mathbb{P}_2, T(\vec{p}(z)) = \vec{p}(z) + 7$.

6. $T : \mathbb{P}_2 \to \mathbb{P}_3, T(\vec{p}(z)) = (z+3)\vec{p}(z)$.

7. $T : \mathbb{P}_2 \to \mathbb{R}^3, T(\vec{p}(z)) = (\vec{p}(0), \vec{p}(1), \vec{p}(2))$.

8. $T : \mathbb{P}_2 \to \mathbb{P}_2, T(a + bz + cz^2) = cz^2$.

9. $T : \mathbb{P}_2 \to \mathbb{R}^3, T(\vec{p}(z)) = (\vec{p}(0) + 8, \vec{p}(1) + 8, \vec{p}(2) + 8)$.

10. Let M be the vector space of all 4×4 matrices, and $\vec{p} = (8, 9, 10, 11)$. $T : M \to \mathbb{R}^4, T(\vec{A}) = \vec{A}\vec{p}$.

11. Let M be the vector space of all 4×4 matrices. $T : M \to M, T(\vec{A}) = \det(\vec{A})$.

12. Let M be the vector space of all 4×4 matrices. $T : M \to \mathbb{R}, T(\vec{A}) = \dim(\text{null } \vec{A})$.

13. $T : \mathbb{P} \to \mathbb{R}, T(\vec{p}(z)) = \int_{-1}^{1} \vec{p}(z)dz$.

14. $T : \mathbb{P} \to \mathbb{P}, T(\vec{p}(z)) = \int \vec{p}(z)dz$, with constant of integration always set to 0. For example, $T(1) = z$, and $T(z + z^3) = \frac{1}{2}z^2 + \frac{1}{4}z^4$.

15. Let $\vec{w} = (3, 4, 5)$, and $W = \text{span}(\vec{w})$. $T : \mathbb{R}^3 \to W, T(\vec{x}) = \text{proj}_W \vec{x}$.

16. (T/F?) If m, b are any real numbers, then $f : \mathbb{R} \to \mathbb{R}$ defined by $f(z) = mz + b$ is a linear transformation because the graph is a straight line.

17. (T/F?) If $T : V \to W$ is a linear transformation, then $S : V \to W$ defined by $S(\vec{v}) = 7T(\vec{v})$ is a linear transformation.

18. (T/F?) If $T : V \to W$ and $S : V \to W$ are linear transformations, then $T + S : V \to W$ defined by $(T + S)(\vec{v}) = T(\vec{v}) + S(\vec{v})$ is a linear transformation.

19. (T/F?) If $T : V \to W$ and $S : W \to U$ are linear transformations, then $TS : V \to U$ defined by $TS(\vec{v}) = T(S(\vec{v}))$ is a linear transformation.

20. Suppose $\dim(V) = 5, \dim(W) = 4$, $\mathscr{B}$ and $\mathscr{D}$ are bases of V and W respectively, and $T : V \to W$ is a linear transformation. Fill in the blanks: The size of the matrix for T relative to $\mathscr{B}$ and $\mathscr{D}$ is ____ × ____.

21. Prove theorem 7.15.

22. Prove the following theorem. [H]

> **Theorem 7.17:** Suppose $T : V \to W$ is a linear transformation. Then $\dim(\text{range}(T)) + \dim(\text{null}(T)) = \dim(V)$.

23. Suppose that $\mathscr{B} = (\vec{b}_1, \dots, \vec{b}_c)$ and $\mathscr{D} = (\vec{d}_1, \dots, \vec{d}_c)$ are two bases of a vector space V. The matrix $P = \left[[\vec{b}_1]_{\mathscr{D}} \ \cdots \ [\vec{b}_c]_{\mathscr{D}} \right]$ is called the *change of coordinates matrix from $\mathscr{B}$ to $\mathscr{D}$*. Show that $[\vec{v}]_{\mathscr{D}} = P[\vec{v}]_{\mathscr{B}}$ for all $\vec{v}$ in V.

24. Define a linear transformation $T : \mathbb{P}_2 \to \mathbb{R}^4$ by $T(\vec{p}(z)) = (\vec{p}(0), \vec{p}(1), \vec{p}(2), \vec{p}(3))$. Let $\mathscr{B} = (1, z, z^2)$ and $\mathscr{D} = (\vec{e}_1, \vec{e}_2, \vec{e}_3, \vec{e}_4)$ be the standard bases.

(a) Find A, the matrix of T relative to $\mathscr{B}$ and $\mathscr{D}$.

(b) Check that theorem 7.14 works with $[T(2 + 3z)]_{\mathscr{D}}$.

(c) How many pivot positions does A have? What does this say about $\dim(\text{null}(T))$ and $\dim(\text{range}(T))$?

25. The first four *Hermite polynomials*, $\mathscr{H} = (1, 2z, 4z^2 - 2, 8z^3 - 12z)$, form a basis of $\mathbb{P}_3$, and are useful in various areas of math and physics. Let $T : \mathbb{P}_3 \to \mathbb{P}_3$ be differentiation: $T(\vec{p}(z)) = \frac{d}{dz}\vec{p}(z)$. Find A, the matrix of T relative to $\mathscr{H}$ and $\mathscr{H}$.

26. Define a linear transformation $T : \mathbb{P}_3 \to \mathbb{P}_4$ by $T(\vec{p}(z)) = \int \vec{p}(z)dz$, with the constant of integration always set to 0. For example, $T(2z + z^3) = z^2 + \frac{1}{4}z^4$. Let $\mathscr{B} = (1, z, z^2, z^3)$ and $\mathscr{D} = (1, z, z^2, z^3, z^4)$ be the standard bases.

(a) Find A, the matrix of T relative to $\mathscr{B}$ and $\mathscr{D}$.

(b) Describe $\text{null}(A)$ and $\text{null}(T)$.

(c) Describe $\text{col}(A)$ and $\text{range}(T)$.

27. Define a linear transformation $T : \mathbb{P}_2 \to \mathbb{P}_3$ by $T(\vec{p}(z)) = (z - 5)\vec{p}(z)$. Let $\mathscr{B} = (1, z, z^2)$ and $\mathscr{D} = (1, z, z^2, z^3)$ be the standard bases.

(a) Find A, the matrix of T relative to $\mathscr{B}$ and $\mathscr{D}$.

(b) Describe $\text{null}(A)$ and $\text{null}(T)$.

(c) Describe $\text{col}(A)$ and $\text{range}(T)$.

28. Let M be the vector space of all 2×2 matrices, with basis:

$$\mathscr{B} = \vec{B}_1, \vec{B}_2, \vec{B}_3, \vec{B}_4 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

[H]22. Hint: Use thm 1.22.

Define $T : M \to \mathbb{R}^2$ by $T(\vec{A}) = A\begin{bmatrix}1\\2\end{bmatrix}$.

(a) Find A, the matrix of T relative to $\mathscr{B}$ and the standard basis of $\mathbb{R}^2$.

(b) Find a basis of null(A).

(c) Find a basis of null(T).

The next problem is about *eigenvectors* and *eigenvalues* of linear transformations, which mean the same thing as they did in chapter 4:

> **Definition.** Suppose $T : V \to V$ is a linear transformation. We say $\vec{u}$ is an **eigenvector** corresponding with an **eigenvalue** λ (a scalar) of T, if:
>
> $$T(\vec{u}) = \lambda\vec{u} \quad \text{and} \quad \vec{u} \neq \vec{0}$$

29. Let V be the set of all differentiable $\mathbb{R} \to \mathbb{R}$ functions.

(a) Define $T(\vec{f}(z)) = \frac{d}{dz}\vec{f}(z)$. Find a couple eigenvectors of T and their corresponding eigenvalues.

(b) Define $S(\vec{f}(z)) = \frac{d^2}{dz^2}\vec{f}(z)$. Find a couple eigenvectors of S that are *not* eigenvectors of T. [H]

[H]29. Hint: It's a basic kind of function that isn't polynomial, logarithmic, or exponential.

Appendix A

Proof-Writing Basics

This optional section covers the basics of writing proofs of "If P, then Q" and "P if and only if Q" statements, and should be covered before section 1.3 or 1.4 for students that have no experience with proof-writing.

A mathematical *proof* is a written (or spoken) argument, which should convince the reader beyond any doubt that a statement (sometimes called a *proposition* or a *theorem*) is true, as long as the reader has the background knowledge required to understand it. Writing and reading proofs is the main way that mathematicians do mathematical research, in the same way that experiments are the main way that scientists do scientific research. When you write a proof for an assignment in a math class, it serves several purposes:

- It shows that you understand why the statement is true.
- It gets you to study the definitions, theorems, and examples related to the statement you're proving.
- It gets you to practice your reasoning and argument-writing skills, which you can apply to other areas of life, and is one of the reasons you're going to college.

Before we get into writing proofs, let's look at the concept of a *definition*. In math, definitions are different than they are in, say, a dictionary. Dictionary definitions are *descriptive*, which means they try to *describe* the different ways people use a word. For example, when I google the word "literally" I get the following definition from Oxford Languages:

1. In a literal manner or sense; exactly. "The driver took it literally when asked to go straight across the traffic circle."
2. *(informal)* Used for emphasis or to express strong feeling while not being literally true. "I was literally blown away by the response I got."

Imagine if you needed to *prove*, beyond any doubt, that something is *literal* or not. It would be hopeless, because, for one, the first definition uses the word in its own definition. Even worse, the second definition is the opposite of the first!

In math, a definition can be *prescriptive*, which means that the definition is telling you exactly what the word means, every time it's used in the paper/book/course, with no room for nuance or interpretation. Sometimes one book, say, book A, will define a term differently than book B; but when you're talking about results in book A you need to use the definitions of that book to avoid confusion. When the authors of A and B get together to talk, they need to start by agreeing on which definition they're using before using the controversial term.

Here's an example:

Definition. An integer m is **even** if it can be written $m = 2k$ for some integer k. An integer n is **odd** if it can be written $n = 2k + 1$ for some integer k.

From this point on, any time you see the term "even integer" you know that's a shorthand way of saying "an integer that can be written as $2k$ for some integer k." And using any other meaning of the word, from everyday English, is a **bad idea**:

- One million is nice and even (*round?*) number but 998,576 is not.
- Zero is an odd (*special, unusual?*) number because it's the only number that can be added to other numbers without changing them.

In everyday English, these statements are ambiguous, but true in some sense. In a math class that uses the above definition, they are completely false.

Now, the purpose of definitions is to make statements shorter and easier to read. For example, consider the two statements:

1. If x, y are odd integers, then $x + y$ is an even integer.
2. If x, y are integers that can be written $x = 2k + 1$ and $y = 2c + 1$ for some integers c and k, then $x + y = 2p$ for some integer p.

These two statements mean exactly the same thing; the first one is easier-to-read shorthand for the second. When you prove the first statement, you need to keep in mind that you're really proving the second one.

Many students, when confronted with their first proof-writing assignments, don't know where to begin; the following strategy can help. Even experienced mathematicians follow similar strategies when they're stuck.

Strategy for proving statements of the form "If P, then Q":

1. On scratch paper, get some intuition for why the proof is true, with examples or pictures. (*Optional.*)
2. Start the proof by writing "Assume P" at the top, but with P written in expanded terms, using the definition.
3. Write "WTS [Want To Show] Q", but with Q in expanded terms, using the definition. *(Optional.)*
4. Use your assumption of P and any facts you know to show Q is true. Also, it's nice to end the proof with a symbol like "■" or "Q.E.D." to let the reader know that you're done.

Now, steps 1 and 3 are optional because a proof can be perfectly valid and convincing without them, but they're a good idea if you're not sure how to write the proof.

Example 1. Prove that if x, y are odd integers, then $x + y$ is an even integer.

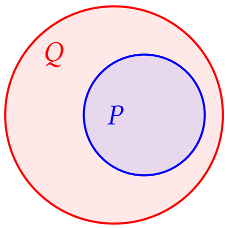
Definition. An integer n is **divisible by** an integer p if $n = pq$ for some integer q.

Example 2. Show that if n is an odd integer, then $n^2 - 1$ is divisible by 8.

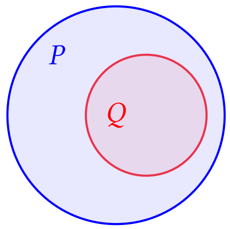
The statement "If not Q, then not P" is called the **contrapositive** of "If P, then Q." The picture below on the left should give you an idea why a statement is equivalent to its contrapositive. Sometimes it's easier to prove the contrapositive than the original statement.

The statement "If Q, then P" is called the **converse** of "If P, then Q." The figures below in the left and middle show converses of each other.

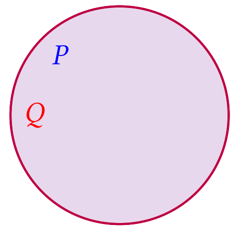
The statement "P if and only if Q" means "If P, then Q, *and* if Q, then P." In other words, if one of them is true, the other is true, and if one is false, the other is false.



If P, then Q.
If not Q, then not P.



If Q, then P.
If not P, then not Q.



P if and only if Q.

Example 3. Suppose n is an integer. Show that n is odd if and only if n^2 is odd.

So far, we've been showing that statements are *true*. But how do you show that a statement of the form "If P, then Q" is *false*? By finding a **counterexample**, which makes P true but Q false.

Example 4. Show that it is not true that if $a - 1$ is divisible by 8, then $a = n^2$ for some odd integer n.

A.0 Exercises

1. Show that if n, m are even integers, then $n + m$ is even.

2. Show that if n, m are odd integers, then $n + m$ is even.

3. Show that if a is even, then $3a$ is even.

4. Show that if n is an integer, then $(n + 1)^2 - n^2$ is odd.

5. Prove or find a counterexample to the statement: If n, m are integers and $n + m$ is odd, then n is odd.

6. Suppose a is an integer. Show that a is even if and only if $3a$ is even.

7. Suppose n is an integer. Show that n is odd if and only if $n + 5$ is odd.

8. Show that if n is an odd integer, then there exist integers p, q such that $p^2 - q^2 = n$.

9. Prove, or find a counterexample to the statement: There exist integers p, q such that $p^2 - q^2 = n$ if and only if n is odd.

Definition. An integer a is **prime** if it's greater than 1, and there do not exist integers p, q greater than 1 such that $a = pq$.

10. (*challenge*) Show that if a is an integer greater than 1, then $a^3 + 1$ is not prime.

11. (*challenge*) Show that if n is a non-prime integer, then $2^n - 1$ is not prime.

12. Suppose n is an integer. Find a counterexample to the statement: "n is prime if and only if $2^n - 1$ is prime."